

A LOCAL PROOF OF LANGLANDS'S FIRST MAIN LEMMA FOR LOCAL EPSILON FACTORS OVER NONARCHIMEDEAN LOCAL FIELDS

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ABSTRACT. We give a complete local proof of Langlands's First Main Lemma for normalized one-dimensional local epsilon factors over every nonarchimedean local field, in both mixed and equal characteristic. The proof reconstructs and completes the local method initiated by Dwork and organized by Langlands; its decisive p -adic calculations and guiding ramification-theoretic and stationary-phase ideas are due to them. The present paper unifies their normalizations and supplies omitted endpoint and non-stable wild calculations. Exact norm–trace correction formulas yield the equal-characteristic extension.

The author is also completing a local proof of Langlands's Second Main Lemma and will verify it in Lean. Together with the present result and Langlands's remaining reductions, that work will give a complete proof of the local construction of epsilon factors envisioned by Langlands, without global functional equations.

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1. INTRODUCTION

Let F be a nonarchimedean local field with finite residue field and Weil group W_F . Langlands's local epsilon factors are determined by their values on one-dimensional representations, multiplicativity, and compatibility with induction. Brauer induction reduces their construction to local constants of quasi-characters, but it also creates an independence problem: distinct Brauer decompositions must give the same result. In his notes on the functional equation of Artin L -functions [3], Langlands reduced this problem to four explicit local identities. The present paper gives a complete local proof of the identity he called the First Main Lemma.

Let K/F be cyclic of prime degree ℓ . For a continuous quasi-character $\chi_F : F^\times \rightarrow \mathbf{C}^\times$ and a nontrivial continuous additive character $\psi_F : F \rightarrow \mathbf{C}^\times$, put

$$\chi_K = \chi_F \circ N_{K/F}, \quad \psi_K = \psi_F \circ \text{Tr}_{K/F}.$$

Let $S(K/F)$ be the full group of continuous characters of $F^\times$ that are trivial on $N_{K/F}(K^\times)$.

Theorem 1.1 (First Main Lemma). *For every cyclic extension K/F of prime degree, every continuous quasi-character χ_F of $F^\times$, and every nontrivial continuous additive character ψ_F of F ,*

$$(1) \quad \Delta_K(\chi_K, \psi_K) \prod_{\mu \in S(K/F)} \Delta_F(\mu, \psi_F) = \prod_{\mu \in S(K/F)} \Delta_F(\mu \chi_F, \psi_F).$$

Here $\Delta_E(\theta, \psi)$ is Langlands's normalized local Gauss factor, recalled in Section 2. This is the identity quoted as Lemma 10.1 in Langlands, equation (12.1), p. 144, and used again on p. 257; see [3, pp. 144, 257].

Historical background. Dwork's 1954 thesis [4] contains the decisive p -adic calculations behind the cyclic-prime relation, while the formulation of the First Main Lemma and the ramification-theoretic and stationary-phase organization are due to Langlands. Dwork worked in mixed characteristic, in different normalizations, and Langlands's local manuscript was never completed: Chapter 10, which contained the conclusion of the First Main Lemma, is missing. After Deligne found a shorter proof using global functional equations, Langlands stopped preparing a complete version of the local argument; this history is described in the electronic edition of his notes [3].

The Second Main Lemma and the complete local construction. The present theorem settles the First Main Lemma, but Langlands's original local construction also requires the Second Main Lemma. The author is presently completing a companion local proof of the Second Main Lemma and will formalize and verify the finished proof in Lean 4. Once this is done, the Lean-verified First and Second Main Lemmas, together with Langlands's remaining reductions and the standard one-dimensional theory, will furnish a complete proof of the local construction and well-definedness of epsilon factors envisioned in [3]. In particular, the proof of the construction will no longer depend on a global functional equation or a globalization argument.

Contributions of the present paper. The proof is a reconstruction and completion of Dwork and Langlands rather than an independent construction. Nevertheless, several ingredients required for a complete theorem are supplied here rather than copied from either source. We first translate Dwork's mixed-characteristic calculations and Langlands's formulation into a common system of normalizations and notation. Langlands's Haar-integral definition is converted into an exact finite sum, and its nonvanishing is proved by Fourier duality, independently of Lamprecht's formula.

We then formulate stationary parameters as quotient classes, prove their transport under norm before choosing representatives, and track the phase change caused by every change of representative. This yields explicit high- and low-conductor tables in which the source representatives, common denominators, and residual coordinates are retained until their use. The paper supplies the missing endpoint and intermediate calculations, including the non-stable wild odd-prime range and the complete wild quadratic calculation, and organizes the unramified, tame, endpoint, stable, and non-stable cases into an exhaustive dispatch.

Among the additional mathematical bridges are the critical norm polynomial, the stable-to-critical trace comparison, the affine-translation vanishing needed to pass from actual source rows to normalized rows, and the branch matrix for characteristic-two critical refinements. Finally, exact norm–trace correction formulas isolate the mixed-characteristic terms and show directly that they vanish in equal characteristic. Thus the equal-characteristic theorem is a proved extension of the local calculation, not a formal assertion by analogy. The proof uses explicit local ramification and norm theory, stationary phase, and finite-field harmonic analysis. It uses neither local class field theory nor a global functional equation.

Logical organization of the proof. The argument is most transparent when the common reductions are made before the individual cases.

First, Langlands’s Haar-integral definition is converted into an exact finite sum on $U_E^0/U_E^{m_E(\theta)}$. The finite sum is proved nonzero by Fourier duality, independently of Lamprecht’s formula. This makes the local constant, the inverse-pairing identity, and the error quotient available without a later nonvanishing dependency.

Second, the stationary parameter attached to a character is constructed as a quotient class. Representatives are chosen only when a later formula needs one, and every representative change is accompanied by the corresponding translation of the critical function. This prevents a stationary phase from being replaced silently by the phase of an unrelated representative.

Third, the abstract transport of stationary classes under norm is proved before the explicit high- and low-conductor tables. The tables then retain the actual source representatives, common denominators, and residual coordinates until their final consumers. A common factor-separation identity isolates the admissible, elementary, critical, residual, and correction terms in the First Main quotient. The odd-prime and quadratic calculations differ only after this common stage.

Finally, twist invariance permits the base character to be chosen with minimal conductor in its $S(K/F)$ -orbit. The complete case tree is

K/F	case used in the proof
unramified	unramified comparison
ramified tame, $\ell = 2$	tame quadratic calculation
ramified tame, ℓ odd	tame odd calculation
wild, $\ell = 2$	wild quadratic calculation
wild, ℓ odd, $m \leq 1$	endpoint conductors
wild, ℓ odd, $2T \leq m$	stable range
wild, ℓ odd, $m > 1$, $m < 2T$	non-stable odd-prime calculation.

Here t is the unique lower ramification break in the ramified wild case, $T = t + 1$, and m is the conductor of the chosen minimal-orbit representative.

Equal characteristic. The local calculation is sufficiently exact to show where mixed characteristic is actually used. In the odd-prime non-stable case the last correction has the closed form

$$X = \frac{pB}{1 - n^{p-1}}.$$

The denominator is proved nonzero. In mixed characteristic the numerator is controlled by a ramification-theoretic valuation estimate; in equal characteristic p , the image of the integer p in F is zero and hence $X = 0$.

For a wild quadratic extension the corresponding exact correction is

$$X = \frac{2 \operatorname{Tr}_{K/F}(u)}{1 + N_{K/F}(u)}.$$

Again the denominator is nonzero, and in characteristic two the correction vanishes because $2 = 0$ in F . The only remaining quadratic boundary is excluded in equal characteristic by the fact that a ramified Artin–Schreier quadratic extension has odd lower break. Thus the equal-characteristic theorem is not obtained by formally extending a mixed-characteristic result: it follows from the exact local correction formulas proved here.

Organization. Section 2 establishes the two realizations of the local constant, proves nonvanishing, introduces the error term, and reduces the theorem to the displayed case tree. The next four sections develop the cyclic-prime ramification, stationary-phase, finite-field, and stationary-parameter tools. We then treat the unramified, tame, endpoint, stable, odd non-stable, and quadratic cases. The last section contains only the exhaustive dispatch.

2. LOCAL CONSTANTS AND THE ELEMENTARY REDUCTION

2.1. Local fields, characters, and exact conductors. Let E be a nonarchimedean local field. Write

$$\mathcal{O}_E = \{x \in E : v_E(x) \geq 0\}, \quad \mathfrak{p}_E = \{x \in E : v_E(x) \geq 1\},$$

and let $k_E = \mathcal{O}_E/\mathfrak{p}_E$ have cardinality q_E . We normalize $v_E(E^\times) = \mathbf{Z}$. Put

$$U_E^0 = \mathcal{O}_E^\times, \quad U_E^r = 1 + \mathfrak{p}_E^r \quad (r \geq 1).$$

All quasi-characters are continuous homomorphisms $E^\times \rightarrow \mathbf{C}^\times$, and all additive characters are continuous homomorphisms from the additive group of E to $\mathbf{C}^\times$.

Remark 2.1 (Characteristic convention). Write $p = \text{char } k_E$. Thus either $\text{char } E = 0$ with residue characteristic p , or $\text{char } E = p$. An integer occurring in an identity inside E denotes its image under $\mathbf{Z} \rightarrow E$; in particular $p = 0$ in equal characteristic p .

Definition 2.2 (Multiplicative conductor). The conductor exponent $m_E(\theta)$ is the least $m \geq 0$ for which θ is trivial on U_E^m . Thus $m_E(\theta) = 0$ precisely when θ is unramified.

Definition 2.3 (Additive conductor). For a nontrivial additive character ψ , the integer $n_E(\psi)$ is characterized by the condition that $\mathfrak{p}_E^{-n_E(\psi)}$ is the largest fractional ideal on which ψ is trivial.

These are Langlands's conventions; see [3, p. 5].

2.2. The Haar-integral and finite realizations of the local constant. Define the total phase

$$\text{ph}(z) = \begin{cases} z/|z|, & z \neq 0, \\ 1, & z = 0. \end{cases} \quad \mathcal{A}(z) := \text{ph}(z).$$

Let $m = m_E(\theta)$ and $n = n_E(\psi)$, and choose $\gamma \in E^\times$ such that

$$(2) \quad \gamma \mathcal{O}_E = \mathfrak{p}_E^{m+n}.$$

For a positive Haar measure du on U_E^0 , Langlands defines

$$(3) \quad \Delta_E(\theta, \psi) = \theta(\gamma) \text{ph} \left(\int_{U_E^0} \psi(u/\gamma) \theta(u)^{-1} du \right).$$

Put $Q_m = U_E^0/U_E^m$. For $\bar{u} \in Q_m$, represented by $u \in U_E^0$, define

$$(4) \quad g_{\theta, \psi, \gamma}(\bar{u}) = \psi(u/\gamma) \theta(u)^{-1}$$

and

$$(5) \quad G_E(\theta, \psi; \gamma) = \sum_{\bar{u} \in Q_m} g_{\theta, \psi, \gamma}(\bar{u}).$$

Proposition 2.4 (Finite realization and nonvanishing). *The summand in (4) is well defined. There is a positive real number c_m , depending only on the chosen Haar measure and m , such that*

$$(6) \quad \int_{U_E^0} \psi(u/\gamma) \theta(u)^{-1} du = c_m G_E(\theta, \psi; \gamma).$$

Moreover

$$(7) \quad G_E(\theta, \psi; \gamma) \neq 0.$$

Consequently

$$(8) \quad \Delta_E(\theta, \psi) = \theta(\gamma) \text{ph}(G_E(\theta, \psi; \gamma)),$$

and the zero branch in the definition of ph is never used for a local constant.

Proof. If $u/v \in U_E^m$, then $\theta(u) = \theta(v)$ and $u - v \in \mathfrak{p}_E^m$. By (2), $(u - v)/\gamma \in \mathfrak{p}_E^{-n}$, so $\psi(u/\gamma) = \psi(v/\gamma)$. Thus the summand descends to Q_m . The integrand is constant on every coset of U_E^m in U_E^0 , and all such cosets have the same positive measure c_m ; this proves (6).

It remains to prove nonvanishing. If $m = 0$, the quotient Q_0 is a singleton and $G_E(\theta, \psi; \gamma) = 1$. Assume $m > 0$ and put $A = \mathcal{O}_E/\mathfrak{p}_E^m$. The pairing

$$A \times A \longrightarrow \mathbf{C}^\times, \quad (\bar{x}, \bar{y}) \longmapsto \psi(xy/\gamma)$$

is perfect. Indeed, its annihilator in either variable is exactly $\mathfrak{p}_E^m$: this follows from (2) and the maximality of $\mathfrak{p}_E^{-n}$ in the definition of $n_E(\psi)$.

Extend the function $u \mapsto \theta(u)^{-1}$ on $A^\times$ by zero to a function $f : A \rightarrow \mathbf{C}$, and write

$$\hat{f}(x) = \sum_{u \in A^\times} \theta(u)^{-1} \psi(xu/\gamma).$$

Then $\hat{f}(1) = G_E(\theta, \psi; \gamma)$. If $x \in A^\times$, the substitution $u \mapsto x^{-1}u$ gives

$$(9) \quad \hat{f}(x) = \theta(x) \hat{f}(1).$$

If x is not a unit, exactness of the conductor provides $a \in U_E^{m-1}$ with $\theta(a) \neq 1$. Since $x(a-1)u/\gamma \in \mathfrak{p}_E^{-n}$ for every $u \in U_E^0$, multiplication by a permutes $A^\times$ and gives

$$\hat{f}(x) = \theta(a)^{-1} \hat{f}(x),$$

whence $\hat{f}(x) = 0$. If $\hat{f}(1)$ were zero, (9) and the preceding vanishing would make every Fourier coefficient of f zero. Perfect finite Fourier duality would then give $f = 0$, contradicting $f(1) = 1$. Thus (7) holds. Finally the positive factor c_m does not change phase, proving (8). $\square$

Corollary 2.5 (Nonvanishing of the local Gauss integral). *For every quasi-character θ and every nontrivial additive character ψ , the integral in (3) is nonzero.*

Lemma 2.6 (Independence of the admissible denominator). *The right side of (3) is independent of the choice of γ satisfying (2).*

Proof. Let $\gamma' = a\gamma$ with $a \in U_E^0$. Multiplication by a permutes Q_m , and (5) gives

$$G_E(\theta, \psi; \gamma') = \theta(a)^{-1} G_E(\theta, \psi; \gamma).$$

The restriction of θ to U_E^0 factors through the finite group Q_m , so $|\theta(a)| = 1$. By nonvanishing,

$$\text{ph}(G_E(\theta, \psi; \gamma')) = \theta(a)^{-1} \text{ph}(G_E(\theta, \psi; \gamma)).$$

Since $\theta(\gamma') = \theta(a)\theta(\gamma)$, the two factors cancel. $\square$

The same comparison shows immediately that the value is independent of the positive normalization of Haar measure.

For later use, let k be a finite field, let $\varphi : k \rightarrow \mathbf{C}^\times$ be nontrivial, and let $\eta : k^\times \rightarrow \mathbf{C}^\times$ be multiplicative. Define

$$(10) \quad \tau_k(\eta, \varphi) = - \sum_{x \in k^\times} \eta(x)^{-1} \varphi(x).$$

Thus $\tau_k(1, \varphi) = 1$.

Lemma 2.7 (Normalization of the residual additive character). *Let ψ_F have conductor exponent n , and fix a uniformizer ϖ_F . The formula*

$$\bar{\psi}(\bar{x}) = \psi_F(\tilde{x} \varpi_F^{-n-1}) \quad (\bar{x} \in k_F)$$

defines a nontrivial additive character of k_F , independently of the lift $\tilde{x}$ modulo $\mathfrak{p}_F$. If ψ_{can} is a prescribed nontrivial additive character of k_F , there is a unit $a \in \mathcal{O}_F^\times$ such that the residual character attached to $\psi_F^a(x) = \psi_F(ax)$ is ψ_{can} .

Proof. Well-definedness follows because two lifts differ by an element of $\mathfrak{p}_F$, and therefore their arguments differ by an element of $\mathfrak{p}_F^{-n}$, on which ψ_F is trivial. Nontriviality follows from the maximality of $\mathfrak{p}_F^{-n}$. Every additive character of the finite field k_F is uniquely of the form

$$x \mapsto \exp\left(\frac{2\pi i}{p} \operatorname{Tr}_{k_F/\mathbf{F}_p}(cx)\right),$$

because the trace pairing $k_F \times k_F \rightarrow \mathbf{F}_p$ is nondegenerate. If $\bar{\psi}$ and ψ_{can} correspond to c and c_0 , choose a unit whose residue is c_0/c . $\square$

If $m_E(\theta) = 0$ and ϖ_E is a uniformizer, then $\gamma = \varpi_E^{n_E(\psi)}$ is admissible and

$$(11) \quad \Delta_E(\theta, \psi) = \theta(\varpi_E)^{n_E(\psi)}.$$

If $m_E(\theta) = 1$, define

$$\bar{\theta}(\bar{u}) = \theta(u), \quad \bar{\psi}(\bar{x}) = \psi(x/\varpi_E^{n_E(\psi)+1}).$$

Then

$$(12) \quad \Delta_E(\theta, \psi) = -\theta(\varpi_E)^{n_E(\psi)+1} \mathcal{A}(\tau_{k_E}(\bar{\theta}, \bar{\psi})).$$

These are the exact local-to-residue-field formulas used in the unramified, tame, and endpoint cases.

2.3. Norm and trace, elementary identities, and the error term. Let K/F be a finite extension. Write

$$N = N_{K/F} : K^\times \rightarrow F^\times, \quad T = \operatorname{Tr}_{K/F} : K \rightarrow F.$$

For a nontrivial additive character ψ_F and a quasi-character χ_F , put

$$\psi_K = \psi_F \circ T, \quad \chi_K = \chi_F \circ N.$$

If K/F is abelian, set

$$(13) \quad S(K/F) = \{\mu : F^\times \rightarrow \mathbf{C}^\times : \mu \text{ continuous and } \mu \circ N = 1\}.$$

By the universal property of the quotient, this is the group of continuous characters of the topological quotient $F^\times/N(K^\times)$. For cyclic K/F of prime degree, the quotient and $S(K/F)$ are proved directly to have order ℓ in Corollary 3.12 below. That calculation uses the norm filtration and not local class field theory.

Lemma 2.8 (Change of additive character). *For $a \in E^\times$, put $\psi^a(x) = \psi(ax)$. Then*

$$(14) \quad \Delta_E(\theta, \psi^a) = \theta(a) \Delta_E(\theta, \psi).$$

Proof. If $v_E(a) = r$, then $n_E(\psi^a) = n_E(\psi) + r$. Hence $a\gamma$ is admissible for (θ, ψ^a) whenever γ is admissible for (θ, ψ) , and $\psi^a(u/(a\gamma)) = \psi(u/\gamma)$. $\square$

Corollary 2.9 (Scaling invariance of the First Main identity). *For every $a \in F^\times$, the First Main Lemma for ψ_F is equivalent to the First Main Lemma for ψ_F^a .*

Proof. By Corollary 3.12, the group $S(K/F)$ has ℓ elements. The two sides acquire the respective factors

$$\chi_K(a) \prod_{\mu \in S(K/F)} \mu(a) \quad \text{and} \quad \chi_F(a)^\ell \prod_{\mu \in S(K/F)} \mu(a).$$

Since $N_{K/F}(a) = a^\ell$ for $a \in F^\times$, these factors agree. $\square$

Lemma 2.10 (The trivial character). *For every nontrivial additive character ψ ,*

$$\Delta_E(1, \psi) = 1.$$

Proof. Here $m_E(1) = 0$. Formula (11) gives the result. $\square$

Lemma 2.11 (Inverse pairing). *Let μ be a finite-order character of $E^\times$. Then*

$$(15) \quad \Delta_E(\mu, \psi) \Delta_E(\mu^{-1}, \psi) = \mu(-1).$$

Proof. Choose a common admissible γ and write

$$I_\mu = \int_{U_E^0} \psi(u/\gamma) \mu(u)^{-1} du.$$

Both I_μ and $I_{\mu^{-1}}$ are nonzero by Corollary 2.5. Since μ and ψ are unitary,

$$\overline{I_\mu} = \int_{U_E^0} \psi(-u/\gamma) \mu(u) du = \mu(-1) I_{\mu^{-1}}.$$

The product of the two phases is therefore $\mu(-1)^{-1} = \mu(-1)$, and the factors $\mu(\gamma)$ and $\mu^{-1}(\gamma)$ cancel. $\square$

Corollary 2.12. *If K/F is cyclic of odd prime degree, then*

$$(16) \quad \prod_{\mu \in S(K/F)} \Delta_F(\mu, \psi_F) = 1.$$

Proof. The finiteness of $S(K/F)$ follows from Corollary 3.12. For $x \in F^\times$, viewed as an element of $K^\times$, one has $N_{K/F}(x) = x^\ell$. Hence, for every $\mu \in S(K/F)$,

$$\mu(x)^\ell = \mu(N_{K/F}(x)) = 1.$$

Thus every nontrivial μ has order ℓ , which is odd, and in particular $\mu(-1) = 1$. Pair μ with μ^{-1} and use Lemma 2.10 for the trivial character. $\square$

Definition 2.13 (Error term). Define

$$(17) \quad \mathcal{E}_{K/F}(\chi_F) = \frac{\Delta_K(\chi_K, \psi_K) \prod_{\mu \in S(K/F)} \Delta_F(\mu, \psi_F)}{\prod_{\mu \in S(K/F)} \Delta_F(\mu \chi_F, \psi_F)}.$$

The denominator is nonzero by Corollary 2.5, and the First Main Lemma is the assertion $\mathcal{E}_{K/F}(\chi_F) = 1$.

Lemma 2.14 (Twist invariance). *For every $\nu \in S(K/F)$,*

$$\mathcal{E}_{K/F}(\nu \chi_F) = \mathcal{E}_{K/F}(\chi_F).$$

Proof. Since $\nu \circ N = 1$, the character on $K^\times$ is unchanged. Multiplication by ν permutes the full group $S(K/F)$, including its trivial character, so the denominator is only reindexed. $\square$

2.4. Reduction to minimal norm-character orbits and the case tree.

Proposition 2.15 (Minimal-orbit reduction). *To prove Theorem 1.1, it is enough, for each $S(K/F)$ -orbit of quasi-characters, to prove $\mathcal{E}_{K/F}(\chi_F) = 1$ for one member of minimal multiplicative conductor. For such a member, the proof reduces to the following mutually exhaustive branches:*

- (i) K/F is unramified;
- (ii) K/F is ramified tame and $\ell = 2$;
- (iii) K/F is ramified tame and ℓ is odd;
- (iv) K/F is wild and $\ell = 2$;
- (v) K/F is wild, ℓ is odd, and $m_F(\chi_F) \leq 1$;
- (vi) K/F is wild, ℓ is odd, $m_F(\chi_F) > 1$, and $2(t+1) \leq m_F(\chi_F)$;
- (vii) K/F is wild, ℓ is odd, $m_F(\chi_F) > 1$, and $m_F(\chi_F) < 2(t+1)$.

Here t is the unique lower break in the ramified wild case.

Proof. The group $S(K/F)$ is finite by Corollary 3.12, so every orbit contains a member of minimal conductor. Twist invariance transports the result from that member to the entire orbit. A ramified extension of prime degree is totally ramified. It is either tame, or wild with $\ell = \text{char } k_F$. Since ℓ is prime, the degree is quadratic or odd. In the wild odd-degree case the conductor is either at most one or greater than one; in the latter range the two final inequalities are complementary. Thus the list is exhaustive. $\square$

3. RAMIFICATION NOTATION AND THE CYCLIC-PRIME NORM THEOREM

Let K/F be a finite Galois extension of nonarchimedean local fields, with Galois group G . For an integer $i \geq -1$, the lower ramification group is

$$G_i = \{\sigma \in G : \sigma(x) \equiv x \pmod{\mathfrak{p}_K^{i+1}} \text{ for every } x \in \mathcal{O}_K\}.$$

For a real $u \geq -1$, put $G_u = G_{\lceil u \rceil}$. Define the Herbrand function and its inverse by

$$(18) \quad \varphi_{K/F}^H(u) = \int_0^u \frac{ds}{[G_0 : G_s]}, \quad \psi_{K/F}^H = (\varphi_{K/F}^H)^{-1}.$$

The superscript H is used throughout to prevent confusion with additive characters. This notation avoids the conflict noted by Langlands on pp. 28 and 62.

When K/F is cyclic of prime degree ℓ and ramified, it is totally ramified and there is a unique integer $t \geq 0$ such that

$$(19) \quad G = G_0 = \cdots = G_t, \quad G_{t+1} = 1.$$

Consequently

$$(20) \quad \psi_{K/F}^H(u) = \begin{cases} u, & -1 \leq u \leq t, \\ t + \ell(u - t), & u \geq t. \end{cases}$$

Indeed the integrand in (18) is 1 up to t and $1/\ell$ after t . This is the cyclic-prime form of [3, Lemma 6.3, p. 29].

The only non-elementary ramification statement needed in this section is the following graded norm theorem. We state it separately in order to avoid a circular argument based on correcting norms successively at levels below the break.

Theorem 3.1 (Graded norm theorem for a cyclic prime extension). *Let K/F be ramified cyclic of prime degree ℓ , with lower break t . For $i \geq 0$ write*

$$\mathrm{gr}^i U_E = U_E^i / U_E^{i+1},$$

where $\mathrm{gr}^0 U_E = k_E^\times$ and, for $i \geq 1$, $\mathrm{gr}^i U_E$ is identified with the additive group of k_E . Then the norm induces maps

$$\overline{N}_i : \mathrm{gr}^i U_K \longrightarrow \mathrm{gr}^i U_F \quad (0 \leq i \leq t)$$

with the following properties.

- (i) For $0 \leq i < t$, $\overline{N}_i$ is an isomorphism.
- (ii) At $i = t$, its kernel and cokernel both have order ℓ . Its kernel is the image of G_t/G_{t+1} under the ramification map

$$\sigma \longmapsto \begin{cases} \overline{\sigma(\pi_K)/\pi_K}, & t = 0, \\ \overline{(\sigma(\pi_K) - \pi_K)/\pi_K^{t+1}}, & t > 0. \end{cases}$$

- (iii) For every integer $r > t$, put $a = \psi_{K/F}^H(r)$. Norm induces an isomorphism

$$U_K^a / U_K^{a+1} \xrightarrow{\sim} U_F^r / U_F^{r+1}.$$

- (iv) The norm image is closed. Consequently the graded isomorphisms in (iii) give surjectivity by successive approximation:

$$N(U_K^{\psi_{K/F}^H(r)}) = U_F^r, \quad N(U_K^{\psi_{K/F}^H(r)+1}) = U_F^{r+1}.$$

This is the cyclic-prime specialization of [2, Chapter V, §3, Propositions 4–5 and Corollaries 1–3].

Proof. The graded maps and their kernels are those of [2, Chapter V, §3, Propositions 4–5]; their surjectivity and the resulting norm equalities are [2, Chapter V, §3, Corollaries 1–3]. For completeness, the passage from the graded statement to (iv) is as follows. Below the unique break the norm maps on successive unit quotients are isomorphisms. At the break, the ramification homomorphism displayed in the statement identifies the kernel; source and target have the same order, so the kernel and cokernel both have order ℓ . Above the break, Herbrand's change of

numbering gives the isomorphism in (iii). The subgroup $N(U_K^a)$ is compact and hence closed in U_F^0 . Successive lifting through the graded isomorphisms therefore converges to a norm and gives the two equalities in (iv). $\square$

Lemma 3.2 (Frobenius differences and trace-zero hyperplanes). *Let $k/\mathbf{F}_p$ be a finite field extension and let $\lambda \in k^\times$. Then the $\mathbf{F}_p$ -linear maps*

$$\wp(z) = z^p - z, \quad \wp_\lambda(z) = z^p - \lambda^{p-1}z$$

satisfy

$$\ker(\wp) = \mathbf{F}_p, \quad \text{im}(\wp) = \ker(\text{Tr}_{k/\mathbf{F}_p}),$$

and

$$(21) \quad \ker(\wp_\lambda) = \lambda \mathbf{F}_p, \quad \text{im}(\wp_\lambda) = \lambda^p \ker(\text{Tr}_{k/\mathbf{F}_p}).$$

Under the perfect trace pairing $(a, x) \mapsto \text{Tr}_{k/\mathbf{F}_p}(ax)$, the annihilator of the last image is $\lambda^{-p}\mathbf{F}_p$.

Proof. The kernel of $\wp$ is the prime field. Hence its image has $\mathbf{F}_p$ -dimension $[k : \mathbf{F}_p] - 1$. Frobenius invariance of the trace gives $\text{Tr}(z^p - z) = 0$, so the image is contained in the trace-zero hyperplane; the two spaces have the same dimension and therefore coincide. Since

$$\wp_\lambda(\lambda y) = \lambda^p(y^p - y),$$

the scaled kernel and image follow. The orthogonal complement of $\ker(\text{Tr}_{k/\mathbf{F}_p})$ is $\mathbf{F}_p$: it contains $\mathbf{F}_p$, and both sides have dimension one. Scaling the second variable by λ^p gives the annihilator $\lambda^{-p}\mathbf{F}_p$. $\square$

Proposition 3.3 (Critical norm polynomial at a positive wild break). *Let K/F be totally ramified cyclic of prime degree $p = \text{char } k_F$, with positive lower break t , and identify the residue fields with k . Choose a uniformizer ϖ_K , put $\varpi_F = N_{K/F}(\varpi_K)$, choose a generator σ of $\text{Gal}(K/F)$, and set*

$$\lambda = \frac{\overline{\sigma(\varpi_K) - \varpi_K}}{\varpi_K^{t+1}} \in k^\times.$$

In the source coordinate $1 + \varpi_K^t \tilde{z}$ and the target coordinate $1 + \varpi_F^t \tilde{w}$, the critical graded norm is the additive polynomial

$$(22) \quad P_\lambda(Z) = Z^p - \lambda^{p-1}Z.$$

Consequently

$$\ker P_\lambda = \lambda \mathbf{F}_p, \quad \text{im } P_\lambda = \lambda^p \ker(\text{Tr}_{k/\mathbf{F}_p}),$$

and its annihilator under the trace pairing is $\lambda^{-p}\mathbf{F}_p$. The linear coefficient is the normalized trace:

$$(23) \quad \frac{\overline{\text{Tr}_{K/F}(\varpi_K^t)}}{\varpi_F^t} = -\lambda^{p-1}.$$

Proof. The degree- p endpoint of the exact norm expansion is

$$N_{K/F}(\tilde{z}\varpi_K^t) = \tilde{z}^p \varpi_F^t,$$

so the leading coefficient of the induced critical polynomial is one. Its linear term is the trace term in the norm expansion, hence its coefficient is $\overline{\text{Tr}_{K/F}(\varpi_K^t)}/\varpi_F^t$. The graded norm theorem shows that the induced map is additive and that its kernel is the ramification line $\lambda \mathbf{F}_p$. Hence the monic additive polynomial is

$$\prod_{j \in \mathbf{F}_p} (Z - j\lambda) = Z^p - \lambda^{p-1}Z.$$

Comparison of the linear coefficients gives (23); the remaining assertions are Lemma 3.2. $\square$

Lemma 3.4 (Strong wild symmetric-coefficient estimate). *Assume that K/F is wildly ramified cyclic of prime degree $\ell = p = \text{char } k_F$, with lower break t . Put*

$$T = t + 1, \quad D = v_K(\mathfrak{D}_{K/F}) = (p - 1)T.$$

For $x \in K$ let $s_j(x)$ be the j th elementary symmetric function of its p conjugates. If $v_K(x) \geq a$, then, for $1 \leq j \leq p - 1$,

$$(24) \quad v_F(s_j(x)) \geq \left\lfloor \frac{ja + (p - 1)T}{p} \right\rfloor.$$

For the last coefficient,

$$(25) \quad v_F(s_p(x)) = v_F(N_{K/F}x) = v_K(x) \geq a.$$

Proof. We first recall the trace-ideal equality in the present normalization. For every integer q ,

$$(26) \quad \text{Tr}_{K/F}(\mathfrak{p}_K^q) = \mathfrak{p}_F^{\lfloor (q+D)/p \rfloor}.$$

Indeed, trace duality identifies the inverse different with $\mathfrak{p}_K^{-D}$. The inclusion $\text{Tr}(\mathfrak{p}_K^q) \subseteq \mathfrak{p}_F^r$ is equivalent to $\text{Tr}(\mathfrak{p}_K^{q-pr}) \subseteq \mathcal{O}_F$, hence to $q - pr \geq -D$. The largest such r is $\lfloor (q + D)/p \rfloor$, proving (26).

Put $P_i(x) = \text{Tr}_{K/F}(x^i)$. Since $x^i \in \mathfrak{p}_K^{ia}$, equation (26) gives

$$(27) \quad v_F(P_i(x)) \geq \left\lfloor \frac{ia + D}{p} \right\rfloor \quad (i \geq 1).$$

We now use Newton's identities

$$(28) \quad js_j(x) = \sum_{i=1}^j (-1)^{i-1} s_{j-i}(x) P_i(x), \quad s_0(x) = 1.$$

Because $j < p$, the integer j is a unit of $\mathcal{O}_F$. We prove (24) by induction on j . For $j = 1$ it is (27). Suppose it is known below j . The term $i = j$ in (28) has the required valuation by (27). For $i < j$, the induction hypothesis and (27) give

$$\left\lfloor \frac{(j-i)a + D}{p} \right\rfloor + \left\lfloor \frac{ia + D}{p} \right\rfloor.$$

For real numbers A, B one has $\lfloor A \rfloor + \lfloor B \rfloor \geq \lfloor A + B \rfloor - 1$. Here the sum before taking floors is

$$\frac{ja + D}{p} + \frac{D}{p}.$$

The extension is wild, so $T \geq 2$ and $D/p = (p - 1)T/p \geq 1$. Hence the last displayed integral lower bound is at least $\lfloor (ja + D)/p \rfloor$. Every term in (28) therefore has the required valuation. Finally $s_p(x) = N_{K/F}(x)$, and total ramification gives $v_F(Nx) = v_K(x)$. $\square$

Corollary 3.5 (Norm truncation at an explicit depth). *Let $q, r \geq 0$ and assume*

$$(29) \quad 2q + (p - 1)T \geq pr.$$

Then, for every $y \in \mathfrak{p}_K^q$,

$$(30) \quad N_{K/F}(1 + y) \equiv 1 + \text{Tr}_{K/F}(y) + N_{K/F}(y) \pmod{\mathfrak{p}_F^r}.$$

Proof. For $2 \leq j \leq p - 1$, Lemma 3.4 gives

$$v_F(s_j(y)) \geq \left\lfloor \frac{jq + (p - 1)T}{p} \right\rfloor \geq \left\lfloor \frac{2q + (p - 1)T}{p} \right\rfloor \geq r.$$

The exact norm expansion then gives (30). $\square$

Proposition 3.6 (Stable-to-critical normalized trace). *Let K/F be totally ramified cyclic of odd prime degree p , with positive lower break t . Choose ϖ_K , put $\varpi_F = N_{K/F}(\varpi_K)$, and define λ as in Proposition 3.3. Suppose $d \geq t + 1$ and put*

$$d' = pd - \frac{p - 1}{2}t.$$

Then

$$(31) \quad \frac{\varpi_F}{\varpi_K^p} \equiv 1 \pmod{\mathfrak{p}_K^{t+1}},$$

$$(32) \quad \varpi_K^{2d'} \equiv \varpi_F^{2d-t} \varpi_K^t \pmod{\mathfrak{p}_K^{2d'+1}},$$

and

$$(33) \quad \frac{\mathrm{Tr}_{K/F}(\varpi_K^{2d'})}{\varpi_F^{2d}} \equiv \frac{\mathrm{Tr}_{K/F}(\varpi_K^t)}{\varpi_F^t} \pmod{\mathfrak{p}_F}.$$

The two normalized traces are units, and their common residue is the critical norm coefficient

$$(34) \quad \frac{\overline{\mathrm{Tr}_{K/F}(\varpi_K^{2d'})}}{\varpi_F^{2d}} = \frac{\overline{\mathrm{Tr}_{K/F}(\varpi_K^t)}}{\varpi_F^t} = -\lambda^{p-1}.$$

Proof. For $j \in \mathbf{F}_p$, put

$$a_j = \frac{\sigma^j(\varpi_K)}{\varpi_K} - 1.$$

The ramification homomorphism gives $a_j \in \mathfrak{p}_K^t$ and $\overline{a_j/\varpi_K^t} = j\lambda$. Products of at least two of the a_j lie in $\mathfrak{p}_K^{t+1}$, and $\sum_{j \in \mathbf{F}_p} j = 0$. Hence

$$\frac{\varpi_F}{\varpi_K^p} = \prod_{j \in \mathbf{F}_p} (1 + a_j) \equiv 1 \pmod{\mathfrak{p}_K^{t+1}},$$

which proves (31). Since $2d' = p(2d - t) + t$, raising the inverse ratio to the power $2d - t$ and multiplying by ϖ_K^t gives (32).

The trace-ideal formula (26) sends the difference in (32) into $\mathfrak{p}_F^{2d+1}$. After division by ϖ_F^{2d} this is exactly (33). In the critical source coordinate the linear coefficient of the graded norm is $\overline{\mathrm{Tr}(\varpi_K^t)/\varpi_F^t}$. Proposition 3.3 identifies it with $-\lambda^{p-1}$, which is nonzero. This proves (34) and the unit assertions. $\square$

Theorem 3.7 (Cyclic-prime norm filtration). *Let K/F be ramified cyclic of prime degree ℓ , with break t . Put $d_{K/F} = v_K(\mathfrak{D}_{K/F})$. Then:*

- (a) $d_{K/F} = (\ell - 1)(t + 1)$.
- (b) For every integer a ,

$$(35) \quad \mathrm{Tr}_{K/F}(\mathfrak{p}_K^a) = \mathfrak{p}_F^{\lfloor (a+d_{K/F})/\ell \rfloor}.$$

- (c) Norm induces isomorphisms

$$(36) \quad K^\times/U_K^t \xrightarrow{\sim} F^\times/U_F^t, \quad U_K^r/U_K^t \xrightarrow{\sim} U_F^r/U_F^t \quad (0 \leq r \leq t).$$

The map $U_K^t/U_K^{t+1} \rightarrow U_F^t/U_F^{t+1}$ has cokernel of order ℓ .

- (d) For every integer $r > t$,

$$(37) \quad N_{K/F}(U_K^{\psi_{K/F}^H(r)}) = U_F^r, \quad N_{K/F}(U_K^{\psi_{K/F}^H(r)+1}) = U_F^{r+1}.$$

Proof. Choose a uniformizer π_K . For every $\sigma \neq 1$ one has $v_K(\sigma(\pi_K) - \pi_K) = t + 1$. The different formula

$$v_K(\mathfrak{D}_{K/F}) = \sum_{\sigma \neq 1} v_K(\sigma(\pi_K) - \pi_K),$$

which is [2, Chapter IV, §1, Proposition 4], therefore gives (a).

For (b), the trace-dual of $\mathfrak{p}_K^a$ under the pairing $(x, y) \mapsto \mathrm{Tr}_{K/F}(xy)$ is $\mathfrak{p}_K^{-a-d_{K/F}}$ by [2, Chapter III, §3, Proposition 7]. Write $\mathrm{Tr}(\mathfrak{p}_K^a) = \mathfrak{p}_F^c$. An element $y \in F$ belongs to the dual of this trace ideal if and only if

$$y\mathfrak{p}_K^a \subseteq \mathfrak{p}_K^{-d_{K/F}}, \quad \text{equivalently} \quad \ell v_F(y) + a \geq -d_{K/F}.$$

Thus the dual ideal is $\mathfrak{p}_F^{[(-a-d_{K/F})/\ell]}$, whereas the dual of $\mathfrak{p}_F^c$ is $\mathfrak{p}_F^{-c}$. Hence

$$-c = \left\lceil \frac{-a - d_{K/F}}{\ell} \right\rceil, \quad c = \left\lfloor \frac{a + d_{K/F}}{\ell} \right\rfloor,$$

which proves (35).

For (c), apply parts (i) and (ii) of Theorem 3.1. Splicing the graded isomorphisms for $0 \leq i < t$ gives the second map in (36). The valuation map is preserved because the residue degree is one: for $x \in K^\times$, $v_F(Nx) = v_K(x)$.

If $t = 0$ (the tame case), then $K^\times/U_K^t = K^\times/U_K^0 \simeq \mathbf{Z}$, and norm is the identity on this valuation quotient. If $t > 0$ (the wild case), the residue norm on $U_K^0/U_K^1 = k_F^\times$ is $x \mapsto x^\ell = x^p$, hence is the Frobenius automorphism; the remaining positive graded pieces below t are isomorphisms by Theorem 3.1. These two separate arguments give the first map in (36). The order- ℓ cokernel at the critical level is exactly Theorem 3.1(ii).

Part (d) is part (iv) of Theorem 3.1. The successive-approximation step is worth spelling out. Given $u_0 \in U_F^r$, choose x_0 whose norm has the same class modulo U_F^{r+1} . If $u_0/N(x_0) \in U_F^{r+1}$, choose x_1 at the next Herbrand level, and continue. The partial products of the x_i are Cauchy, their norms converge to u_0 , and continuity of norm gives the desired preimage. Closedness of the norm image ensures that no limit leaves the image. $\square$

Corollary 3.8 (Different and additive conductors). *Let $\psi_K = \psi_F \circ \text{Tr}_{K/F}$. If K/F is totally ramified of degree ℓ , then*

$$(38) \quad n_K(\psi_K) = \ell n_F(\psi_F) + d_{K/F}.$$

Proof. The largest fractional ideal I on which $\psi_F \circ \text{Tr}$ is trivial is characterized by $\text{Tr}(I) \subseteq \mathfrak{p}_F^{-n_F(\psi_F)}$. By the trace duality used in the proof of (35), this ideal is $\mathfrak{p}_K^{-\ell n_F(\psi_F) - d_{K/F}}$. $\square$

Corollary 3.9 (Ramified norm quotient and norm characters). *Let K/F be ramified cyclic of prime degree ℓ . Then $F^\times/N(K^\times)$ is cyclic of order ℓ , the group $S(K/F)$ has ℓ elements, and every nontrivial $\mu \in S(K/F)$ has conductor exponent $t + 1$.*

Proof. The first isomorphism in Theorem 3.7(c) shows that

$$F^\times = N(K^\times)U_F^t.$$

Its injectivity also shows that

$$N(K^\times) \cap U_F^t = N(U_K^t).$$

Consequently

$$(39) \quad F^\times/N(K^\times) \simeq U_F^t/N(U_K^t).$$

Taking $r = t + 1$ in (37) gives

$$U_F^{t+1} = N(U_K^{\psi_{K/F}^H(t+1)}) \subseteq N(U_K^t),$$

because $\psi_{K/F}^H(t+1) > t$. Hence

$$U_F^t/N(U_K^t) \simeq \text{coker} \left(U_K^t/U_K^{t+1} \longrightarrow U_F^t/U_F^{t+1} \right).$$

By Theorem 3.7(c), this cokernel has order ℓ . Thus $F^\times/N(K^\times)$ has order ℓ , and is cyclic because ℓ is prime.

Moreover, $N(K^\times)$ contains the open subgroup U_F^{t+1} , so the quotient is finite and discrete. The elements of $S(K/F)$ are therefore exactly the complex characters of this cyclic quotient. A character is determined by the image of a generator, which may be any ℓ th root of unity. Hence $S(K/F)$ has ℓ elements.

Finally, (39) shows that U_F^t maps onto the norm quotient, whereas U_F^{t+1} maps to the identity. A nontrivial norm character is therefore nontrivial on U_F^t and trivial on U_F^{t+1} . Its conductor exponent is exactly $t + 1$. $\square$

Proposition 3.10 (Conductor of norm pullback). *Let $m = m_F(\chi_F)$ and $m' = m_K(\chi_F \circ N)$.*

(i) If $m > t + 1$, then

$$(40) \quad m' - 1 = \psi_{K/F}^H(m - 1).$$

(ii) If $m \leq t$, then $m' = m$.

(iii) If $m = t + 1$, then $m' \leq m$; moreover $m' < m$ if and only if there is a $\mu \in S(K/F)$ for which $m_F(\mu\chi_F) < t + 1$.

Proof. For (i), put $a = \psi_{K/F}^H(m - 1)$. By (37), $N(U_K^{a+1}) = U_F^m$ and $N(U_K^a) = U_F^{m-1}$. Since m is the exact conductor of χ_F , the pullback is trivial on U_K^{a+1} and nontrivial on U_K^a . Thus $m' = a + 1$, which is (40).

For (ii), first separate $m = 0$: an unramified character is trivial on U_F^0 , hence its norm pullback is unramified and $m' = 0$. Now assume $1 \leq m \leq t$. The character χ_F is nontrivial on U_F^{m-1}/U_F^m . The graded norm map is an isomorphism at every level $< t$, so its pullback is nontrivial on U_K^{m-1}/U_K^m and trivial on U_K^m . Hence $m' = m$.

For (iii), the pullback is always trivial on U_K^{t+1} , so $m' \leq t + 1 = m$. It has smaller conductor precisely when the restriction of χ_F to the image

$$I = \text{im}(U_K^t/U_K^{t+1} \longrightarrow U_F^t/U_F^{t+1})$$

is trivial. By Theorem 3.7(c), the quotient $(U_F^t/U_F^{t+1})/I$ has order ℓ . The proof of Corollary 3.9 identifies this quotient with $F^\times/N(K^\times)$, so its character group is exactly the restriction of $S(K/F)$ to the last unit layer. Therefore $\chi_F|_I = 1$ if and only if there is a unique restriction $\mu|_{U_F^t/U_F^{t+1}}$ with $(\mu\chi_F)|_{U_F^t/U_F^{t+1}} = 1$, equivalently $m_F(\mu\chi_F) < t + 1$. $\square$

This is [3, Lemmas 8.8 and 8.12, pp. 73 and 78–79].

Lemma 3.11 (Unramified norm, trace, and conductor compatibility). *Let K/F be an unramified extension of degree f , and let π be a uniformizer of F . Then π is also a uniformizer of K , $[k_K : k_F] = f$, and the following statements hold.*

(a) *Reduction commutes with trace and norm:*

$$(41) \quad \overline{\text{Tr}_{K/F}(x)} = \text{Tr}_{k_K/k_F}(\bar{x}) \quad (x \in \mathcal{O}_K), \quad \overline{N_{K/F}(u)} = N_{k_K/k_F}(\bar{u}) \quad (u \in \mathcal{O}_K^\times).$$

(b) *For every integer a ,*

$$(42) \quad \text{Tr}_{K/F}(\mathfrak{p}_K^a) = \mathfrak{p}_F^a.$$

(c) *For every $r \geq 0$,*

$$(43) \quad N_{K/F}(U_K^r) = U_F^r, \quad N_{K/F}(\pi) = \pi^f.$$

(d) *If $\psi_K = \psi_F \circ \text{Tr}_{K/F}$ and $\chi_K = \chi_F \circ N_{K/F}$, then*

$$(44) \quad n_K(\psi_K) = n_F(\psi_F), \quad m_K(\chi_K) = m_F(\chi_F).$$

(e) *The quotient $F^\times/N_{K/F}(K^\times)$ is cyclic of order f , generated by the class of π .*

Proof. Since K/F is unramified, its ramification index is 1. Thus $\mathfrak{p}_K = \pi\mathcal{O}_K$, and the residue extension has degree f . The conjugates of an integral element reduce to the conjugates of its residue class. Taking the first and last coefficients of the characteristic polynomial therefore gives (41).

We use two elementary finite-field facts. The map Tr_{k_K/k_F} is onto: if $q = |k_F|$, its polynomial

$$X + X^q + \cdots + X^{q^{f-1}}$$

is nonzero and has degree $q^{f-1} < q^f = |k_K|$, so it cannot vanish on all of k_K ; a nonzero k_F -linear map to the one-dimensional space k_F is surjective. The map $N_{k_K/k_F} : k_K^\times \rightarrow k_F^\times$ is onto because $k_K^\times$ is cyclic and

$$N_{k_K/k_F}(x) = x^{(q^f-1)/(q-1)}$$

has image of order $q - 1$.

Choose $x \in \mathcal{O}_K$ whose residue has trace 1. Then $\text{Tr}_{K/F}(x)$ is a unit of $\mathcal{O}_F$; after multiplying x by its inverse, we obtain an integral element of trace 1. Hence $\text{Tr}_{K/F}(\mathcal{O}_K) = \mathcal{O}_F$. Since $\mathfrak{p}_K^a = \pi^a \mathcal{O}_K$ for every $a \in \mathbf{Z}$, this proves (42).

For $r \geq 1$ and $x \in \mathcal{O}_K$, multiplication of the conjugates gives

$$(45) \quad N_{K/F}(1 + \pi^r x) \equiv 1 + \pi^r \text{Tr}_{K/F}(x) \pmod{\mathfrak{p}_F^{r+1}}.$$

Thus the map induced by norm on $U_K^r/U_K^{r+1} \rightarrow U_F^r/U_F^{r+1}$ is the finite-field trace and is surjective. Starting with $v \in U_F^r$, choose a lift of its class, divide v by the norm of that lift, and repeat at levels $r+1, r+2, \dots$. Completeness makes the product of the successive lifts converge to an element of U_K^r with norm v . This proves $N(U_K^r) = U_F^r$ for $r \geq 1$. Surjectivity of the finite-field norm, followed by the case $r=1$, gives $N(U_K^0) = U_F^0$. Finally, because $\pi \in F$, its norm is π^f . This proves (43).

Put $n = n_F(\psi_F)$. By (42), $\psi_F \circ \text{Tr}$ is trivial on $\mathfrak{p}_K^{-n}$ and is not trivial on $\mathfrak{p}_K^{-n-1}$, so $n_K(\psi_K) = n$. By (43), for every $r \geq 0$ the character $\chi_F \circ N$ is trivial on U_K^r if and only if χ_F is trivial on U_F^r . Hence the multiplicative conductors are equal. This proves (44).

Every $y \in K^\times$ has the form $y = \pi^a u$ with $u \in U_K^0$, so

$$N_{K/F}(K^\times) = \pi^{f\mathbf{Z}} U_F^0$$

by (43). The last assertion follows. $\square$

Corollary 3.12 (Norm characters in cyclic prime degree). *Let K/F be cyclic of prime degree ℓ . Then $F^\times/N_{K/F}(K^\times)$ is cyclic of order ℓ , and $S(K/F)$ has ℓ elements. In particular, $S(K/F)$ is finite.*

Proof. If K/F is ramified, this is Corollary 3.9. Suppose that it is unramified. Lemma 3.11(e) gives a cyclic norm quotient of order ℓ , and (43) shows that the norm group contains the open subgroup U_F^0 . Thus the quotient is finite and discrete. Its continuous complex characters are all of its abstract characters, and a character is determined by the image of a generator. There are exactly ℓ possible images, the ℓ th roots of unity. Hence $S(K/F)$ has ℓ elements. $\square$

By Lemma 2.14, we shall always replace χ_F by a member of minimal conductor in its $S(K/F)$ -orbit. For such a representative, Proposition 3.10(iii) implies

$$(46) \quad m \leq t+1 \implies m' = m.$$

4. LAMPRECHT'S FORMULA, WITH PROOF

Let E be nonarchimedean, let θ be a quasi-character of conductor $m = 2d + \varepsilon > 1$, $\varepsilon \in \{0, 1\}$, and let $\gamma \mathcal{O}_E = \mathfrak{p}_E^{m+n_E(\psi)}$.

Lemma 4.1 (Finite stationary-phase duality). *Let $n = n_E(\psi)$, let $M, q, r \in \mathbf{Z}$ with $r \leq q$, and let $\Gamma \in E^\times$ satisfy*

$$\Gamma \mathcal{O}_E = \mathfrak{p}_E^{M+n}.$$

Then

$$(47) \quad \mathfrak{p}_E^{M-q}/\mathfrak{p}_E^{M-r} \times \mathfrak{p}_E^r/\mathfrak{p}_E^q \longrightarrow \mathbf{C}^\times, \quad (\bar{c}, \bar{x}) \longmapsto \psi(cx/\Gamma),$$

is a perfect pairing.

Proof. The pairing is well defined: changing c by an element of $\mathfrak{p}_E^{M-r}$ or x by an element of $\mathfrak{p}_E^q$ changes cx/Γ by an element of $\mathfrak{p}_E^{-n}$, on which ψ is trivial. The annihilator of $\mathfrak{p}_E^r/\mathfrak{p}_E^q$ in the first factor is zero, because

$$c\mathfrak{p}_E^r/\Gamma \subseteq \mathfrak{p}_E^{-n} \iff c \in \mathfrak{p}_E^{M-r}.$$

Similarly, the annihilator of the first factor in the second is zero, because

$$\mathfrak{p}_E^{M-q}x/\Gamma \subseteq \mathfrak{p}_E^{-n} \iff x \in \mathfrak{p}_E^q.$$

Thus the pairing is nondegenerate in both variables. Both finite additive groups have cardinality q_E^{q-r} , hence the induced maps to the respective character groups are isomorphisms. $\square$

Lemma 4.2 (Stationary numerator class at a chosen depth). *Let $q = m_E(\theta) > 1$, let $n = n_E(\psi)$, choose an integer M and an element $\Gamma \in E^\times$ such that*

$$\Gamma \mathcal{O}_E = \mathfrak{p}_E^{M+n},$$

and let r satisfy

$$\lceil q/2 \rceil \leq r \leq q-1.$$

There is a unique class

$$(48) \quad \text{St}_{\Gamma,r}(\theta) \in \mathfrak{p}_E^{M-q}/\mathfrak{p}_E^{M-r}$$

such that every representative c of this class satisfies

$$(49) \quad \theta(1+x) = \psi(cx/\Gamma) \quad (x \in \mathfrak{p}_E^r).$$

Every representative c has exact valuation

$$(50) \quad v_E(c) = M - q.$$

Proof. For $x, y \in \mathfrak{p}_E^r$, the quotient

$$\frac{(1+x)(1+y)}{1+x+y} = 1 + \frac{xy}{1+x+y}$$

belongs to U_E^q , because $2r \geq q$. Hence $x \mapsto \theta(1+x)$ is an additive character of $\mathfrak{p}_E^r/\mathfrak{p}_E^q$. Lemma 4.1, applied with the present M, q, r, Γ , identifies the full character group of $\mathfrak{p}_E^r/\mathfrak{p}_E^q$ with $\mathfrak{p}_E^{M-q}/\mathfrak{p}_E^{M-r}$. This gives the unique class (48) satisfying (49).

Every representative belongs to $\mathfrak{p}_E^{M-q}$. If it belonged to $\mathfrak{p}_E^{M-q+1}$, then for every $x \in \mathfrak{p}_E^{q-1}$ one would have $cx/\Gamma \in \mathfrak{p}_E^{-n}$. Since $r \leq q-1$, such x lies in the linearization domain, and (49) would make θ trivial on U_E^{q-1} , contrary to the exact conductor q . Thus $v_E(c) = M - q$. Finally $r \leq q-1$ implies $\mathfrak{p}_E^{M-r} \subseteq \mathfrak{p}_E^{M-q+1}$, so adding any element of the ambiguity ideal cannot change this valuation. Hence every representative has valuation $M - q$. $\square$

Lemma 4.3 (Stationary-class calculus). *Use the notation of Lemma 4.2.*

- (a) *If $r \leq s \leq q-1$, restriction from $\mathfrak{p}_E^r$ to $\mathfrak{p}_E^s$ sends $\text{St}_{\Gamma,r}(\theta)$ to the class of the same representative modulo $\mathfrak{p}_E^{M-s}$.*
- (b) *If $\Gamma' = a\Gamma$, then ac represents the stationary class relative to Γ' whenever c represents the class relative to Γ .*
- (c) *Let θ_1, θ_2 be characters and suppose that, on a common depth r , coefficients c_1, c_2 satisfy*

$$\theta_i(1+x) = \psi(c_i x/\Gamma) \quad (x \in \mathfrak{p}_E^r).$$

Then $c_1 + c_2$ represents the restriction of $\theta_1\theta_2$ on that depth, and jc_1 represents the restriction of the integer power θ_1^j . These assertions are equalities of classes modulo $\mathfrak{p}_E^{M-r}$.

- (d) *Suppose $m_E(\theta_1) = m_E(\theta_2) = q$, take $\lceil q/2 \rceil \leq r \leq q-1$, and choose representatives c_1, c_2 relative to the same Γ . Then*

$$(51) \quad m_E(\theta_1\theta_2) < q \iff c_1 + c_2 \in \mathfrak{p}_E^{M-q+1}.$$

If the conductor drops to $q' < q$ and $q' > 1$, put $r' = \lceil q'/2 \rceil$ and choose a representative c' of the stationary class at the new depth r' by Lemma 4.2. Then

$$(52) \quad c' \equiv c_1 + c_2 \pmod{\mathfrak{p}_E^{M-r}}.$$

Thus the old sum records only the restriction of the newly defined class on the old layer; it is not automatically a representative at the new depth. If $q' \leq 1$, there is no stationary layer: the conductor-zero or conductor-one formula must be used, and the old sum is not retained as a parameter.

Proof. Parts (a) and (b) follow immediately from (49) and uniqueness. Part (c) follows by multiplying the two character identities; uniqueness at the common depth identifies the resulting class modulo $\mathfrak{p}_E^{M-r}$.

For (d), the product has conductor at most q . It has conductor less than q precisely when it is trivial on U_E^{q-1} . By part (c), its restriction there is

$$x \mapsto \psi((c_1 + c_2)x/\Gamma).$$

The annihilator of $\mathfrak{p}_E^{q-1}$ is $\mathfrak{p}_E^{M-q+1}$, which proves (51). If the new conductor is $q' > 1$, a representative c' of its stationary class at depth r' induces the product character on every deeper unit layer. Hence c' and $c_1 + c_2$ induce the same additive character on $\mathfrak{p}_E^r$, even when $r \geq q'$ and that character is already trivial. By the annihilator computation in (47), their difference belongs to $\mathfrak{p}_E^{M-r}$, which proves (52). Thus a sum on the old stationary layer is used at a lower conductor only through this restriction of the newly defined class; it is not treated as a canonical representative at the new depth. When the new conductor is 0 or 1, the last assertion of the statement is simply the definition of the applicable endpoint formula. $\square$

For an admissible element γ in Lamprecht's formula one has $M = q$. Taking $q = 2d + \varepsilon$ and $r = d + \varepsilon$ therefore gives

$$\text{St}_{\gamma, d+\varepsilon}(\theta) \in \mathcal{O}_E/\mathfrak{p}_E^d,$$

and (50) says that every representative of this class is a unit.

Lemma 4.4 (Magnitude of a nondegenerate finite quadratic sum). *Let V be a finite abelian group and let $q : V \rightarrow \mathbf{C}^\times$ satisfy $|q(x)| = 1$, $q(0) = 1$, and*

$$B(x, y) = \frac{q(x+y)}{q(x)q(y)}.$$

Assume that B is a nondegenerate bicharacter. Then

$$(53) \quad \left| \sum_{x \in V} q(x) \right|^2 = |V|.$$

In particular, the sum is nonzero.

Proof. Put $S = \sum_x q(x)$. Since $\overline{q(y)} = q(y)^{-1}$, the change of variables $x = y + h$ gives

$$\begin{aligned} |S|^2 &= \sum_{x, y \in V} q(x) \overline{q(y)} = \sum_{h, y \in V} q(y+h) q(y)^{-1} \\ &= \sum_{h \in V} q(h) \sum_{y \in V} B(h, y). \end{aligned}$$

For $h = 0$ the inner sum is $|V|$. For $h \neq 0$, nondegeneracy says that $B(h, \cdot)$ is a nontrivial character, whose sum is zero. Hence only $h = 0$ survives and $|S|^2 = |V|$. $\square$

Theorem 4.5 (Lamprecht). *Put*

$$\beta = \beta(\theta) := \text{St}_{\gamma, d+\varepsilon}(\theta) \in \mathcal{O}_E/\mathfrak{p}_E^d.$$

Equivalently, β is the unique class satisfying

$$(54) \quad \theta(1+x) = \psi(\beta x/\gamma) \quad (x \in \mathfrak{p}_E^{d+\varepsilon}).$$

Every representative of this class is a unit. Fix any representative, again denoted $\beta \in \mathcal{O}_E^\times$. If $\varepsilon = 0$, then

$$(55) \quad \Delta_E(\theta, \psi) = \theta(\gamma) \psi(\beta/\gamma) \theta(\beta)^{-1}.$$

If $\varepsilon = 1$, choose $\delta \mathcal{O}_E = \mathfrak{p}_E^d$ and put

$$(56) \quad \phi_\theta(\bar{x}) = \psi(\delta \beta x/\gamma) \theta(1 + \delta x)^{-1}, \quad \bar{x} \in k_E.$$

Then this function is well defined, nowhere zero, and

$$(57) \quad \Delta_E(\theta, \psi) = \theta(\gamma)\psi(\beta/\gamma)\theta(\beta)^{-1}\mathcal{A}\left(\sum_{x \in k_E} \phi_\theta(x)\right), \quad \mathcal{A}(z) := \text{ph}(z).$$

Moreover

$$(58) \quad \phi_\theta(x+y) = \phi_\theta(x)\phi_\theta(y)\psi_0(xy),$$

where

$$(59) \quad \psi_0(\bar{z}) = \psi(\beta\delta^2\bar{z}/\gamma)$$

is a well-defined nontrivial additive character of k_E . Consequently

$$(60) \quad \left| \sum_{x \in k_E} \phi_\theta(x) \right| = q_E^{1/2}.$$

Proof. Apply Lemma 4.2 with

$$q = M = m = 2d + \varepsilon, \quad \Gamma = \gamma, \quad r = d + \varepsilon.$$

Since $M - q = 0$ and $M - r = d$, it gives precisely the class $\beta \in \mathcal{O}_E/\mathfrak{p}_E^d$ characterized by (54); (50) gives $v_E(\beta) = 0$ for every representative. This proves the first assertions of the theorem.

Put

$$I = \int_{U_E^0} \psi(u/\gamma)\theta(u)^{-1} du, \quad A = \mathfrak{p}_E^{d+\varepsilon}/\mathfrak{p}_E^m.$$

For each $x \in A$, substitute $u \mapsto u(1+x)$ in the integral. Haar measure on U_E^0 is translation invariant, and (54) gives

$$I = \int_{U_E^0} \psi(u/\gamma)\theta(u)^{-1}\psi((u-\beta)x/\gamma) du.$$

Average this identity over $x \in A$. Character orthogonality and Lemma 4.1, specialized to $M = q = m$, $r = d + \varepsilon$, and $\Gamma = \gamma$, give

$$(61) \quad \frac{1}{|A|} \sum_{x \in A} \psi((u-\beta)x/\gamma) = \begin{cases} 1, & u - \beta \in \mathfrak{p}_E^d, \\ 0, & u - \beta \notin \mathfrak{p}_E^d. \end{cases}$$

Hence only the compact open subset $\beta(1 + \mathfrak{p}_E^d)$ survives.

Assume first that $m = 2d$. Write $u = \beta(1+z)$ with $z \in \mathfrak{p}_E^d$. Since $z \in \mathfrak{p}_E^d$, the linearization identity with the variable z gives directly $\theta(1+z) = \psi(\beta z/\gamma)$. Therefore on the surviving set

$$\psi(u/\gamma)\theta(u)^{-1} = \psi(\beta/\gamma)\theta(\beta)^{-1}.$$

The integral is this nonzero constant times the positive measure of the surviving set. Taking its phase proves (55).

Now assume that $m = 2d + 1$. Since $m > 1$, one has $d \geq 1$. Partition $\beta(1 + \mathfrak{p}_E^d)$ into its q_E cosets modulo $\beta(1 + \mathfrak{p}_E^{d+1})$. They are represented by $u_x = \beta(1 + \delta\bar{x})$, $\bar{x} \in k_E$. If $w \in \mathfrak{p}_E^{d+1}$, then the quotient of the integrand at $u_x(1+w)$ by its value at u_x is

$$\psi(u_x w/\gamma)\theta(1+w)^{-1}.$$

Linearization gives $\theta(1+w) = \psi(\beta w/\gamma)$, while $(u_x - \beta)w \in \mathfrak{p}_E^{2d+1} = \mathfrak{p}_E^m$; hence the displayed quotient is one. Thus the integrand is constant on each of the q_E cosets and its value at u_x is

$$\psi(\beta/\gamma)\theta(\beta)^{-1}\phi_\theta(\bar{x}).$$

This proves (57), once the sum is known to be nonzero.

Changing a lift of $\bar{x}$ changes δx by an element of $\mathfrak{p}_E^{d+1}$, and the same cancellation proves that ϕ_θ is well defined. It is nowhere zero because both factors in its definition lie in $\mathbf{C}^\times$.

For lifts $x, y \in \mathcal{O}_E$, put

$$w = \frac{\delta^2 xy}{1 + \delta(x+y)}.$$

Then

$$(1 + \delta x)(1 + \delta y) = (1 + \delta(x + y))(1 + w).$$

The additive linear factors in (56) cancel, so

$$\frac{\phi_\theta(x + y)}{\phi_\theta(x)\phi_\theta(y)} = \theta(1 + w) = \psi(\beta w/\gamma).$$

Since $w - \delta^2 xy \in \mathfrak{p}_E^{3d}$ and $3d \geq 2d + 1 = m$, the last expression equals $\psi(\beta \delta^2 xy/\gamma)$. This proves (58) and (59). The character ψ_0 is well defined because its coefficient has valuation $-n_E(\psi) - 1$; it is nontrivial for the same reason. The bicharacter $(x, y) \mapsto \psi_0(xy)$ is nondegenerate: for $x \neq 0$, multiplication by x is a bijection of k_E , so $y \mapsto \psi_0(xy)$ is nontrivial. Lemma 4.4 now gives (60); in particular the finite sum and the original integral are nonzero. $\square$

This is [3, Lemma 8.1, pp. 60–62, and Lemma 9.3, pp. 86–87].

Lemma 4.6 (Independence of the stationary representative). *The complete stationary expression in Lamprecht’s formula is independent of the representative of the class $\beta \in \mathcal{O}_E/\mathfrak{p}_E^d$. More precisely, if $\beta' = \beta + h$ with $h \in \mathfrak{p}_E^d$, then the right sides of (55) and (57), respectively, are unchanged when β is replaced by β' and the Hasse function is formed with β' .*

Proof. Suppose first that $m = 2d$. Since β is a unit, $z = h/\beta$ lies in $\mathfrak{p}_E^d$, and the linearization formula gives $\theta(1 + z) = \psi(\beta z/\gamma) = \psi(h/\gamma)$. Hence

$$\frac{\psi(\beta'/\gamma)\theta(\beta')^{-1}}{\psi(\beta/\gamma)\theta(\beta)^{-1}} = \psi(h/\gamma)\theta(1 + h/\beta)^{-1} = 1.$$

Now let $m = 2d + 1$. Choose $\delta\mathcal{O}_E = \mathfrak{p}_E^d$ and write

$$a = \frac{h}{\beta\delta} \in \mathcal{O}_E.$$

Let ϕ_β and $\phi_{\beta'}$ denote the Hasse functions formed with β and β' . Directly from (56),

$$(62) \quad \phi_{\beta'}(x) = \phi_\beta(x) \psi(\delta h \tilde{x}/\gamma).$$

The functional equation (58) gives

$$\phi_\beta(x + a) = \phi_\beta(x) \phi_\beta(a) \psi(\beta \delta^2 a \tilde{x}/\gamma).$$

Since $\beta \delta a = h$, the last additive factor is the one in (62). Therefore

$$\phi_{\beta'}(x) = \phi_\beta(a)^{-1} \phi_\beta(x + a), \quad \sum_{x \in k_E} \phi_{\beta'}(x) = \phi_\beta(a)^{-1} \sum_{x \in k_E} \phi_\beta(x).$$

On the other hand,

$$\frac{\psi(\beta'/\gamma)\theta(\beta')^{-1}}{\psi(\beta/\gamma)\theta(\beta)^{-1}} = \psi(h/\gamma)\theta(1 + \delta a)^{-1} = \phi_\beta(a).$$

Because the Hasse sums are nonzero, taking phases shows that these two multipliers cancel. Thus the complete odd-conductor stationary factor is also unchanged. $\square$

For a chosen representative β of the stationary class, set

$$\Delta_{2,E}(\theta) = \psi(\beta/\gamma)\theta(\beta)^{-1}, \quad \Delta_{3,E}(\theta) = \begin{cases} 1, & m_E(\theta) \text{ even,} \\ \mathcal{A}(\sum \phi_\theta), & m_E(\theta) \text{ odd.} \end{cases}$$

Then

$$(63) \quad \Delta_E(\theta, \psi) = \theta(\gamma) \Delta_{2,E}(\theta) \Delta_{3,E}(\theta).$$

When the conductor is odd, $\Delta_{2,E}(\theta)$ and $\Delta_{3,E}(\theta)$ separately depend on the chosen representative; their product is independent by Lemma 4.6. Every later comparison therefore either keeps these two factors together or applies the corresponding translation to the critical function.

Lemma 4.7 (Exact stable twist). *If $m_E(\theta) = 2d + \varepsilon > 1$ and $m_E(\nu) \leq d$, then, for any representative β of the stationary class of θ ,*

$$(64) \quad \Delta_E(\nu\theta, \psi) = \nu(\gamma/\beta)\Delta_E(\theta, \psi).$$

The right-hand side is independent of the chosen representative.

Proof. The character ν is trivial on $1 + \mathfrak{p}_E^d$, so the stationary classes of θ and $\nu\theta$ coincide; choose the same representative β for both. Their Hasse functions are then identical. Taking the ratio in (63) gives $\nu(\gamma)\nu(\beta)^{-1}$. If $\beta' \equiv \beta \pmod{\mathfrak{p}_E^d}$, then $\beta'/\beta \in U_E^d$ and $\nu(\beta'/\beta) = 1$, proving the final independence assertion. $\square$

The formulation above is the form of [3, Lemma 9.4, pp. 87–88] used below.

Lemma 4.8 (Conductors and leading classes in a minimal norm-character orbit). *Let K/F be ramified cyclic of prime degree ℓ , let t be its break, put $T = t + 1$, and let τ be a nontrivial element of $S(K/F)$. Choose χ_F of minimal conductor in its $S(K/F)$ -orbit and put $m = m_F(\chi_F)$. Then, for every nontrivial power τ^j ,*

$$(65) \quad m_F(\tau^j \chi_F) = \max\{m, T\}.$$

If $T > 1$ and $m = T$, choose a common denominator Γ , write

$$\Gamma \mathcal{O}_F = \mathfrak{p}_F^{M+n_F(\psi_F)},$$

choose a common depth

$$\lceil T/2 \rceil \leq r \leq T - 1,$$

and let a and b represent the stationary restrictions of τ and χ_F . Then the leading classes of $b + ja$ do not vanish:

$$(66) \quad b + ja \notin \mathfrak{p}_F^{M-T+1}.$$

Thus $b + ja$ represents a character of exact conductor T ; no lower-conductor Lamprecht parameter is obtained by reusing a cancelled sum on the old layer.

Proof. A nontrivial norm character has conductor T by Corollary 3.9. If $m < T$, then χ_F is trivial on U_F^{T-1} , whereas τ^j is not, so the product has conductor T . If $m > T$, then τ^j is trivial on U_F^{m-1} , whereas χ_F is not, so the product has conductor m . If $m = T$, every member of the orbit has conductor at least T by the minimal choice of χ_F , and at most T because both factors are trivial on U_F^T . This proves (65).

Assume $T > 1$ and $m = T$. Lemma 4.3(c) identifies $b + ja$ with the stationary restriction of $\tau^j \chi_F$ on the common layer. If its leading class vanished, equation (51) would give conductor less than T , contradicting (65). This proves (66) and the final assertion. $\square$

5. FINITE-FIELD IDENTITIES

Let k be a finite field. We use the normalized Gauss sum $\tau_k(\chi, \psi)$ defined in (10); in particular $\tau_k(1, \psi) = 1$. The local conductor-zero and conductor-one formulas were already isolated in (11) and (12). We now establish the finite-field identities used to compare those formulas and the higher stationary phases.

Lemma 5.1 (Finite-field Gauss-sum magnitude). *For*

$$G_k(\chi, \psi) = \sum_{x \in k^\times} \chi(x)\psi(x)$$

one has $G_k(1, \psi) = -1$. If χ is nontrivial, then

$$(67) \quad |G_k(\chi, \psi)|^2 = |k|.$$

Consequently every $G_k(\chi, \psi)$ and every $\tau_k(\chi, \psi)$ is nonzero.

Proof. The trivial-character formula is $\sum_{x \in k^\times} \psi(x) = -1$, because the sum over all of k is zero. Assume χ nontrivial and write $q = |k|$. Since the values of χ and ψ have absolute value one,

$$\begin{aligned} |G_k(\chi, \psi)|^2 &= \sum_{x, y \in k^\times} \chi(x/y) \psi(x - y) \\ &= \sum_{t \in k^\times} \chi(t) \sum_{y \in k^\times} \psi((t-1)y). \end{aligned}$$

The inner sum is $q - 1$ when $t = 1$ and is -1 when $t \neq 1$. Therefore

$$|G_k(\chi, \psi)|^2 = (q - 1) - \sum_{t \neq 1} \chi(t).$$

Character orthogonality gives $\sum_{t \in k^\times} \chi(t) = 0$, so $\sum_{t \neq 1} \chi(t) = -1$ and the last expression is q . $\square$

Theorem 5.2 (Hasse–Davenport lift). *For a finite extension k'/k of degree f ,*

$$(68) \quad \tau_{k'}(\chi \circ N_{k'/k}, \psi \circ \text{Tr}_{k'/k}) = \tau_k(\chi, \psi)^f.$$

This is the classical lifting relation of Davenport and Hasse [1]. We include the Euler-product proof in the normalization used here.

Proof. It is convenient first to use the ordinary Gauss sum

$$G_k(\alpha, \psi) = \sum_{x \in k^\times} \alpha(x) \psi(x).$$

For a monic polynomial

$$P(X) = X^r + a_1 X^{r-1} + \cdots + a_r \in k[X], \quad a_r \neq 0,$$

put

$$w(P) = \chi((-1)^r a_r) \psi(-a_1).$$

If P, Q are monic with nonzero constant terms, then the constant terms multiply and the negative of the coefficient of degree $\deg(PQ) - 1$ is the sum of the roots; hence $w(PQ) = w(P)w(Q)$. Therefore

$$\begin{aligned} L(T) &= \prod_{\substack{P \text{ monic irreducible} \\ P \neq X}} (1 - w(P)T^{\deg P})^{-1} \\ &= \sum_{\substack{P \text{ monic} \\ P(0) \neq 0}} w(P)T^{\deg P}. \end{aligned}$$

For degree $r \geq 2$, the coefficient is zero, because a_1 ranges freely through k and $\sum_{a_1 \in k} \psi(-a_1) = 0$. The coefficient in degree 1 is

$$\sum_{a_1 \neq 0} \chi(-a_1) \psi(-a_1) = G_k(\chi, \psi).$$

Thus

$$(69) \quad L(T) = 1 + G_k(\chi, \psi)T.$$

On the other hand, taking logarithms of the Euler product and grouping closed points after extension of scalars gives

$$(70) \quad \log L(T) = \sum_{r \geq 1} \frac{T^r}{r} \sum_{x \in k_r^\times} \chi(N_{k_r/k} x) \psi(\text{Tr}_{k_r/k} x),$$

where k_r/k is the degree- r extension. Indeed, an element of k_r of degree $d \mid r$ contributes through its monic minimal polynomial with multiplicity d , exactly as in the usual Euler-product proof for the zeta function of the affine line. Comparing (69) and (70) yields

$$(71) \quad G_{k_r}(\chi \circ N, \psi \circ \text{Tr}) = (-1)^{r-1} G_k(\chi, \psi)^r.$$

Now apply (71) to χ^{-1} . Since $\tau_k(\chi, \psi) = -G_k(\chi^{-1}, \psi)$, the signs cancel:

$$\tau_{k_r}(\chi \circ N, \psi \circ \text{Tr}) = -(-1)^{r-1} G_k(\chi^{-1}, \psi)^r = \tau_k(\chi, \psi)^r.$$

Taking $r = f$ proves the theorem. $\square$

Theorem 5.3 (Hasse–Davenport multiplication formula). *Let $\ell \mid |k^\times|$, let η have order ℓ , and let χ be any multiplicative character. Then*

$$(72) \quad \chi(\ell^\ell) \tau_k(\chi^\ell, \psi) \prod_{j=1}^{\ell-1} \tau_k(\eta^j, \psi) = \tau_k(\chi, \psi) \prod_{j=1}^{\ell-1} \tau_k(\chi \eta^j, \psi).$$

Equivalently, for $G(\alpha) = \sum_{x \in k^\times} \alpha(x) \psi(x)$ with $G(1) = -1$,

$$(73) \quad \prod_{j=0}^{\ell-1} G(\alpha \eta^j) = \alpha(\ell^{-\ell}) G(\alpha^\ell) \prod_{j=1}^{\ell-1} G(\eta^j).$$

Proof. The ordinary identity (73) is the classical Davenport–Hasse multiplication formula [1]. Apply it with $\alpha = \chi^{-1}$. Inversion permutes the nontrivial powers of η , and $G(\alpha) = -\tau_k(\alpha^{-1}, \psi)$. Both sides contain exactly ℓ ordinary Gauss sums when the factor $G(\chi^{-\ell})$ is counted, so the signs cancel. Finally $\chi^{-1}(\ell^{-\ell}) = \chi(\ell^\ell)$, which gives (72) in Langlands’s normalization. $\square$

Lemma 5.4 (Quadratic phase). *Assume $\text{char } k \neq 2$. Let ν be the quadratic character and put*

$$Q_\psi(a, b) = \mathcal{A} \left(\sum_{x \in k} \psi \left(\frac{a}{2} x^2 + bx \right) \right), \quad a \neq 0.$$

Then

$$(74) \quad Q_\psi(a, b) = \nu(a) \psi(-b^2/2a) Q_\psi(1, 0), \quad Q_\psi(1, 0)^2 = \nu(-1).$$

The defining sum has absolute value $|k|^{1/2}$.

Proof. Let

$$S(a, b) = \sum_{x \in k} \psi \left(\frac{a}{2} x^2 + bx \right).$$

Completing the square gives

$$S(a, b) = \psi(-b^2/2a) S(a, 0).$$

For $c \neq 0$, every $y \in k^\times$ has $1 + \nu(y)$ square roots, while zero has one square root. Hence

$$(75) \quad \begin{aligned} \sum_{x \in k} \psi(cx^2) &= \sum_{y \in k} (1 + \nu(y)) \psi(cy) \\ &= \sum_{y \in k^\times} \nu(y) \psi(cy) = \nu(c) G_k(\nu, \psi). \end{aligned}$$

The additive-character part disappeared because $\sum_{y \in k} \psi(cy) = 0$. Taking $c = a/2$ and comparing with $a = 1$ proves the first phase identity. Lemma 5.1 shows at once that $|S(a, b)| = |k|^{1/2}$.

It remains to compute the square of the phase. Put $G = G_k(\nu, \psi)$. Changing variables $x = ty$ in the product of the two Gauss sums gives

$$G^2 = \sum_{t \in k^\times} \nu(t) \sum_{y \in k^\times} \psi((t+1)y).$$

Here the inner sum is $|k| - 1$ for $t = -1$ and is -1 for $t \neq -1$. Therefore

$$\begin{aligned} G^2 &= \nu(-1)(|k| - 1) - \sum_{t \neq -1} \nu(t) \\ &= \nu(-1)(|k| - 1) + \nu(-1) = \nu(-1)|k|, \end{aligned}$$

because $\sum_{t \in k^\times} \nu(t) = 0$. Since $S(1, 0) = \nu(1/2)G$ and $\nu(1/2)^2 = 1$, dividing by absolute values gives $Q_\psi(1, 0)^2 = \nu(-1)$. $\square$

Theorem 5.5 (Hasse-function lift). *Let k'/k be finite and let a nowhere-zero function $\phi : k \rightarrow \mathbf{C}^\times$ satisfy*

$$\phi(x+y) = \phi(x)\phi(y)\psi(xy).$$

Put

$$\omega_{k'/k}(x) = \sum_{\{\sigma, \tau\}} x^\sigma x^\tau, \quad \phi_{k'/k}(x) = \phi(\text{Tr } x)\psi(-\omega_{k'/k}(x)).$$

Here the sum defining $\omega_{k'/k}$ is over the unordered pairs of distinct k -embeddings $k' \hookrightarrow \bar{k}$. Then $\phi_{k'/k}$ satisfies the analogous functional equation over k' and

$$(76) \quad - \sum_{x \in k'} \phi_{k'/k}(x) = \left(- \sum_{x \in k} \phi(x) \right)^{[k':k]}.$$

Proof. The identity

$$\omega(x+y) = \omega(x) + \omega(y) + \text{Tr}(x)\text{Tr}(y) - \text{Tr}(xy)$$

together with the functional equation for ϕ proves the corresponding functional equation for $\phi_{k'/k}$.

For the sum identity, let

$$P(X) = X^r + a_1 X^{r-1} + a_2 X^{r-2} + \dots$$

be monic and put

$$w(P) = \phi(-a_1)\psi(-a_2),$$

where $a_2 = 0$ in degree 1. If Q has first coefficients b_1, b_2 , then the first two coefficients of PQ are $a_1 + b_1$ and $a_2 + b_2 + a_1 b_1$. Hence

$$\begin{aligned} w(PQ) &= \phi(-a_1 - b_1)\psi(-a_2 - b_2 - a_1 b_1) \\ &= \phi(-a_1)\phi(-b_1)\psi(a_1 b_1)\psi(-a_2 - b_2 - a_1 b_1) = w(P)w(Q). \end{aligned}$$

Thus

$$L(T) = \prod_{P \text{ monic irreducible}} (1 - w(P)T^{\deg P})^{-1} = \sum_{P \text{ monic}} w(P)T^{\deg P}.$$

For degree $r \geq 2$, the coefficient is zero because a_2 ranges freely and ψ is nontrivial. In degree 1 it is $\sum_{x \in k} \phi(x)$. Therefore

$$(77) \quad L(T) = 1 + \left(\sum_{x \in k} \phi(x) \right) T.$$

We next justify the closed-point expansion, including compatibility with repeated closed points. Let P be monic irreducible of degree d , choose a root $\xi \in k_d$, and put

$$s = \text{Tr}_{k_d/k}(\xi), \quad e = \omega_{k_d/k}(\xi).$$

The first two coefficients of P are $-s$ and e , so

$$w(P) = \phi(s)\psi(-e).$$

If $r = hd$ and ξ is viewed in k_r , each of its d conjugates occurs h times. Consequently

$$(78) \quad \text{Tr}_{k_r/k}(\xi) = hs, \quad \omega_{k_r/k}(\xi) = he + \binom{h}{2}s^2.$$

Iteration of the functional equation for ϕ gives

$$(79) \quad \phi(hs) = \phi(s)^h \psi\left(\binom{h}{2}s^2\right).$$

Combining the last two displays yields

$$(80) \quad \phi(\text{Tr}_{k_r/k} \xi)\psi(-\omega_{k_r/k}(\xi)) = (\phi(s)\psi(-e))^h = w(P)^h.$$

Thus the d roots belonging to the closed point P contribute $dw(P)^h$ to the sum over k_r . Expanding the logarithm of the Euler product and regrouping by $r = h \deg P$ therefore gives

$$(81) \quad \log L(T) = \sum_{r \geq 1} \frac{T^r}{r} \sum_{x \in k_r} \phi(\mathrm{Tr}_{k_r/k} x) \psi(-\omega_{k_r/k}(x)).$$

Comparing the logarithm of (77) with (81) gives

$$\sum_{x \in k_r} \phi_{k_r/k}(x) = (-1)^{r-1} \left(\sum_{x \in k} \phi(x) \right)^r.$$

Multiplying by -1 is exactly (76). $\square$

This is [3, Lemma 9.1, pp. 81–84].

Proposition 5.6 (Ramified Hasse-phase comparison). *Let K/F be totally ramified cyclic of odd prime degree ℓ , with single break t . Suppose $m_F(\chi_F) = 2d + 1 > 1$, $2d + 1 \geq t + 1$, and $m_K(\chi_K) = 2d' + 1$, where*

$$d' = \ell d - \frac{\ell - 1}{2} t.$$

Assume compatible representatives of the stationary classes in Proposition 6.8 have been chosen. In either of the following situations,

$$t = 0 \quad \text{or} \quad d \geq t + 1,$$

the residual factors satisfy

$$(82) \quad \Delta_{3,K}(\chi_K) = \Delta_{3,F}(\chi_F)^\ell.$$

Proof. Put $n = n_F(\psi_F)$ and choose uniformizers with $\varpi_F = N_{K/F}(\varpi_K)$. The different exponent is $(\ell - 1)(t + 1)$, and therefore

$$n_K(\psi_K) = \ell n + (\ell - 1)(t + 1), \quad (2d' + 1) + n_K(\psi_K) = \ell(2d + 1 + n).$$

Consequently

$$\Gamma := \varpi_F^{2d+1+n} \in F^\times$$

is admissible both downstairs and upstairs. By Proposition 6.8(b), the downstairs and upstairs stationary classes admit the same representative $\beta \in \mathcal{O}_F^\times$; choose this representative in both Lamprecht factorizations. If the conductor is odd, the corresponding critical functions are understood with the simultaneous translation required by Lemma 4.6.

Let k be the common residue field and, for $x \in k$, choose a lift $\tilde{x} \in \mathcal{O}_F \subset \mathcal{O}_K$. Write $E_2(y)$ for the second elementary symmetric function of the conjugates of y . The lower critical function is

$$(83) \quad \phi(x) = \psi_F \left(\frac{\beta \varpi_F^d \tilde{x}}{\Gamma} \right) \chi_F(1 + \varpi_F^d \tilde{x})^{-1},$$

and its polar character is

$$\psi_0(z) = \psi_F \left(\frac{\beta \varpi_F^{2d} \tilde{z}}{\Gamma} \right).$$

Put $Y = \varpi_K^{d'} \tilde{x}$. We claim that the upper critical function is

$$(84) \quad \phi_K(x) = \psi_F \left(\frac{\beta \mathrm{Tr}_{K/F}(Y)}{\Gamma} \right) \chi_F(1 + \mathrm{Tr}_{K/F}(Y) + E_2(Y))^{-1}.$$

Indeed, it is enough to prove

$$(85) \quad N_{K/F}(1 + Y) \equiv 1 + \mathrm{Tr}_{K/F}(Y) + E_2(Y) \pmod{\mathfrak{p}_F^{2d+1}}.$$

If $t = 0$, then $d' = \ell d$. Every term of degree at least three in the norm has valuation at least $3d \geq 2d + 1$, and the norm term has valuation $\ell d \geq 2d + 1$. If $t > 0$, then $\ell = p$ is the residue characteristic and $d \geq t + 1$. For $3 \leq j \leq p - 1$, Lemma 3.4 gives

$$v_F(s_j(Y)) \geq \left\lfloor \frac{jd' + (p - 1)(t + 1)}{p} \right\rfloor \geq 2d + 1,$$

while

$$v_F(NY) = d' = pd - \frac{p-1}{2}t \geq 2d+1.$$

This proves (85). The same estimates, with the valuation of the lift increased by one, show that (84) is independent of the lift of x . They also give

$$\mathrm{Tr}(Y) \in \mathfrak{p}_F^d, \quad E_2(Y) \in \mathfrak{p}_F^{2d},$$

and one extra power of $\mathfrak{p}_F$ when $\tilde{x} \in \mathfrak{p}_K$.

Set

$$(86) \quad \epsilon = \frac{\mathrm{Tr}_{K/F}(\varpi_K^{2d'})}{\varpi_F^{2d}} \in \mathcal{O}_F.$$

The preceding valuation estimates show that this quotient is integral. We now show that it is a unit and determine its residue. If $t = 0$, then

$$c = \overline{\varpi_K^{d'}/\varpi_F^d} \in k^\times$$

is fixed by $\mathrm{Gal}(K/F)$, because $d' = \ell d$. Hence

$$(87) \quad \bar{\epsilon} = \ell c^2.$$

If $t > 0$, Proposition 3.6 gives

$$(88) \quad \bar{\epsilon} = -\lambda^{p-1}.$$

In particular, ϵ is a unit in both cases.

For $x \in k$ write

$$Y_x = \varpi_K^{d'} \tilde{x}, \quad A_x = \mathrm{Tr}(Y_x), \quad B_x = E_2(Y_x), \quad U_x = 1 + A_x + B_x.$$

For each fixed pair x, y , lift $x + y$ by $\tilde{x} + \tilde{y}$; the lift-independence proved above permits this choice, and then $Y_{x+y} = Y_x + Y_y$. The identity

$$B_{x+y} = B_x + B_y + A_x A_y - \mathrm{Tr}(Y_x Y_y)$$

gives

$$U_x U_y \equiv U_{x+y} + \mathrm{Tr}(Y_x Y_y) \pmod{\mathfrak{p}_F^{2d+1}}.$$

Indeed, the omitted products $A_x B_y$, $A_y B_x$, and $B_x B_y$ have valuation at least $3d \geq 2d+1$. Since $U_{x+y} \equiv 1 \pmod{\mathfrak{p}_F^d}$ and $\mathrm{Tr}(Y_x Y_y) \in \mathfrak{p}_F^{2d}$, division by U_{x+y} gives

$$(89) \quad \frac{U_x U_y}{U_{x+y}} \equiv 1 + \mathrm{Tr}(Y_x Y_y) \pmod{\mathfrak{p}_F^{2d+1}}.$$

The additive factors in (84) cancel in the polar quotient, whereas its multiplicative factors give $\chi_F(U_x U_y / U_{x+y})$. Lamprecht linearization on $\mathfrak{p}_F^{d+1}$ and (89) therefore imply

$$\begin{aligned} \frac{\phi_K(x+y)}{\phi_K(x)\phi_K(y)} &= \psi_F\left(\frac{\beta \mathrm{Tr}(Y_x Y_y)}{\Gamma}\right) \\ &= \psi_0(\bar{\epsilon}xy), \end{aligned}$$

because $\mathrm{Tr}(Y_x Y_y) = \epsilon \varpi_F^{2d} \tilde{x} \tilde{y}$. Thus

$$(90) \quad \phi_K(x+y) = \phi_K(x)\phi_K(y)\psi_0(\bar{\epsilon}xy),$$

and ϕ_K is a nondegenerate quadratic phase on k .

We compare its normalized sum with that of ϕ . First assume that k has odd characteristic. The polar character of ϕ is $(x, y) \mapsto \psi_0(xy)$, so for a unique $\alpha_0 \in k$ one has

$$\phi(x) = \psi_0(x^2/2 + \alpha_0 x).$$

Put

$$T_x = \overline{\mathrm{Tr}(Y)/\varpi_F^d}, \quad Q_x = \overline{\mathrm{Tr}(Y^2)/\varpi_F^{2d}}.$$

Using $\mathrm{Tr}(Y)/\varpi_F^d$ itself as a lift of T_x in (83), one has

$$\phi(T_x) = \psi_F\left(\frac{\beta \mathrm{Tr}(Y)}{\Gamma}\right) \chi_F(1 + \mathrm{Tr}(Y))^{-1}.$$

Because $E_2(Y) \in \mathfrak{p}_F^{2d}$,

$$\frac{1 + \text{Tr}(Y)}{1 + \text{Tr}(Y) + E_2(Y)} \equiv 1 - E_2(Y) \pmod{\mathfrak{p}_F^{2d+1}}.$$

A second application of Lamprecht linearization therefore gives the exact residual identity

$$(91) \quad \phi_K(x) = \phi(T_x) \psi_0 \left(-\frac{\overline{E_2(Y)}}{\varpi_F^{2d}} \right).$$

Now $2E_2(Y) = (\text{Tr } Y)^2 - \text{Tr}(Y^2)$, so

$$-\frac{\overline{E_2(Y)}}{\varpi_F^{2d}} = \frac{Q_x - T_x^2}{2}.$$

Substituting $\phi(T_x) = \psi_0(T_x^2/2 + \alpha_0 T_x)$ in (91) yields

$$(92) \quad \phi_K(x) = \psi_0(Q_x/2 + \alpha_0 T_x).$$

If $t = 0$, then $T_x = \ell c x$ and $Q_x = \ell c^2 x^2$; after replacing x by $c^{-1}x$,

$$(93) \quad \phi_K(x) = \psi_0(\ell x^2/2 + \ell \alpha_0 x).$$

If $t > 0$, the trace-ideal formula gives $T_x = 0$, while $Q_x = \bar{\epsilon} x^2$, and hence

$$(94) \quad \phi_K(x) = \psi_0(\bar{\epsilon} x^2/2).$$

Let ν be the quadratic character of $k^\times$ and put $Q = Q_{\psi_0}(1, 0)$. In the tame case, the phase of the lower sum is $Q\psi_0(-\alpha_0^2/2)$ and that of the upper sum is

$$\nu(\ell)Q\psi_0(-\ell\alpha_0^2/2).$$

Since $Q^2 = \nu(-1)$, the required power identity is equivalent to

$$(95) \quad \nu(\ell) = \nu(-1)^{(\ell-1)/2}.$$

Let $|k| = p^f$. If f is even, both sides are 1. If f is odd, the order of p modulo ℓ is odd because $\ell \mid p^f - 1$; therefore $(p/\ell) = 1$. Quadratic reciprocity now gives exactly (95).

In the wild case, $\ell = p$ and $\bar{\epsilon} = -\lambda^{p-1}$, so $\nu(\bar{\epsilon}) = \nu(-1)$. Formula (94) and $Q^2 = \nu(-1)$ reduce the desired identity to

$$\nu(-1)^{(p-3)/2} = 1.$$

If $p \equiv 1 \pmod{4}$, then $\nu(-1) = 1$; if $p \equiv 3 \pmod{4}$, the exponent $(p-3)/2$ is even. This proves the comparison in odd residue characteristic.

It remains to treat residue characteristic two. Then $t = 0$, because ℓ is odd. With the same c as above, the norm calculation gives

$$(96) \quad \phi_K(c^{-1}x) = \phi(\ell x) \psi_0 \left(-\binom{\ell}{2} x^2 \right) = \phi(x)^\ell.$$

The second equality follows by iterating $\phi(x+y) = \phi(x)\phi(y)\psi_0(xy)$. Put $S = \sum_x \phi(x)$ and $H = \mathcal{A}(S)$. Choose $c_0 \in k^\times$ so that $\psi_0(x^2) = \psi_0(c_0 x)$. Then

$$\overline{\phi(x)} = \phi(x)\psi_0(c_0 x), \quad \overline{S} = \phi(c_0)^{-1}S, \quad H^2 = \phi(c_0).$$

As $\phi(x)^4 = 1$, equation (96) shows that the desired power identity is equivalent to

$$\phi(c_0)^{(\ell-1)/2} = 1 \quad (\ell \equiv 1 \pmod{4}), \quad \phi(c_0)^{(\ell+1)/2} = 1 \quad (\ell \equiv 3 \pmod{4}).$$

Write $\psi_0(x) = (-1)^{\text{Tr}_{k/\mathbb{F}_2}(ax)}$. One may take $c_0 = a^{2^{f-1}-1}$, where $f = [k : \mathbb{F}_2]$; hence

$$\phi(c_0)^2 = \psi_0(c_0^2) = (-1)^f.$$

The two displayed powers are automatic for $\ell \equiv 1, 7 \pmod{8}$. If $\ell \equiv 3, 5 \pmod{8}$, then $(2/\ell) = -1$; since $\ell \mid 2^f - 1$, the order of 2 modulo ℓ and therefore f are even. Thus $\phi(c_0)^2 = 1$, and the remaining two residue classes also follow.

In every case the phase of the upper sum is the ℓ th power of the phase of the lower sum. This is precisely (82). $\square$

Lemma 5.7 (Affine pencil of quadratic phases). *Let k have odd characteristic p , normalize ψ so that $\psi(x^p) = \psi(x)$, and take $\rho \neq 0$ with $1 + j\rho \neq 0$ for every $j \in \mathbf{F}_p$. Choose $a_0, b_0 \in k$ with*

$$a_0^p = 1 - \rho^{1-p}, \quad b_0^p = \sigma - \rho^{-1}\tau.$$

Then

$$(97) \quad Q_\psi(a_0, b_0) \prod_{j \in \mathbf{F}_p^\times} Q_\psi(j\rho, j\tau) = \prod_{j \in \mathbf{F}_p} Q_\psi(1 + j\rho, \sigma + j\tau).$$

Proof. Apply Lemma 5.4. The quadratic-character parts agree because

$$\prod_{j \in \mathbf{F}_p} (1 + j\rho) = 1 - \rho^{p-1}, \quad \prod_{j \neq 0} j = -1,$$

and $\nu(a_0) = \nu(1 - \rho^{1-p})$. The additive parts agree by

$$(98) \quad \sum_{j \in \mathbf{F}_p} \frac{(\sigma + j\tau)^2}{1 + j\rho} = \frac{\rho^{p-3}(\sigma\rho - \tau)^2}{\rho^{p-1} - 1}.$$

To prove (98), use

$$\sum_j \frac{1}{1 + j\rho} = \frac{\rho^{p-1}}{\rho^{p-1} - 1}, \quad \sum_j \frac{j}{1 + j\rho} = -\frac{\rho^{p-2}}{\rho^{p-1} - 1}, \quad \sum_j \frac{j^2}{1 + j\rho} = \frac{\rho^{p-3}}{\rho^{p-1} - 1},$$

which follow from the logarithmic derivative of $\prod_j (X + j) = X^p - X$. Finally the right side of (98) is $(b_0^2/a_0)^p$, and the normalization $\psi(z^p) = \psi(z)$ finishes the proof. $\square$

Proposition 5.8 (Homogeneous affine-pencil quotient). *Under the hypotheses and notation of Lemma 5.7, let $c \in k^\times$ and choose $c_0 \in k^\times$ with $c_0^p = c$. Then*

$$(99) \quad \frac{Q_\psi(ca_0^p, cb_0^p) \prod_{j \in \mathbf{F}_p^\times} Q_\psi(cj\rho, cj\tau)}{\prod_{j \in \mathbf{F}_p} Q_\psi(c(1 + j\rho), c(\sigma + j\tau))} = \frac{Q_\psi(a_0, b_0) \prod_{j \in \mathbf{F}_p^\times} Q_\psi(j\rho, j\tau)}{\prod_{j \in \mathbf{F}_p} Q_\psi(1 + j\rho, \sigma + j\tau)}.$$

Thus a common nonzero scale on all actual rows does not change the normalized affine-pencil quotient.

Proof. For $a, c \neq 0$, completion of the square gives the scaling identity

$$(100) \quad Q_\psi(ca, cb) = \nu(c) \psi\left(-\frac{(c-1)b^2}{2a}\right) Q_\psi(a, b),$$

where ν is the quadratic character. Frobenius invariance also gives $Q_\psi(a^p, b^p) = Q_\psi(a, b)$. Apply (100) to every row. The product of the quadratic-character factors cancels because numerator and denominator contain the same number of rows and $p-1$ is even. The unit-indexed additive corrections have total exponent zero because $\sum_{j \in \mathbf{F}_p^\times} j = 0$. For the denominator, the rational-sum identity (98) shows that its total correction is the Frobenius of the upper correction. The relation $c_0^p = c$ and $\psi(z^p) = \psi(z)$ identify these two corrections. The remaining common factor therefore cancels from the quotient, proving (99). $\square$

Lemma 5.9 (Artin-Schreier phase). *Let k have characteristic 2 and let $\psi_0(x) = (-1)^{\text{Tr}_{k/\mathbf{F}_2}(x)}$. Then*

$$\psi_0(c + c^2) = 1 \quad (c \in k).$$

Proof. The absolute trace is Frobenius-invariant, so $\text{Tr}(c + c^2) = 0$. $\square$

6. STATIONARY PARAMETERS UNDER NORM

We now establish the stationary-parameter comparison under norm. The formulation is specialized to cyclic prime degree.

Throughout this section a symbol such as $\beta(\theta)$ denotes the stationary *class* furnished by Lemma 4.2, unless a representative is explicitly chosen. Thus a formula

$$\beta(\theta) \equiv c \pmod{\mathfrak{p}_E^r}$$

means that c represents that class at the indicated precision; it does not assert that the character canonically determines the element $c \in E$. When we say that one may take $\beta(\theta) = c$, we are making a deliberate choice of representative. Products and powers of characters are handled by Lemma 4.3; if a conductor drops but remains greater than one, the stationary class is redefined at the new conductor and only its restriction to the old layer is represented by the old sum. If it drops to conductor zero or one, the corresponding endpoint formula is used and there is no stationary class. Finally, changing a representative is always accompanied by the translation of the odd critical function prescribed by Lemma 4.6. Hence every complete Lamprecht factor used below is independent of these choices.

Let $m = m_F(\chi_F) = 2d + \varepsilon > 1$ and $m' = m_K(\chi_F \circ N) = 2d' + \varepsilon'$. We use one of the following depth conventions.

- (i) If K/F is unramified, then $m' = m$.
- (ii) If K/F is ramified with break t and $m > t + 1$, then

$$m' = \psi_{K/F}^H(m - 1) + 1.$$

- (iii) If K/F is ramified, $2 \leq m \leq t + 1$, and χ_F has minimal conductor in its $S(K/F)$ -orbit, then $m' = m$; this is (46).

For $x \in \mathfrak{p}_K^{d'+\varepsilon'}$ put

$$(101) \quad P_{K/F}(x) = N(1 + x) - 1 \pmod{\mathfrak{p}_F^m}.$$

For every finite separable extension, multiplication of the conjugates of $1 + x$ gives the exact identity

$$(102) \quad N_{K/F}(1 + x) = 1 + \sum_{j=1}^{[K:F]} s_j(x), \quad s_1(x) = \text{Tr}_{K/F}(x), \quad s_{[K:F]}(x) = N_{K/F}(x),$$

where $s_j(x)$ is the j th elementary symmetric function of the F -conjugates of x . This identity is polynomial and remains valid regardless of the residue characteristic because K/F itself is separable.

Lemma 6.1 (Norm filtration at stationary depth). *In each of the three depth conventions above,*

$$(103) \quad N(U_K^{d'+\varepsilon'}) \subseteq U_F^{d+\varepsilon},$$

$$(104) \quad N(U_K^{m'}) \subseteq U_F^m,$$

$$(105) \quad N(U_K^{d'}) \subseteq U_F^d.$$

In the unramified case these inclusions are equalities. Consequently, if $b, b' \in \mathcal{O}_K^\times$ and $b' \equiv b \pmod{\mathfrak{p}_K^{d'}}$, then

$$(106) \quad N(b') \equiv N(b) \pmod{\mathfrak{p}_F^d}.$$

Thus the norm of a stationary numerator class modulo $\mathfrak{p}_K^{d'}$ is a well-defined class modulo $\mathfrak{p}_F^d$.

Proof. Put

$$q = \left\lceil \frac{m}{2} \right\rceil = d + \varepsilon, \quad q' = \left\lceil \frac{m'}{2} \right\rceil = d' + \varepsilon'.$$

In the unramified case Lemma 3.11(c) gives equality at every unit depth.

Assume that K/F is ramified. We first prove (103) in the high range. Set $r = m - 1 \geq t$ and $s = \lfloor r/2 \rfloor = q - 1$. Then

$$m' = \psi_{K/F}^H(r) + 1, \quad q' - 1 = \left\lfloor \frac{\psi_{K/F}^H(r)}{2} \right\rfloor.$$

We claim that

$$(107) \quad \left\lfloor \frac{\psi_{K/F}^H(r)}{2} \right\rfloor \geq \psi_{K/F}^H(s).$$

If $s \leq t$, then the right side is s , whereas $\psi_{K/F}^H(r) \geq r \geq 2s$. If $s > t$, write $r = 2s + e$ with $e \in \{0, 1\}$. The explicit formula (20) gives

$$\psi_{K/F}^H(r) - 2\psi_{K/F}^H(s) = (\ell - 1)t + \ell e \geq 0,$$

which again proves (107). Hence

$$(108) \quad q' \geq \psi_{K/F}^H(q - 1) + 1.$$

If $q - 1 < t$, norm preserves the filtration below the break, so $N(U_K^q) \subseteq U_F^q$. If $q - 1 = t$, the critical norm map gives $N(U_K^{t+1}) \subseteq U_F^{t+1}$. If $q - 1 > t$, part (iv) of Theorem 3.1 gives $N(U_K^{\psi_{K/F}^H(q-1)+1}) = U_F^q$. Together with (108) and the decreasing unit filtration, these three cases prove (103).

For (104), if $m > t + 1$, part (iv) of the graded norm theorem, applied with $r = m - 1$, gives $N(U_K^{m'}) = U_F^m$. If $m = t + 1$, then $m' = t + 1$ and the critical norm map gives $N(U_K^{m'}) \subseteq U_F^m$.

We next prove (105) in the high range. Since $\psi_{K/F}^H(u) \geq u$, one has $m' \geq m$ and hence $d' \geq d$. If $d \leq t$, norm preserves the filtration at depth d , so

$$N(U_K^{d'}) \subseteq N(U_K^d) \subseteq U_F^d.$$

If $d = t + 1$, the same conclusion follows from the critical inclusion $N(U_K^{t+1}) \subseteq U_F^{t+1}$. Finally suppose $d \geq t + 2$ and put

$$R = \psi_{K/F}^H(d - 1) + 1.$$

Using (20) and $m = 2d + \varepsilon$, one obtains

$$\begin{aligned} m' - 2R &= \psi_{K/F}^H(2d + \varepsilon - 1) + 1 - 2(\psi_{K/F}^H(d - 1) + 1) \\ &= (\ell - 1)(t + 1) + \ell\varepsilon \geq 0. \end{aligned}$$

Thus $d' = \lfloor m'/2 \rfloor \geq R$. Part (iv) of Theorem 3.1, applied to $d - 1 > t$, now gives

$$N(U_K^{d'}) \subseteq N(U_K^R) = U_F^d.$$

This proves the third inclusion in the high range.

In the low range $m' = m \leq t + 1$, so $q' = q$ and $d' = d \leq t$. Below the break, norm maps U_K^i into U_F^i for every $i \leq t$; at the last possible step $m = t + 1$ one uses the critical inclusion $N(U_K^{t+1}) \subseteq U_F^{t+1}$. Applying these statements at $i = q$, $i = m$, and $i = d$ proves all three inclusions.

For the final assertion, write $b' = b(1 + z)$ with $z \in \mathfrak{p}_K^{d'}$. Then $1 + z \in U_K^{d'}$, so (105) gives $N(b')/N(b) \in U_F^d$, which is equivalent to (106). $\square$

Lemma 6.2 (Truncated norm homomorphism and adjoint). *The formula (101) induces an additive homomorphism*

$$P_{K/F} : \mathfrak{p}_K^{d'+\varepsilon}/\mathfrak{p}_K^{m'} \longrightarrow \mathfrak{p}_F^{d+\varepsilon}/\mathfrak{p}_F^m.$$

For each admissible pair (γ_F, γ_K) , the pairings

$$(\bar{x}, \bar{y}) \longmapsto \psi_F(xy/\gamma_F), \quad (\bar{x}', \bar{y}') \longmapsto \psi_K(x'y'/\gamma_K)$$

between $\mathcal{O}_F/\mathfrak{p}_F^d$ and $\mathfrak{p}_F^{d+\varepsilon}/\mathfrak{p}_F^m$, and between $\mathcal{O}_K/\mathfrak{p}_K^{d'}$ and $\mathfrak{p}_K^{d'+\varepsilon}/\mathfrak{p}_K^{m'}$, are perfect. Consequently there is a unique additive adjoint

$$(109) \quad P_{K/F; \gamma_F, \gamma_K}^* : \mathcal{O}_F/\mathfrak{p}_F^d \longrightarrow \mathcal{O}_K/\mathfrak{p}_K^{d'}$$

satisfying

$$(110) \quad \psi_F \left(\frac{x P_{K/F}(y)}{\gamma_F} \right) = \psi_K \left(\frac{P_{K/F; \gamma_F, \gamma_K}^*(x) y}{\gamma_K} \right).$$

When the denominator pair is fixed, we abbreviate this map to $P_{K/F}^*$. Thus the adjoint notation always includes the chosen pair, even when it is suppressed. If $P_{K/F} = \text{Tr}_{K/F}$ at the required precision and $\gamma_K/\gamma_F \in \mathcal{O}_K^\times$, then

$$(111) \quad P_{K/F; \gamma_F, \gamma_K}^*(x) \equiv \frac{\gamma_K}{\gamma_F} x \pmod{\mathfrak{p}_K^{d'}}.$$

In particular, when $\gamma_F = \gamma_K = \gamma \in F^\times$, the adjoint is the natural inclusion $\mathcal{O}_F/\mathfrak{p}_F^d \rightarrow \mathcal{O}_K/\mathfrak{p}_K^{d'}$.

Proof. The first inclusion in Lemma 6.1 places $P_{K/F}(x)$ in the stated target. If x' differs from x by an element of $\mathfrak{p}_K^{m'}$, then

$$\frac{1+x'}{1+x} \in U_K^{m'},$$

so the second inclusion shows that their norms are congruent modulo U_F^m ; hence $P_{K/F}$ is well defined.

Let $x, y \in \mathfrak{p}_K^{d'+\varepsilon'}$. Since $2(d' + \varepsilon') \geq m'$, one has $xy \in \mathfrak{p}_K^{m'}$. Multiplicativity of norm gives

$$1 + P(x + y + xy) = (1 + P(x))(1 + P(y)).$$

The preceding well-definedness permits deletion of xy . Moreover $P(x), P(y) \in \mathfrak{p}_F^{d+\varepsilon}$ and $2(d+\varepsilon) \geq m$, so $P(x)P(y) \in \mathfrak{p}_F^m$. Reducing the displayed identity modulo $\mathfrak{p}_F^m$ gives $P(x+y) = P(x) + P(y)$.

Perfectness of the two pairings is Lemma 4.1, once over F and once over K . For fixed $x \in \mathcal{O}_F/\mathfrak{p}_F^d$, the left side of (110) is a character of the source of P . Perfectness over K supplies a unique representing class in $\mathcal{O}_K/\mathfrak{p}_K^{d'}$. Character multiplication and uniqueness show that this representing class depends additively on x . This is the desired adjoint.

Finally suppose that $P_{K/F} = \text{Tr}_{K/F}$ at the required precision and that $\gamma_K/\gamma_F \in \mathcal{O}_K^\times$. Trace transitivity gives

$$\psi_F \left(\frac{x \text{Tr}_{K/F}(y)}{\gamma_F} \right) = \psi_K \left(\frac{xy}{\gamma_F} \right) = \psi_K \left(\frac{(\gamma_K/\gamma_F)xy}{\gamma_K} \right).$$

Perfectness over K proves (111); the common-denominator case is its specialization. $\square$

Corollary 6.3 (Stationary class under norm pullback). *Let*

$$\beta_F = \text{St}_{\gamma_F, d+\varepsilon}(\chi_F) \in \mathcal{O}_F/\mathfrak{p}_F^d, \quad \beta_K = \text{St}_{\gamma_K, d'+\varepsilon'}(\chi_F \circ N) \in \mathcal{O}_K/\mathfrak{p}_K^{d'}.$$

Then, with $P_{K/F}^ = P_{K/F; \gamma_F, \gamma_K}^*$,*

$$(112) \quad \beta_K = P_{K/F}^*(\beta_F).$$

In particular, every explicit formula for $P_{K/F}^(\beta_F)$ below is an equality of stationary classes. Any element representing its right-hand side may be chosen as the upstairs stationary numerator; its complete Lamprecht factor is independent of that choice by Lemma 4.6.*

Proof. For $y \in \mathfrak{p}_K^{d'+\varepsilon'}$, the definition of $P_{K/F}$ and Lemma 6.1 give

$$\begin{aligned} (\chi_F \circ N)(1+y) &= \chi_F(1 + P_{K/F}(y)) \\ &= \psi_F \left(\frac{\beta_F P_{K/F}(y)}{\gamma_F} \right) \\ &= \psi_K \left(\frac{P_{K/F}^*(\beta_F) y}{\gamma_K} \right). \end{aligned}$$

The uniqueness assertion in Lemma 4.2, applied over K , identifies this class with β_K . $\square$

In the unramified case, $d' = d$ and $\varepsilon' = \varepsilon$. If $y \in \mathfrak{p}_K^{d+\varepsilon}$, the j th elementary symmetric term in $N(1+y)$ has valuation at least $j(d+\varepsilon)$; hence every term with $j \geq 2$ lies in $\mathfrak{p}_F^m$, and

$$(113) \quad P_{K/F}(y) \equiv \text{Tr}_{K/F}(y) \pmod{\mathfrak{p}_F^m}.$$

Lemma 6.4 (Integer powers and Teichmüller representatives). *Assume that K/F is wildly ramified of degree p and put $T = t + 1$ and $s_0 = \lceil T/2 \rceil$. For $j \in \mathbf{F}_p$, let $j\mathbf{z}$ be its integer representative in $\{0, \dots, p-1\}$ and let $[j] \in \mathcal{O}_F$ be its Teichmüller lift. If*

$$\tau(1+x) = \psi_F(Ax) \quad (x \in \mathfrak{p}_F^{s_0}),$$

then

$$(114) \quad \psi_F(j\mathbf{z}Ax) = \psi_F([j]Ax) \quad (x \in \mathfrak{p}_F^{s_0}).$$

Thus every stationary numerator for the integer power τ^j may be represented by the Teichmüller scalar $[j]$ at all depths occurring below. The Teichmüller lifts satisfy

$$(115) \quad \prod_{j \in \mathbf{F}_p} (X - [j]) = X^p - X, \quad \prod_{j \in \mathbf{F}_p} (X + [j]) = X^p + (-1)^p X,$$

and

$$(116) \quad \sum_{j \in \mathbf{F}_p} [j] = \begin{cases} 0, & p \text{ odd}, \\ 1, & p = 2. \end{cases}$$

Proof. In equal characteristic, the prime-field element $j\mathbf{z}$ is already its Teichmüller lift. Suppose that F has characteristic zero and put $e = v_F(p)$. Since $p = \text{Tr}_{K/F}(1)$, the trace-ideal formula (35) with $a = 0$ gives

$$e \geq \left\lfloor \frac{(p-1)T}{p} \right\rfloor \geq \left\lfloor \frac{T}{2} \right\rfloor.$$

Moreover $[j] - j\mathbf{z} \in p\mathcal{O}_F$. Hence for $x \in \mathfrak{p}_F^{s_0}$,

$$([j] - j\mathbf{z})x \in \mathfrak{p}_F^{e+s_0} \subseteq \mathfrak{p}_F^T.$$

The additive character $x \mapsto \psi_F(Ax)$ represents the last nontrivial unit layer of the conductor- T character τ , and is therefore trivial on $\mathfrak{p}_F^T$. This proves (114).

The elements $[j]$ are precisely the p roots of $X^p - X$ in the Teichmüller copy of the prime field. This proves the first polynomial identity; replacing X by $-X$ gives the second. The coefficient of X^{p-1} gives the sum formula. $\square$

Remark 6.5 (Notation for powers of the norm character). From this point onward, an exponent j in τ^j denotes the integer representative of a class in $\mathbf{F}_p$, while the same symbol inside a field-valued formula denotes its Teichmüller lift. The preceding lemma proves that these two uses give the same stationary character at every precision used below.

Lemma 6.6 (Norm representatives at subcritical precision). *Let K/F be ramified cyclic of prime degree with break t . Let $h \in \mathbf{Z}$, let $0 \leq s \leq t$, and let $c \in F^\times$ have $v_F(c) = h$. There is $c_1 \in K^\times$ with $v_K(c_1) = h$ such that*

$$(117) \quad c - N(c_1) \in \mathfrak{p}_F^{h+s}.$$

Consequently every stationary coefficient class of the form $c + \mathfrak{p}_F^{h+s}$, with $s \leq t$, has a representative which is an exact norm. Any finite collection of such classes admits norm representatives chosen independently and hence simultaneously.

Proof. The norm isomorphism

$$K^\times / U_K^t \xrightarrow{\sim} F^\times / U_F^t$$

of Theorem 3.7(c) supplies c_1 with $c/N(c_1) \in U_F^t$. Total ramification gives $v_K(c_1) = v_F(N(c_1)) = h$. Since c and $N(c_1)$ have the same valuation and their quotient lies in U_F^t , their difference belongs to $\mathfrak{p}_F^{h+t} \subseteq \mathfrak{p}_F^{h+s}$. Applying this argument separately to each member of a finite collection proves the last assertion. $\square$

6.1. High-conductor parameters. We first give the wild non-stable calculation in odd prime degree. Let K/F be totally and wildly ramified of odd prime degree p , put

$$T = t + 1, \quad D = (p - 1)T,$$

and suppose

$$T \leq m = m_F(\chi_F) = 2d + \varepsilon \leq 2T - 1, \quad m = T + v \quad (0 \leq v \leq t).$$

As throughout the ramified analysis, assume that χ_F has minimal conductor in its $S(K/F)$ -orbit. Then

$$(118) \quad m' = m_K(\chi_K) = T + pv = pm - D = 2d' + \varepsilon.$$

For $v > 0$ this follows from Proposition 3.10(i), while for $v = 0$ it is (46). In particular, m and m' have the same parity. Put

$$r_0 = \left\lfloor \frac{T}{2} \right\rfloor, \quad s_0 = \left\lceil \frac{T}{2} \right\rceil.$$

Relative to the denominator γ_F , the stationary class of a generator τ of $S(K/F)$ on $\mathfrak{p}_F^{s_0}$ is

$$(119) \quad \text{St}_{\gamma_F, s_0}(\tau) \in \mathfrak{p}_F^v / \mathfrak{p}_F^{v+r_0},$$

because $m = T + v$ and $r_0 + s_0 = T$. Since $r_0 \leq t$, Lemma 6.6 permits the choice of a representative

$$(120) \quad \alpha = N(\alpha_1), \quad v_K(\alpha_1) = v,$$

for which

$$(121) \quad \tau(1 + x) = \psi_F(\alpha x / \gamma_F) \quad (x \in \mathfrak{p}_F^{s_0}).$$

Likewise

$$\text{St}_{\gamma_F, d+\varepsilon}(\chi_F) \in \mathcal{O}_F / \mathfrak{p}_F^d,$$

and $d \leq t$. The same lemma therefore permits an exact norm representative

$$(122) \quad \beta = N(\beta_1), \quad \beta_1 \in U_K,$$

satisfying

$$(123) \quad \chi_F(1 + x) = \psi_F(\beta x / \gamma_F) \quad (x \in \mathfrak{p}_F^{d+\varepsilon}).$$

The two norm representatives are chosen independently and hence simultaneously. If $m = T$, then Lemma 4.8 shows that none of the leading classes $\beta + j\alpha$ for $j \in \mathbf{F}_p$ cancels; thus every twist in the table below still has exact conductor T .

Lemma 6.7 (Norm-character conversion). *For $y \in \mathfrak{p}_K^{s_0}$,*

$$(124) \quad \psi_F\left(\frac{\alpha N(y)}{\gamma_F}\right) = \psi_F\left(-\frac{\alpha \text{Tr}(y)}{\gamma_F}\right).$$

Proof. Corollary 3.5, with $q = s_0$ and $r = T$, gives

$$N(1 + y) \equiv 1 + \text{Tr}(y) + N(y) \pmod{\mathfrak{p}_F^T},$$

because $2s_0 \geq T$. The trace-ideal formula and total ramification give

$$v_F(\text{Tr}(y)) \geq \left\lfloor \frac{s_0 + D}{p} \right\rfloor \geq s_0, \quad v_F(N(y)) = v_K(y) \geq s_0.$$

Thus the sum lies in the layer on which (121) holds. Since τ is trivial on norms and has conductor T , applying τ to $N(1 + y)$ proves (124). $\square$

Proposition 6.8 (High-conductor stationary-class table). *In each case choose an admissible denominator γ_F downstairs and put $\gamma_K = \gamma_F$, with the same element of $F^\times$ viewed in $K^\times$. The conductor identities in the proof show that this is admissible upstairs. Throughout the statement, $P_{K/F}^*$ denotes $P_{K/F; \gamma_F, \gamma_K}^*$ for this common-denominator pair. The following statements hold.*

(a) *If K/F is unramified, then $P_{K/F}^*(\beta) = \beta$ in $\mathcal{O}_K / \mathfrak{p}_K^{d'}$.*

(b) If K/F is totally ramified of odd prime degree and $d \geq t + 1$, then $P_{K/F}^*(\beta) = \beta$ in $\mathcal{O}_K/\mathfrak{p}_K^{d'}$.

(c) In the odd wild non-stable range above, the upstairs stationary class is

$$(125) \quad \beta_K := \beta(\chi_K) \equiv \beta - \frac{\beta_1 \alpha}{\alpha_1} \pmod{\mathfrak{p}_K^{d'}}.$$

For $j \in \mathbf{F}_p$, represented by its Teichmüller lift, the stationary class of the twist is

$$(126) \quad \beta_j := \beta(\tau^j \chi_F) \equiv N(\beta_1 + j\alpha_1) \equiv \beta + j\alpha \pmod{\mathfrak{p}_F^d}.$$

Every right-hand side in these congruences may be selected as the corresponding stationary representative. Moreover the norm of the upstairs class is well defined modulo $\mathfrak{p}_F^d$ and satisfies

$$(127) \quad N(\beta_K) \equiv \prod_{j \in \mathbf{F}_p} (\beta + j\alpha) = \beta^p - \beta\alpha^{p-1} \pmod{\mathfrak{p}_F^d}.$$

(d) In the tamely ramified quadratic case, $P_{K/F}^*(\beta) = \beta$ in $\mathcal{O}_K/\mathfrak{p}_K^{d'}$.

All assertions in (a), (b), and (d) are equalities of stationary classes as in Corollary 6.3; no canonical field representative is asserted.

Proof. We first verify the denominator compatibility used in the statement. In the unramified case, $m' = m$ and $n_K(\psi_K) = n_F(\psi_F)$, so an admissible γ_F remains admissible in K . In every ramified case in this high-conductor table, of degree ℓ , one has

$$m' = \ell m - D, \quad n_K(\psi_K) = \ell n_F(\psi_F) + D,$$

and therefore

$$m' + n_K(\psi_K) = \ell(m + n_F(\psi_F)) = v_K(\gamma_F).$$

Thus $\gamma_K = \gamma_F$ is admissible upstairs in all four parts.

In the unramified case, equation (113), trace transitivity, and the common denominator identify the adjoint with the inclusion $F \hookrightarrow K$. Corollary 6.3 then proves (a).

For (b), put $q = d' + \varepsilon' = \lceil m'/2 \rceil$ and separate the tame and wild cases. If K/F is tamely ramified of odd degree ℓ , then

$$m' - 1 = \ell(m - 1).$$

For $x \in \mathfrak{p}_K^q$ and $2 \leq j \leq \ell - 1$, every summand defining $s_j(x)$ has K -valuation at least jq . Since $s_j(x) \in F$ and $v_K|_F = \ell v_F$, it follows that

$$v_F(s_j(x)) \geq \left\lceil \frac{jq}{\ell} \right\rceil \geq \left\lceil \frac{2q}{\ell} \right\rceil \geq m.$$

Also $v_F(N(x)) = v_K(x) \geq q \geq m$. Thus all nonlinear norm terms vanish modulo $\mathfrak{p}_F^m$.

If K/F is wildly ramified, its degree is the residue characteristic p , $D = (p - 1)T$, and

$$m' = p(m - T) + T.$$

The assumption $d \geq T$ gives $q \geq m$. Moreover $2q + D \geq m' + D = pm$, so the strong symmetric estimate gives $v_F(s_j(x)) \geq m$ for $2 \leq j \leq p - 1$. Again the norm term has valuation at least m . Hence in either case $P = \text{Tr}$ at the required precision and $P^*(\beta) = \beta$. The class interpretation again follows from Corollary 6.3.

For (c), put $q = \lceil m'/2 \rceil$. Since $2q + D \geq pm$, Corollary 3.5 gives

$$(128) \quad N(1 + x) - 1 \equiv \text{Tr}(x) + N(x) \pmod{\mathfrak{p}_F^q} \quad (x \in \mathfrak{p}_K^q).$$

The elementary identity $q - v \geq d + \varepsilon \geq s_0$ shows that $y = \beta_1 x / \alpha_1$ lies in the domain of Lemma 6.7. Because $\beta = N(\beta_1)$ exactly, the conversion identity gives

$$\psi_F\left(\frac{\beta N(x)}{\gamma_F}\right) = \psi_F\left(-\frac{\alpha \text{Tr}(\beta_1 x / \alpha_1)}{\gamma_F}\right).$$

Combining this with the trace term in (128) proves that the class on the right of (125) is $P_{K/F}^*(\beta)$. The element γ_F , viewed in K , is admissible upstairs because

$$m' + n_K(\psi_K) = p(m + n_F(\psi_F)).$$

Corollary 6.3 now proves (125) as an equality of stationary classes.

On the common stationary layer, Lemma 4.3(c) adds the classes of χ_F and τ^j , giving the class of $\beta + j\alpha$. Its conductor is exactly m by Lemma 4.8; in particular, the boundary case $m = T$ involves no unrecorded conductor drop. Factoring $N(\beta_1 + j\alpha_1) = \beta N(1 + j\alpha_1/\beta_1)$, the strong symmetric estimate gives, for every intermediate term, valuation at least d , since

$$iv + (p-1)T \geq pd \quad (1 \leq i \leq p-1).$$

The two endpoint terms are β and $j\alpha$, because $j^p = j$ for the chosen Teichmüller lift. This proves (126) as a class congruence.

For the product congruence, choose the normalized representative

$$\beta_K^0 = \beta - \frac{\beta_1\alpha}{\alpha_1}$$

of the class in (125). By (106), the class of $N(\beta_K)$ modulo $\mathfrak{p}_F^d$ is the class of $N(\beta_K^0)$, so this choice entails no loss of information. Put $y = \beta_1\alpha/\alpha_1$. Then $v_K(y) = (p-1)v$ and $N(y) = \beta\alpha^{p-1}$. In the expansion of $N(\beta - y)$ every intermediate term has valuation at least d , because

$$i(p-1)v + (p-1)T \geq pd \quad (1 \leq i \leq p-1).$$

The endpoints give $\beta^p - \beta\alpha^{p-1}$, which equals $\prod_{j \in \mathbf{F}_p} (\beta + j\alpha)$. This proves (127) independently of the representative of β_K .

Finally, in the tame quadratic case the stationary source is $\mathfrak{p}_K^m$ and the quadratic norm term is already in $\mathfrak{p}_F^m$; hence $P = \text{Tr}$ and $P^*(\beta) = \beta$. Corollary 6.3 again supplies the class interpretation. $\square$

Proposition 6.9 (Compatible wild-quadratic stationary classes and representatives). *Assume $p = 2$, K/F is wildly ramified, and the character is in the high-conductor range. Representatives of the relevant stationary classes may be chosen in the form*

$$\alpha = N(\alpha_1), \quad \beta = \delta_h N(\beta_1), \quad \delta_h \in U_F^t,$$

with $\tau(1+x) = \psi_F(\alpha\delta_h x/\gamma_F)$ on the required unit layer. For these choices, the stationary classes satisfy

$$(129) \quad \beta_K \equiv \beta - \frac{\beta_1\alpha\delta_h}{\alpha_1} \pmod{\mathfrak{p}_K^{d'}}, \quad \beta(\tau\chi_F) \equiv \beta + \alpha\delta_h \pmod{\mathfrak{p}_F^d},$$

and their norm-product class satisfies

$$(130) \quad N(\beta_K) \equiv \beta(\beta + \alpha\delta_h) \pmod{\mathfrak{p}_F^d}.$$

The congruences are equalities of stationary classes. The displayed right-hand sides are selected representatives, not canonically determined field elements.

Proof. Put

$$T = t + 1, \quad m = T + v = 2d + \varepsilon, \quad m' = 2m - T = 2d' + \varepsilon',$$

and set

$$r = \left\lfloor \frac{T}{2} \right\rfloor, \quad s = \left\lceil \frac{T}{2} \right\rceil.$$

Then

$$(131) \quad d' + \varepsilon' = m - r, \quad d' + \varepsilon' - v = s, \quad r + s = T.$$

Choose any representative β of the stationary class of χ_F . By Theorem 3.7(c), choose $\beta_1 \in \mathcal{O}_K^\times$ such that $N(\beta_1)/\beta \in U_F^t$, and put

$$\delta_h = \frac{\beta}{N(\beta_1)} \in U_F^t.$$

Thus $\beta = \delta_h N(\beta_1)$ exactly. In general one need not have $d \leq t$, so no congruence $N(\beta_1) \equiv \beta \pmod{\mathfrak{p}_F^d}$ is asserted. The factor δ_h is essential and will be retained in every subsequent stationary-class formula.

Let c_τ be any representative of the stationary coefficient class of τ relative to γ_F on $\mathfrak{p}_F^s$. That class lies in

$$\mathfrak{p}_F^v / \mathfrak{p}_F^{v+r}.$$

Apply Lemma 6.6 to the element $c_\tau \delta_h^{-1}$ with $(h, s) = (v, r)$. It supplies $\alpha_1 \in K^\times$ with $v_K(\alpha_1) = v$ such that, for $\alpha = N(\alpha_1)$,

$$\alpha \delta_h \equiv c_\tau \pmod{\mathfrak{p}_F^{v+r}}.$$

Thus $\alpha \delta_h$ is a permitted representative and

$$(132) \quad \tau(1+x) = \psi_F\left(\frac{\alpha \delta_h x}{\gamma_F}\right) \quad (x \in \mathfrak{p}_F^s).$$

The choices of β_1 and α_1 are independent; this proves simultaneous existence without asserting uniqueness.

For $y \in \mathfrak{p}_K^s$, the quadratic norm identity is exact:

$$N(1+y) = 1 + \text{Tr}(y) + N(y).$$

The trace-ideal formula puts both $\text{Tr}(y)$ and $N(y)$ in $\mathfrak{p}_F^s$. Since τ is trivial on norms, applying (132) gives the conversion identity

$$(133) \quad \psi_F\left(\frac{\alpha \delta_h N(y)}{\gamma_F}\right) = \psi_F\left(-\frac{\alpha \delta_h \text{Tr}(y)}{\gamma_F}\right).$$

Let $x \in \mathfrak{p}_K^{d'+\varepsilon'}$ and put $y = \beta_1 x / \alpha_1$. By (131), $y \in \mathfrak{p}_K^s$, and the identities $\beta = \delta_h N(\beta_1)$ and $\alpha = N(\alpha_1)$ give

$$\beta N(x) = \alpha \delta_h N(y).$$

Using (133) in the exact expansion $P_{K/F}(x) = \text{Tr}(x) + N(x)$ yields

$$\begin{aligned} \psi_F\left(\frac{\beta P_{K/F}(x)}{\gamma_F}\right) &= \psi_F\left(\frac{\beta \text{Tr}(x) - \alpha \delta_h \text{Tr}(\beta_1 x / \alpha_1)}{\gamma_F}\right) \\ &= \psi_K\left(\frac{(\beta - \beta_1 \alpha \delta_h / \alpha_1)x}{\gamma_F}\right). \end{aligned}$$

Here γ_F , viewed in K , is admissible because $m' + n_K(\psi_K) = 2(m + n_F(\psi_F))$. By Corollary 6.3, this proves the first class congruence in (129).

Since $d + \varepsilon \geq s$, the two stationary restrictions of τ and χ_F are defined on a common layer. Lemma 4.3(c) therefore gives the second class congruence in (129). At $m = T$, its leading class is nonzero by Lemma 4.8, so no hidden conductor drop occurs.

Finally choose the normalized representative

$$\beta_K^0 = \beta - \frac{\beta_1 \alpha \delta_h}{\alpha_1}$$

of the upstairs class, and put $c = \beta_1 \alpha \delta_h / \alpha_1$. By (106), it is enough to compute $N(\beta_K^0)$ modulo $\mathfrak{p}_F^d$. The exact quadratic norm formula gives

$$N(\beta - c) = \beta^2 - \beta \text{Tr}(c) + N(c), \quad N(c) = \beta \alpha \delta_h.$$

Now $v_K(c) = v$, and the trace-ideal formula, with different exponent T , gives

$$v_F(\text{Tr}(c)) \geq \left\lfloor \frac{v+T}{2} \right\rfloor = d.$$

Consequently

$$N(\beta_K) \equiv N(\beta_K^0) \equiv \beta^2 + \beta \alpha \delta_h = \beta(\beta + \alpha \delta_h) \pmod{\mathfrak{p}_F^d},$$

which proves (130) independently of the chosen upstairs representative. $\square$

6.2. Low-conductor parameters. Assume

$$2 \leq m = 2d + \varepsilon \leq T, \quad v = T - m.$$

Choose χ_F of minimal conductor in its $S(K/F)$ -orbit; then $m_K(\chi_K) = m$. Put

$$r_0 = \left\lfloor \frac{T}{2} \right\rfloor, \quad s_0 = \left\lceil \frac{T}{2} \right\rceil.$$

Choose

$$\delta \mathcal{O}_F = \mathfrak{p}_F^{T+n_F(\psi_F)}, \quad \varepsilon_1 \mathcal{O}_K = \mathfrak{p}_K^v, \quad \varepsilon = N(\varepsilon_1),$$

and

$$(134) \quad \gamma_F = \delta/\varepsilon, \quad \gamma_K = \delta/\varepsilon_1.$$

Relative to δ , the stationary class of τ on $\mathfrak{p}_F^{s_0}$ lies in $\mathcal{O}_F/\mathfrak{p}_F^{r_0}$; relative to γ_F , the stationary class of χ_F lies in $\mathcal{O}_F/\mathfrak{p}_F^d$. Since $r_0, d \leq t$, Lemma 6.6 permits independent, hence simultaneous, choices $\alpha_1, \beta_1 \in U_K$ and exact norm representatives

$$(135) \quad \alpha = N(\alpha_1), \quad \beta = N(\beta_1),$$

such that

$$(136) \quad \tau(1+x) = \psi_F(\alpha x/\delta) \quad (x \in \mathfrak{p}_F^{s_0}), \quad \chi_F(1+x) = \psi_F(\beta x/\gamma_F) \quad (x \in \mathfrak{p}_F^{d+\varepsilon}).$$

For $j \neq 0$, Lemma 4.8 gives $m_F(\tau^j \chi_F) = T$. In particular, when $m = T$, the leading class $j\alpha + \beta$ does not cancel.

Proposition 6.10 (Low-conductor stationary-class table). *The upstairs stationary class is*

$$(137) \quad \beta_K := \beta(\chi_K) \equiv \frac{\beta\varepsilon}{\varepsilon_1} - \frac{\beta_1\alpha}{\alpha_1} \pmod{\mathfrak{p}_K^d}.$$

For $j \in \mathbf{F}_p$, let $\tilde{\beta}_j$ denote the stationary numerator class of $\tau^j \chi_F$ after all factors are written with the common additive denominator δ . Thus $\tilde{\beta}_0 = \varepsilon\beta$, while for $j \neq 0$ it is the ordinary conductor- T stationary numerator class relative to δ . Then

$$(138) \quad \tilde{\beta}_j \equiv N(j\alpha_1 + \varepsilon_1\beta_1) \equiv j\alpha + \varepsilon\beta \pmod{\mathfrak{p}_F^{r_0}}.$$

Every displayed right-hand side may be chosen as a representative of the indicated class. Moreover the norm of the upstairs class is well defined modulo $\mathfrak{p}_F^d$ and

$$(139) \quad N(\beta_K) \equiv \beta \prod_{j \in \mathbf{F}_p^\times} (\varepsilon\beta + j\alpha) \pmod{\mathfrak{p}_F^d}.$$

Proof. Put $q = d + \varepsilon = \lceil m/2 \rceil$. Since $2q + (p-1)T \geq pm$, Corollary 3.5 gives

$$(140) \quad N(1+x) - 1 \equiv \text{Tr}(x) + N(x) \pmod{\mathfrak{p}_F^m} \quad (x \in \mathfrak{p}_K^q).$$

For $y = \varepsilon_1\beta_1 x/\alpha_1$ one has

$$v_K(y) \geq q + v = T - \lfloor m/2 \rfloor \geq s_0.$$

The trace-ideal formula and total ramification give

$$v_F(\text{Tr}(y)) \geq \left\lfloor \frac{v_K(y) + (p-1)T}{p} \right\rfloor \geq T - \left\lceil \frac{\lfloor m/2 \rfloor}{p} \right\rceil \geq \left\lceil \frac{T}{2} \right\rceil = s_0, \quad v_F(N(y)) = v_K(y) \geq s_0.$$

Moreover

$$2v_K(y) + (p-1)T \geq 2(T - \lfloor m/2 \rfloor) + (p-1)T \geq pT,$$

so Corollary 3.5, now used with target precision T , gives

$$N(1+y) \equiv 1 + \text{Tr}(y) + N(y) \pmod{\mathfrak{p}_F^T}.$$

The character τ is trivial on norms and on U_F^T , while its linearization is valid on $\mathfrak{p}_F^{s_0}$. Applying τ to this congruence therefore gives

$$(141) \quad \psi_F\left(-\frac{\alpha \text{Tr}(y)}{\delta}\right) = \psi_F\left(\frac{\alpha N(y)}{\delta}\right) = \psi_F\left(\frac{\varepsilon\beta N(x)}{\delta}\right).$$

For the element on the right of (137),

$$\begin{aligned}\psi_K\left(\frac{\beta_K x}{\gamma_K}\right) &= \psi_F\left(\frac{\text{Tr}(\varepsilon_1 \beta_K x)}{\delta}\right) \\ &= \psi_F\left(\frac{\varepsilon \beta \text{Tr}(x) - \alpha \text{Tr}(\varepsilon_1 \beta_1 x / \alpha_1)}{\delta}\right).\end{aligned}$$

Using (141) and (140), this is exactly the pullback of the stationary character of χ_F . Corollary 6.3 proves (137) as an equality of stationary classes.

For $j = 0$, the first congruence in (138) is the exact equality $N(\varepsilon_1 \beta_1) = \varepsilon \beta$. For $j \neq 0$, put $w = \varepsilon_1 \beta_1 / (j \alpha_1)$, so $v_K(w) = v$. The strong symmetric estimate gives

$$v_F(s_i(w)) \geq \left\lfloor \frac{iv + (p-1)T}{p} \right\rfloor \geq r_0 \quad (1 \leq i \leq p-1),$$

and therefore $N(j \alpha_1 + \varepsilon_1 \beta_1) \equiv j \alpha + \varepsilon \beta \pmod{\mathfrak{p}_F^{r_0}}$. By Lemma 4.3(c), this is precisely the sum of the two stationary classes with common denominator δ . Its exact conductor is T for $j \neq 0$ by Lemma 4.8; this explicitly excludes a boundary conductor drop.

Put

$$u = \frac{\varepsilon_1 \beta_1}{\alpha_1}, \quad n = N(u) = \frac{\varepsilon \beta}{\alpha},$$

and choose the normalized representative

$$\beta_K^0 = \frac{\alpha}{\varepsilon_1}(n - u)$$

of the class in (137). By (106), $N(\beta_K)$ modulo $\mathfrak{p}_F^d$ may be computed from this representative. Thus

$$N(\beta_K) \equiv N(\beta_K^0) = \frac{\alpha^p}{\varepsilon} N(n - u) \pmod{\mathfrak{p}_F^d}.$$

In the expansion of $N(n - u)$ the two endpoints are $n^p + (-1)^p n$. For $1 \leq i \leq p-1$, the intermediate term $n^{p-i} s_i(u)$, after multiplication by α^p / ε , has valuation at least

$$L_i = (p-i-1)v + \left\lfloor \frac{iv + (p-1)T}{p} \right\rfloor.$$

Because T is an integer,

$$\left\lfloor \frac{iv + (p-1)T}{p} \right\rfloor = T - \left\lceil \frac{T - iv}{p} \right\rceil.$$

Moreover $i \geq 1$ and $v = T - m \geq 0$, so $T - iv \leq T - v = m = 2d + \varepsilon$. Since $p \geq 2$,

$$\left\lceil \frac{T - iv}{p} \right\rceil \leq \left\lceil \frac{m}{2} \right\rceil = d + \varepsilon.$$

Consequently

$$\begin{aligned}L_i &\geq (p-i-1)v + T - d - \varepsilon \\ &= (p-i)v + d \\ &\geq d.\end{aligned}$$

Thus every intermediate term vanishes modulo $\mathfrak{p}_F^d$, and only the endpoints survive. Since the Teichmüller representatives satisfy

$$\prod_{j \in \mathbf{F}_p} (X + j) = X^p + (-1)^p X,$$

we obtain

$$N(\beta_K) \equiv \frac{\alpha^p}{\varepsilon} (n^p + (-1)^p n) = \beta \prod_{j \neq 0} (\varepsilon \beta + j \alpha) \pmod{\mathfrak{p}_F^d}.$$

This proves (139) independently of the chosen representative of the upstairs class. $\square$

Remark 6.11 (Characteristic independence of the stationary-class tables). The stationary-class calculus, the proofs of Propositions 6.8 and 6.10, and the representative-independence argument use only the cyclic-prime norm filtration, trace duality, the exact norm polynomial, and Lemma 3.4. Hence the stationary-class tables, the compatible representatives selected from them, and the residual calculations based on the complete stationary factors are valid in both mixed and equal characteristic.

7. UNRAMIFIED AND TAME CASES

Proposition 7.1 (Unramified case). *The First Main Lemma holds when K/F is unramified.*

Proof. Put

$$k = k_F, \quad k' = k_K, \quad n = n_F(\psi_F), \quad m = m_F(\chi_F).$$

Choose a uniformizer π of F . By Lemma 3.11, it is also a uniformizer of K , the residue extension k'/k has degree ℓ , and

$$(142) \quad n_K(\psi_K) = n, \quad m_K(\chi_K) = m.$$

Step 1: reduction to a power identity. Lemma 3.11(e) identifies $F^\times/N(K^\times)$ with the cyclic group generated by the class of π , of order ℓ . Hence every $\mu \in S(K/F)$ is unramified, and the values $\mu(\pi)$ run through all ℓ th roots of unity. Twisting by such a μ does not change the conductor of χ_F . Use the same admissible element $\gamma_F = \pi^{m+n}$ for χ_F and $\mu\chi_F$. Since μ is trivial on U_F^0 , their unit integrals are equal, and therefore

$$(143) \quad \Delta_F(\mu\chi_F, \psi_F) = \mu(\pi)^{m+n} \Delta_F(\chi_F, \psi_F).$$

For μ itself, (11) gives

$$(144) \quad \Delta_F(\mu, \psi_F) = \mu(\pi)^n.$$

Both formulas remain valid when $m = 0$. The values $\mu(\pi)$ are the roots of $X^\ell - 1$, whose product is $(-1)^{\ell-1}$; hence

$$\prod_{\mu \in S(K/F)} \mu(\pi) = (-1)^{\ell-1}.$$

The quotient of the product on the right side of (1) by the product of norm-character factors on its left side is

$$(-1)^{m(\ell-1)} \Delta_F(\chi_F, \psi_F)^\ell.$$

Thus the First Main Lemma is equivalent to

$$(145) \quad \Delta_K(\chi_K, \psi_K) = (-1)^{m(\ell-1)} \Delta_F(\chi_F, \psi_F)^\ell.$$

Step 2: conductor $m = 0$. Formula (11) gives

$$\Delta_F(\chi_F, \psi_F) = \chi_F(\pi)^n, \quad \Delta_K(\chi_K, \psi_K) = \chi_K(\pi)^n.$$

By Lemma 3.11(c), $N_{K/F}(\pi) = \pi^\ell$, so $\chi_K(\pi) = \chi_F(\pi)^\ell$. This proves (145) for $m = 0$.

Step 3: conductor $m = 1$. Let

$$\bar{\chi}(\bar{u}) = \chi_F(u), \quad \bar{\psi}(\bar{x}) = \psi_F(x/\pi^{n+1})$$

be the residual multiplicative and additive characters on k . By (41), the corresponding characters on k' are

$$(146) \quad \bar{\chi}_K = \bar{\chi} \circ N_{k'/k}, \quad \bar{\psi}_K = \bar{\psi} \circ \text{Tr}_{k'/k}.$$

The conductor-one formula (12) gives

$$\Delta_F(\chi_F, \psi_F) = -\chi_F(\pi)^{n+1} \mathcal{A}(\tau_k(\bar{\chi}, \bar{\psi}))$$

and

$$\Delta_K(\chi_K, \psi_K) = -\chi_K(\pi)^{n+1} \mathcal{A}(\tau_{k'}(\bar{\chi}_K, \bar{\psi}_K)).$$

Theorem 5.2 and (146) imply

$$\tau_{k'}(\bar{\chi}_K, \bar{\psi}_K) = \tau_k(\bar{\chi}, \bar{\psi})^\ell.$$

Since $\mathcal{A}(z^\ell) = \mathcal{A}(z)^\ell$ for $z \neq 0$, and since $\chi_K(\pi) = \chi_F(\pi)^\ell$, this yields

$$\Delta_K(\chi_K, \psi_K) = (-1)^{\ell-1} \Delta_F(\chi_F, \psi_F)^\ell,$$

which is (145) because $m = 1$.

Step 4: conductor $m > 1$. Write $m = 2d + \varepsilon$, with $\varepsilon \in \{0, 1\}$, and put $\gamma = \pi^{m+n}$. Proposition 6.8(a) says that the stationary classes of χ_F and χ_K admit the same representative $\beta \in \mathcal{O}_F^\times$. Choose this representative for both characters and define

$$B_E = \chi_E(\gamma) \psi_E(\beta/\gamma) \chi_E(\beta)^{-1} \quad (E = F, K).$$

The factorization (63) reads

$$(147) \quad \Delta_E(\chi_E, \psi_E) = B_E \Delta_{3,E}(\chi_E).$$

Since $\beta, \gamma \in F$, one has

$$N_{K/F}(\beta) = \beta^\ell, \quad N_{K/F}(\gamma) = \gamma^\ell, \quad \text{Tr}_{K/F}(\beta/\gamma) = \ell\beta/\gamma.$$

Moreover $\psi_F(\ell z) = \psi_F(z)^\ell$ for every $z \in F$, because ψ_F is an additive character. Therefore

$$(148) \quad B_K = B_F^\ell.$$

If $\varepsilon = 0$, both residual factors are 1 by definition, and (145) follows from (147)–(148); its sign is 1 because m is even.

It remains to treat $\varepsilon = 1$. Then $m = 2d + 1$ and $d \geq 1$. Take $\delta = \pi^d$. Let $\phi_F : k \rightarrow \mathbf{C}^\times$ and $\phi_K : k' \rightarrow \mathbf{C}^\times$ be the Hasse functions from (56):

$$\begin{aligned} \phi_F(\bar{x}) &= \psi_F(\delta\beta x/\gamma) \chi_F(1 + \delta x)^{-1}, \\ \phi_K(\bar{x}) &= \psi_K(\delta\beta x/\gamma) \chi_K(1 + \delta x)^{-1}. \end{aligned}$$

The residual additive character in (58) is

$$(149) \quad \psi_0(\bar{z}) = \psi_F(\beta\delta^2 z/\gamma).$$

Indeed, expanding $(1 + \delta x)(1 + \delta y)$ modulo $\mathfrak{p}_F^m$ in the proof of Theorem 4.5 gives this character.

For $x \in \mathcal{O}_K$, let $E_j(x)$ be the j th elementary symmetric function of its F -conjugates. Thus $E_1(x) = T(x) := \text{Tr}_{K/F}(x)$ and

$$E_2(x) = \sum_{\{\sigma, \tau\}} x^\sigma x^\tau,$$

the sum being over unordered pairs of distinct F -embeddings of K . The norm expansion is

$$N(1 + \delta x) = 1 + \delta T(x) + \delta^2 E_2(x) + \sum_{j=3}^{\ell} \delta^j E_j(x),$$

where the last sum is empty if $\ell = 2$. Each term in that sum lies in $\mathfrak{p}_F^{3d} \subseteq \mathfrak{p}_F^{2d+1}$, and the same is true of the cross term $\delta^3 T(x) E_2(x)$. Hence

$$(150) \quad N(1 + \delta x) \equiv (1 + \delta T(x))(1 + \delta^2 E_2(x)) \pmod{\mathfrak{p}_F^m}.$$

Because $\delta^2 E_2(x) \in \mathfrak{p}_F^{2d} \subseteq \mathfrak{p}_F^{d+1}$, Lamprecht linearization (54) gives

$$\chi_F(1 + \delta^2 E_2(x))^{-1} = \psi_F(-\beta\delta^2 E_2(x)/\gamma) = \psi_0(-\overline{E_2(x)}).$$

Substituting this and (150) in the definition of ϕ_K yields

$$(151) \quad \phi_K(\bar{x}) = \phi_F\left(\text{Tr}_{k'/k}(\bar{x})\right) \psi_0\left(-\omega_{k'/k}(\bar{x})\right).$$

Here (41) identifies $\overline{T(x)}$ with the residue-field trace. The same reduction of the characteristic polynomial, now using its second coefficient, identifies $\overline{E_2(x)}$ with the function $\omega_{k'/k}$ in Theorem 5.5. Thus (151) says exactly that ϕ_K is the Hasse-function lift of ϕ_F .

Theorem 5.5 gives

$$-\sum_{x \in k'} \phi_K(x) = \left(-\sum_{x \in k} \phi_F(x) \right)^\ell,$$

or equivalently

$$\sum_{x \in k'} \phi_K(x) = (-1)^{\ell-1} \left(\sum_{x \in k} \phi_F(x) \right)^\ell.$$

Taking phases gives

$$(152) \quad \Delta_{3,K}(\chi_K) = (-1)^{\ell-1} \Delta_{3,F}(\chi_F)^\ell.$$

Combining (147), (148), and (152) proves (145). Since m is odd in this last case, $(-1)^{\ell-1} = (-1)^{m(\ell-1)}$. This completes the proof. $\square$

Proposition 7.2 (Tame conductor-one local-to-residue comparison). *Let K/F be tamely ramified cyclic of prime degree ℓ and let χ_F have conductor one and minimal conductor in its $S(K/F)$ -orbit. Then $m_K(\chi_F \circ N) = 1$. After choosing compatible uniformizers and reducing the conductor-one local constants by (12), the First Main Lemma is exactly the Hasse–Davenport multiplication identity (72) for the residue character of χ_F and the order- ℓ residue character attached to K/F . In particular the First Main Lemma holds in this case.*

Proof. The break is $t = 0$. Proposition 3.10(iii) shows that $m_K(\chi_F \circ N) \leq 1$; if it were 0, some twist of χ_F by a member of $S(K/F)$ would have conductor 0, contrary to the minimal choice of χ_F . Thus the upper conductor is exactly one.

Choose ϖ_K and put $\varpi_F = N(\varpi_K)$. Let $n = n_F(\psi_F)$. Since the tame different has exponent $\ell - 1$,

$$n_K(\psi_K) = \ell n + \ell - 1.$$

The element

$$\Gamma = \varpi_F^{n+1}$$

has K -valuation $\ell(n+1) = n_K(\psi_K) + 1$, so it is an admissible element for every conductor-one factor over both fields. Define

$$\bar{\psi}(x) = \psi_F(\tilde{x}/\Gamma), \quad \bar{\chi}(x) = \chi_F(\tilde{x}) \quad (x \in k^\times),$$

where k is the common residue field. On residue classes the norm and trace are

$$\overline{N_{K/F}(u)} = \bar{u}^\ell, \quad \overline{\text{Tr}_{K/F}(u)} = \ell \bar{u}.$$

Hence the upper residual characters are

$$(153) \quad \bar{\chi}_K = \bar{\chi}^\ell, \quad \bar{\psi}_K(x) = \bar{\psi}(\ell x).$$

Let $S(K/F) = \{\mu_j : 0 \leq j < \ell\}$, with $\mu_0 = 1$, and let η be the residue character of μ_1 . Then η has order ℓ and the residue character of μ_j is η^j . Moreover $\mu_j(\varpi_F) = 1$, because $\varpi_F = N(\varpi_K)$. Every $\mu_j \chi_F$ has conductor one by minimality.

The coset calculation proving (12), used with the common admissible element Γ , gives

$$\begin{aligned} \Delta_K(\chi_K, \psi_K) &= -\chi_F(\Gamma)^\ell \mathcal{A}(\tau_k(\bar{\chi}^\ell, \bar{\psi}(\ell \cdot))), \\ \Delta_F(\mu_j \chi_F, \psi_F) &= -\chi_F(\Gamma) \mathcal{A}(\tau_k(\bar{\chi} \eta^j, \bar{\psi})), \\ \Delta_F(\mu_j, \psi_F) &= -\mathcal{A}(\tau_k(\eta^j, \bar{\psi})) \quad (1 \leq j < \ell), \end{aligned}$$

while $\Delta_F(1, \psi_F) = 1$. The powers of $\chi_F(\Gamma)$ and the signs are the same on the two sides of the First Main Lemma and cancel.

For a multiplicative character α and $c \in k^\times$, the definition (10) gives

$$\tau_k(\alpha, \bar{\psi}(c \cdot)) = \alpha(c) \tau_k(\alpha, \bar{\psi}).$$

Thus

$$\tau_k(\bar{\chi}^\ell, \bar{\psi}(\ell \cdot)) = \bar{\chi}(\ell^\ell) \tau_k(\bar{\chi}^\ell, \bar{\psi}).$$

After the preceding cancellations, the First Main Lemma is therefore exactly

$$\bar{\chi}(\ell^\ell) \tau_k(\bar{\chi}^\ell, \bar{\psi}) \prod_{j=1}^{\ell-1} \tau_k(\eta^j, \bar{\psi}) = \tau_k(\bar{\chi}, \bar{\psi}) \prod_{j=1}^{\ell-1} \tau_k(\bar{\chi} \eta^j, \bar{\psi}),$$

which is the Hasse–Davenport multiplication formula (72). This proves the proposition. $\square$

Proposition 7.3 (Tamely ramified odd degree). *The First Main Lemma holds for a tamely ramified cyclic extension of odd prime degree.*

Proof. Here $t = 0$ and every nontrivial norm character has conductor 1. Put $n = n_F(\psi_F)$. For $m = 0$, choose a uniformizer ϖ_K of K and set $\varpi_F = N(\varpi_K)$; this is a uniformizer of F . Since $n_K = \ell n + (\ell - 1)$, formula (11) gives

$$\Delta_K(\chi_K) = \chi_F(\varpi_F)^{\ell n + \ell - 1}.$$

Every nontrivial $\mu \in S(K/F)$ is trivial on ϖ_F and has conductor one, so (12) gives

$$\Delta_F(\mu\chi_F) = \chi_F(\varpi_F)^{n+1} \Delta_F(\mu).$$

Multiplying this for the $\ell - 1$ nontrivial μ and including $\Delta_F(\chi_F) = \chi_F(\varpi_F)^n$ gives exponent $n + (\ell - 1)(n + 1) = \ell n + \ell - 1$; the factors $\prod_{\mu \neq 1} \Delta_F(\mu)$ agree on both sides. This proves the unramified-character case.

For $m = 1$, apply Proposition 7.2; it identifies every local scaling and reduces the equality exactly to Theorem 5.3. If $m = 2d + \varepsilon > 1$, then $1 \leq d$ and Lemma 4.7 gives

$$\prod_{\mu \in S} \Delta_F(\mu\chi_F, \psi_F) = \Delta_F(\chi_F, \psi_F)^\ell,$$

because the product of all characters in the odd-order group S is trivial. Corollary 2.12 removes the other norm characters. The parameter comparison in Proposition 6.8(b) and Theorem 5.6 give $\Delta_K(\chi_K, \psi_K) = \Delta_F(\chi_F, \psi_F)^\ell$. $\square$

Proposition 7.4 (Tamely ramified quadratic case). *The First Main Lemma holds for a tamely ramified quadratic extension.*

Proof. Put $n = n_F(\psi_F)$ and write $S(K/F) = \{1, \mu\}$. The residue characteristic is odd. Choose a uniformizer ϖ_K with $\sigma(\varpi_K) = -\varpi_K$ and put $\varpi_F = N(\varpi_K) = -\varpi_K^2$. Then $\mu(\varpi_F) = 1$, and the restriction of μ to $\mathcal{O}_F^\times$ is the quadratic character ν of the common residue field.

If $m_F(\chi_F) = 0$, then $n_K(\psi_K) = 2n + 1$ and

$$\Delta_K(\chi_K) = \chi_F(\varpi_F)^{2n+1},$$

while

$$\Delta_F(\chi_F) \Delta_F(\mu\chi_F) = \chi_F(\varpi_F)^n \chi_F(\varpi_F)^{n+1} \Delta_F(\mu).$$

Multiplying the first display by the remaining factor $\Delta_F(\mu)$ on the left-hand side of (1) gives exactly the second display. Thus this is the desired formula. If $m_F(\chi_F) = 1$, Proposition 7.2, specialized to $\ell = 2$, supplies the exact uniformizer, different, norm, trace, and residual-character identifications and reduces the assertion to Theorem 5.3.

Now let $m = 2d + \varepsilon \geq 2$ and put

$$\gamma = \varpi_F^{m+n}.$$

This element is admissible for the two downstairs characters. It is also admissible upstairs, because in the tame quadratic case

$$m_K(\chi_K) = 2m - 1, \quad n_K(\psi_K) = 2n + 1,$$

and hence

$$v_K(\gamma) = 2(m + n) = m_K(\chi_K) + n_K(\psi_K).$$

Since μ is trivial on U_F^d , Lemma 4.3(c) identifies the stationary classes of χ_F and $\mu\chi_F$ on the Lamprecht layer; Proposition 6.8(d) identifies the pullback class upstairs. Choose the same representative β for all three classes, translating an odd critical function when required by Lemma 4.6. Then Lemma 4.7 gives

$$(154) \quad \Delta_F(\mu\chi_F) = \mu(\gamma/\beta) \Delta_F(\chi_F).$$

Since γ is a power of the norm ϖ_F , one has $\mu(\gamma) = 1$. Since $\mu|_{\mathcal{O}_F^\times} = \nu$ and ν is quadratic,

$$\mu(\gamma/\beta) = \mu(\beta)^{-1} = \nu(\bar{\beta})^{-1} = \nu(\bar{\beta}).$$

By (63), the nonresidual factor upstairs is exactly the square of the corresponding one downstairs:

$$\begin{aligned}\chi_K(\gamma)\Delta_{2,K}(\chi_K) &= \chi_F(\gamma^2)\psi_F(2\beta/\gamma)\chi_F(\beta^2)^{-1} \\ &= (\chi_F(\gamma)\Delta_{2,F}(\chi_F))^2.\end{aligned}$$

Indeed, $N_{K/F}(a) = a^2$ and $\text{Tr}_{K/F}(a) = 2a$ for $a \in F$. Thus (154), together with $\Delta_F(1) = 1$, cancels all nonresidual factors in (1). It remains to prove

$$(155) \quad \Delta_{3,K}(\chi_K)\Delta_F(\mu) = \nu(\bar{\beta})\Delta_{3,F}(\chi_F)^2.$$

Let

$$\psi_0(\bar{x}) = \psi_F(x/\varpi_F^{n+1}), \quad G = \sum_{x \in k^\times} \nu(x)\psi_0(x).$$

Then $\Delta_F(\mu) = \mathcal{A}(G)$ and $\mathcal{A}(G)^2 = \nu(-1)$.

If $m = 2d$, then $m_K(\chi_K) = 4d - 1$. Taking $\delta_K = \varpi_K^{2d-1}$, the trace term in the Hasse function is zero and

$$N(1 + \delta_K x) = 1 + \varpi_F^{2d-1}x^2.$$

Consequently

$$\Delta_{3,K} = \mathcal{A}\left(\sum_{x \in k} \psi_0(-\bar{\beta}x^2)\right) = \nu(-\bar{\beta})\mathcal{A}(G).$$

Multiplying by $\Delta_F(\mu)$ gives $\nu(-\bar{\beta})\nu(-1) = \nu(\bar{\beta})$, proving (155).

If $m = 2d + 1$, the Hasse function over F has the form

$$\phi_F(x) = \psi_0(\bar{\beta}x^2/2 + cx)$$

for a uniquely determined $c \in k$. Here $m_K(\chi_K) = 4d + 1$. With $\delta_K = \varpi_K^{2d} = (-1)^d \varpi_F^d$, direct substitution gives

$$\phi_K(x) = \phi_F((-1)^d x)^2 = \psi_0(\bar{\beta}x^2 + 2(-1)^d cx).$$

Lemma 5.4 now gives

$$\begin{aligned}\Delta_{3,K} &= \nu(2\bar{\beta})\psi_0(-c^2/\bar{\beta})Q_{\psi_0}(1,0), \\ \Delta_{3,F}^2 &= \psi_0(-c^2/\bar{\beta})\nu(-1), \\ \Delta_F(\mu) &= \nu(2)Q_{\psi_0}(1,0).\end{aligned}$$

Since $Q_{\psi_0}(1,0)^2 = \nu(-1)$, these three formulas imply (155). Combining with (154) completes the proof. $\square$

8. WILD ODD DEGREE: STABLE RANGE

Proposition 8.1. *Let K/F be wildly ramified of odd prime degree p , with break t . If $m_F(\chi_F) = 2d + \varepsilon > 1$ and $d \geq t + 1$, then the First Main Lemma holds.*

Proof. Every nontrivial norm character has conductor $t + 1 \leq d$. The stable twist formula and Corollary 2.12 reduce the theorem to

$$\Delta_K(\chi_K, \psi_K) = \Delta_F(\chi_F, \psi_F)^p.$$

The elementary factors agree by Proposition 6.8(b); note that $m' + n_K = p(m + n_F)$ by (40) and (38). If the conductor is even there is no residual factor. If it is odd, Theorem 5.6 gives the required p th-power identity for the residual factors. $\square$

9. THE NON-STABLE WILD CALCULATION

The only remaining cases are wild and non-stable. We now give the complete local calculation. Propositions 6.8 and 6.10 determine the stationary classes and exhibit compatible representatives in the high- and low-conductor ranges, with the quadratic high range governed by Proposition 6.9; the resulting finite phases and valuation estimates are then evaluated explicitly.

9.1. The endpoint conductors in wild prime degree. Before separating odd and quadratic degree, we dispose of the two conductors for which there is no stationary layer. The calculation is valid for every wildly ramified cyclic extension K/F of prime degree p , including $p = 2$. Let t be its break, put $n = n_F(\psi_F)$, and choose uniformizers with $\varpi_F = N_{K/F}(\varpi_K)$. Then

$$(156) \quad n_K(\psi_K) = pn + (p-1)(t+1)$$

by Corollary 3.8.

Conductor 0. If χ_F is unramified, then

$$\Delta_K(\chi_K, \psi_K) = \chi_F(\varpi_F)^{n_K(\psi_K)}.$$

For $j \in \mathbf{F}_p$, let τ^j run through $S(K/F)$, with $\tau^0 = 1$. The character τ^j has conductor $t+1$ for $j \neq 0$, whereas τ^0 has conductor 0. Since an unramified character is constant on the unit group, direct substitution in (3) gives

$$\Delta_F(\chi_F \tau^j, \psi_F) = \chi_F(\varpi_F)^{m_F(\tau^j)+n} \Delta_F(\tau^j, \psi_F).$$

Multiplication over j gives the exponent

$$n + (p-1)(t+1+n) = pn + (p-1)(t+1) = n_K(\psi_K),$$

so the First Main Lemma follows.

Conductor 1. We now give the finite-sum comparison, retaining every local-to-residue scaling factor. Choose an admissible element γ_F for (χ_F, ψ_F) , so $v_F(\gamma_F) = n+1$, and use it to write the residual additive character as

$$\psi_0(\bar{x}) = \psi_F(x/\gamma_F), \quad x \in \mathcal{O}_F.$$

After the normalization of Lemma 2.7, $\psi_0(x^p) = \psi_0(x)$.

Let $f(T)$ be the minimal polynomial of ϖ_K and put

$$(157) \quad \gamma_K = \gamma_F \frac{f'(\varpi_K)}{\varpi_K^{p-1}}.$$

The different formula gives $v_K(f'(\varpi_K)) = (p-1)(t+1)$, so $v_K(\gamma_K) = 1 + n_K(\psi_K)$; hence γ_K is admissible for (χ_K, ψ_K) . For $0 \leq i \leq p-1$, partial fractions for $T^i/f(T)$ at infinity give

$$(158) \quad \mathrm{Tr}_{K/F} \left(\frac{\varpi_K^i}{f'(\varpi_K)} \right) = \begin{cases} 0, & i < p-1, \\ 1, & i = p-1. \end{cases}$$

If $x \in \mathcal{O}_K$ has residue $a \in k$, then $x - a \in \mathfrak{p}_K$ and the trace-ideal formula (35) shows that

$$\mathrm{Tr}_{K/F} \left(\frac{(x-a)\varpi_K^{p-1}}{f'(\varpi_K)} \right) \in \mathfrak{p}_F.$$

Consequently the residual additive character determined by γ_K is again ψ_0 . If $\bar{\chi}$ denotes the residue-field character of χ_F , then that of χ_K is $x \mapsto \bar{\chi}(x^p)$. Frobenius invariance of ψ_0 therefore identifies the two finite sums by the change of variable $x \mapsto x^p$. Thus

$$(159) \quad \frac{\Delta_K(\chi_K, \psi_K)}{\Delta_F(\chi_F, \psi_F)} = \chi_F \left(\frac{N(\gamma_K)}{\gamma_F} \right).$$

It remains to identify the residue class of the element on the right. Let σ generate $\mathrm{Gal}(K/F)$ and set

$$\lambda = \frac{\overline{\sigma(\varpi_K) - \varpi_K}}{\varpi_K^{t+1}} \in k^\times.$$

The ramification map sends σ^j to $j\lambda$. Proposition 3.3 gives, in these source and target coordinates,

$$(160) \quad P(Z) = Z^p - \lambda^{p-1}Z, \quad \mathrm{im} P = \lambda^p \ker(\mathrm{Tr}_{k/\mathbf{F}_p}).$$

Choose an admissible γ_τ for τ and a representative b_τ of its stationary numerator class such that

$$\tau(1+x) = \psi_F(b_\tau x / \gamma_\tau)$$

on the Lamprecht layer, and put $c_\tau = \gamma_\tau / b_\tau$.

Lemma 9.1 (Endpoint annihilator and discriminant congruence). *With the notation of the conductor-1 endpoint calculation, the associated reciprocal stationary element $c_\tau = \gamma_\tau/b_\tau$ and the ramification coefficient λ satisfy*

$$(161) \quad \left(\frac{\overline{c_\tau}}{\varpi_F^t \gamma_F} \right)^{p-1} = \lambda^{p(p-1)}.$$

Moreover, for $D = f'(\varpi_K)/\varpi_K^{(p-1)(t+1)}$, one has

$$\bar{D} = (-1)^p \lambda^{p-1}, \quad \overline{N(D)} = (-1)^p \lambda^{p(p-1)}.$$

Proof. The critical norm polynomial is $P(Z) = Z^p - \lambda^{p-1}Z$. Its image is $\lambda^p \ker(\text{Tr}_{k/\mathbf{F}_p})$, and the annihilator of this image under the trace pairing is $\lambda^{-p}\mathbf{F}_p$. On the critical unit quotient write $x = \varpi_F^t \tilde{z}$. Since $\tau(1+x) = \psi_F(x/c_\tau)$ and $\psi_0(\bar{y}) = \psi_F(y/\gamma_F)$, the residual coefficient of τ is

$$a_\tau = \frac{\overline{\varpi_F^t \gamma_F}}{c_\tau}.$$

The character τ is trivial on the norm image. Hence $\psi_0(a_\tau P(z)) = 1$ for every $z \in k$, so $a_\tau \in \lambda^{-p}\mathbf{F}_p$. It is nonzero because τ has conductor $t+1$; consequently $a_\tau = \lambda^{-p}u$ for some $u \in \mathbf{F}_p^\times$. Taking inverses and then $(p-1)$ st powers gives

$$\left(\frac{\overline{c_\tau}}{\varpi_F^t \gamma_F} \right)^{p-1} = (\lambda^p u^{-1})^{p-1} = \lambda^{p(p-1)},$$

which is (161).

For the second assertion, write

$$f'(\varpi_K) = \prod_{j=1}^{p-1} (\varpi_K - \sigma^j \varpi_K).$$

After division by $\varpi_K^{(p-1)(t+1)}$, the residues of the factors are the nonzero elements $-j\lambda$ of the ramification line. Their product is $(-1)^p \lambda^{p-1}$. Taking the norm to F raises this residue to the p th power and gives the last identity. $\square$

Put

$$D = \frac{f'(\varpi_K)}{\varpi_K^{(p-1)(t+1)}} \in \mathcal{O}_K^\times.$$

Since the nontrivial ramification classes are $j\lambda$,

$$(162) \quad \bar{D} = (-1)^p \lambda^{p-1}, \quad \overline{N(D)} = (-1)^p \lambda^{p(p-1)}.$$

Using (157) and $N(\varpi_K) = \varpi_F$, we obtain

$$\frac{N(\gamma_K)}{\gamma_F} = (\gamma_F \varpi_F^t)^{p-1} N(D).$$

Equations (161) and (162) give the congruence

$$(163) \quad \frac{N(\gamma_K)}{\gamma_F} \equiv (-1)^p c_\tau^{p-1} \pmod{U_F^1}.$$

Because $m_F(\chi_F) = 1$, χ_F depends only on valuation and residue, so the congruence may be inserted in (159).

Finally, Lemma 4.3(c) shows that $j b_\tau$ represents the stationary numerator class of τ^j on the same layer. Its conductor remains $t+1$ because τ^j is a nontrivial norm character. Choose this representative; changing b_τ within its permitted class changes neither the residue congruence (163) nor the complete Lamprecht factor. Since $m_F(\chi_F) = 1 \leq \lfloor (t+1)/2 \rfloor$, the exact stable-twist formula gives

$$\Delta_F(\tau^j \chi_F, \psi_F) = \chi_F \left(\frac{c_\tau}{j} \right) \Delta_F(\tau^j, \psi_F).$$

The product of the Teichmüller representatives of $\mathbf{F}_p^\times$ is -1 for odd p and is 1 for $p = 2$; hence

$$\prod_{j \in \mathbf{F}_p^\times} \frac{c_\tau}{j} = (-1)^p c_\tau^{p-1}.$$

After cancelling the common factors $\prod_{j \neq 0} \Delta_F(\tau^j, \psi_F)$, the First Main Lemma is exactly (159) together with (163). Thus the endpoint conductors 0 and 1 follow from the internal trace, finite-field, and stable-twist arguments and Lemma 9.1.

Lemma 9.2 (Characteristic-independent polar coefficient and stationary translation). *Let E be a nonarchimedean local field, let ψ_E be a nontrivial additive character, put $p = \text{char } k_E$, let $m_E(\theta) = 2d + 1 > 1$, and choose an admissible element γ_θ with*

$$\gamma_\theta \mathcal{O}_E = \mathfrak{p}_E^{2d+1+n_E(\psi_E)}.$$

Choose $c \in \mathfrak{p}_E^d \setminus \mathfrak{p}_E^{d+1}$ and a representative $\beta_\theta \in \mathcal{O}_E^\times$ of the stationary numerator class. Fix a nontrivial Frobenius-invariant additive character $\psi_0 : k_E \rightarrow \mathbf{C}^\times$, so that $\psi_0(z^p) = \psi_0(z)$, and put

$$\Phi_{\theta, \beta_\theta, c}(\bar{x}) = \psi_E(\beta_\theta c \tilde{x} / \gamma_\theta) \theta(1 + c \tilde{x})^{-1} \quad (\bar{x} \in k_E).$$

There is a unique element $\kappa_\theta(c) \in k_E^\times$ such that

$$(164) \quad \psi_E\left(\frac{\beta_\theta c^2 \tilde{z}}{\gamma_\theta}\right) = \psi_0(\kappa_\theta(c)z) \quad (z \in k_E).$$

It is the polar coefficient of the complete critical function:

$$(165) \quad \frac{\Phi_{\theta, \beta_\theta, c}(x+y)}{\Phi_{\theta, \beta_\theta, c}(x)\Phi_{\theta, \beta_\theta, c}(y)} = \psi_0(\kappa_\theta(c)xy).$$

If $\lambda \in k_E^\times$ and $\tilde{\lambda} \in \mathcal{O}_E^\times$ is any lift, then

$$(166) \quad \kappa_\theta(\tilde{\lambda}c) = \lambda^2 \kappa_\theta(c).$$

Precomposition with Frobenius $x \mapsto x^p$ replaces the polar coefficient by $\kappa_\theta(c)^{1/p}$, and precomposition with inverse Frobenius replaces it by $\kappa_\theta(c)^p$.

If $\beta'_\theta = \beta_\theta + c_0$ is another permitted representative, define $\Delta B_\theta(c_0, c) \in k_E$ uniquely by

$$(167) \quad \psi_E\left(\frac{c_0 c \tilde{z}}{\gamma_\theta}\right) = \psi_0(\Delta B_\theta(c_0, c)z) \quad (z \in k_E).$$

Then

$$(168) \quad \Phi_{\theta, \beta'_\theta, c}(x) = \Phi_{\theta, \beta_\theta, c}(x) \psi_0(\Delta B_\theta(c_0, c)x).$$

Thus changing the representative leaves the polar coefficient fixed and changes only the affine character; the corresponding elementary stationary factor must be translated at the same time as in Lemma 4.6.

Proof. The coefficient $\beta_\theta c^2 / \gamma_\theta$ has valuation $-n_E(\psi_E) - 1$. Hence the left side of (164) is a well-defined nontrivial additive character of k_E . The trace pairing of the finite field is perfect, so every additive character has a unique expression $z \mapsto \psi_0(\kappa z)$; exactness of the valuation makes $\kappa \neq 0$. This proves existence and uniqueness of $\kappa_\theta(c)$.

For lifts of $x, y \in k_E$, one has

$$\frac{(1+cx)(1+cy)}{1+c(x+y)} = 1 + \frac{c^2 xy}{1+c(x+y)}.$$

The argument on the right lies in $\mathfrak{p}_E^{2d} \subseteq \mathfrak{p}_E^{d+1}$, where the stationary linearization is valid, and it differs from $c^2 xy$ by an element of $\mathfrak{p}_E^{3d} \subseteq \mathfrak{p}_E^{2d+1}$. Therefore

$$\frac{\Phi_{\theta, \beta_\theta, c}(x+y)}{\Phi_{\theta, \beta_\theta, c}(x)\Phi_{\theta, \beta_\theta, c}(y)} = \psi_E\left(\frac{\beta_\theta c^2 \tilde{x}\tilde{y}}{\gamma_\theta}\right),$$

and (165) follows from (164). Replacing c by $\tilde{\lambda}c$ proves (166). Since $\psi_0(z^p) = \psi_0(z)$, the two Frobenius assertions follow by reindexing the polar identity.

For another representative, c_0 belongs to $\mathfrak{p}_E^d$. Hence $z \mapsto \psi_E(c_0 c \tilde{z} / \gamma_\theta)$ is a well-defined additive character of k_E , possibly trivial, and so determines the unique element in (167). Substitution in the definition of Φ gives (168). The last assertion is exactly the representative-change compatibility of Lemma 4.6. $\square$

Corollary 9.3 (One-layer-deeper representative changes). *In the notation of Lemma 9.2, suppose that $\beta'_\theta = \beta_\theta + c_0$ with $c_0 \in \mathfrak{p}_E^{d+1}$. Then*

$$\Delta B_\theta(c_0, c) = 0, \quad \Phi_{\theta, \beta'_\theta, c} = \Phi_{\theta, \beta_\theta, c}.$$

Proof. Since $v_E(c) = d$ and $v_E(\gamma_\theta) = 2d + 1 + n_E(\psi_E)$,

$$v_E(c_0 c / \gamma_\theta) \geq -n_E(\psi_E).$$

The additive character is therefore trivial on this quotient. The translation character in (167) is trivial, so its coefficient is zero; equation (168) gives the function equality. $\square$

9.2. Odd residue characteristic: critical phases and residual coefficients. We first isolate the exact coefficient extraction from Lamprecht's critical sum and then compute the resulting stationary coefficients.

Let k have odd characteristic. By Corollary 2.9 and Lemma 2.7, we normalize the residual additive character once and for all as

$$\psi_0(z) = \exp\left(\frac{2\pi i}{p} \operatorname{Tr}_{k/\mathbf{F}_p}(z)\right).$$

In particular $\psi_0(z^p) = \psi_0(z)$. For $A \neq 0$ and $B \in k$ put

$$(169) \quad Q_{\psi_0}(A, B) = |k|^{-1/2} \sum_{x \in k} \psi_0(Ax^2/2 + Bx).$$

Completing the square gives

$$(170) \quad Q_{\psi_0}(A, B) = q_0 \nu(A) \psi_0\left(-\frac{B^2}{2A}\right),$$

where ν is the quadratic character and $q_0 = |k|^{-1/2} \sum_x \psi_0(x^2/2)$.

Lemma 9.4 (Powers of the basic quadratic phase). *One has*

$$(171) \quad q_0^2 = \nu(-1), \quad q_0^{p-1} = \nu(-1), \quad q_0^p = q_0 \nu(-1).$$

Proof. The first identity is Lemma 5.4. In particular $q_0^4 = 1$. If $p \equiv 1 \pmod{4}$, then -1 is a square in $\mathbf{F}_p \subset k$, so $\nu(-1) = 1$, and $p-1$ is divisible by 4; hence $q_0^{p-1} = 1 = \nu(-1)$. If $p \equiv 3 \pmod{4}$, then $p-1 \equiv 2 \pmod{4}$, and therefore $q_0^{p-1} = q_0^2 = \nu(-1)$. Multiplication by q_0 gives the last identity. $\square$

For every local character θ occurring below, we use the parity convention

$$(172) \quad \mathcal{Q}(\theta) = \begin{cases} Q_{\psi_0}(A_\theta, B_\theta), & m(\theta) \text{ odd}, \\ 1, & m(\theta) \text{ even}. \end{cases}$$

Thus a coefficient pair is required only in the odd-conductor rows; all even-conductor residual factors are literally 1.

For the exact transport identities below, we also use a finite function in an even-conductor row. If $m(\theta) = 2d > 1$, choose $c\mathcal{O}_E = \mathfrak{p}_E^d$ and a representative β of the stationary class, and put

$$(173) \quad \Phi_\theta(\bar{x}) = \psi(\beta c \tilde{x} / \gamma) \theta(1 + c \tilde{x})^{-1} \quad (\bar{x} \in k_E).$$

The even-conductor Lamprecht linearization is valid on $\mathfrak{p}_E^d$ and shows that

$$\Phi_\theta(\bar{x}) = 1 \quad (\bar{x} \in k_E).$$

Thus in every even row the symbol Φ_θ denotes the literal constant function 1, and its normalized residue class may be taken to be 0. This convention introduces no Hasse factor; it only makes the transport identities valid without referring to an undefined odd critical function.

For an odd-conductor character θ , choose a critical coordinate c and define its coefficient pair (A_θ, B_θ) by the exact identity

$$(174) \quad \psi(\beta cx/\gamma)\theta(1+cx)^{-1} = \psi_0(A_\theta x^2/2 + B_\theta x) \quad (x \in k).$$

This identity is simply the finite function occurring in Lamprecht's formula, written in the unique quadratic-polynomial form. Multiplying characters adds the pairs whenever their critical layers coincide. If one character has smaller conductor, its second finite difference vanishes on the deeper critical layer; its effect is then a translation of the stationary point and has no quadratic coefficient there. More precisely, for $x, y \in \mathcal{O}_E$ one has the exact identity

$$\frac{(1+cx)(1+cy)}{1+c(x+y)} = 1 + \frac{c^2 xy}{1+c(x+y)}.$$

If the smaller character has conductor at most $2v_E(c)$, it is trivial on the unit on the right. Hence its polar quotient on the deeper critical line is identically one. The quotient of its finite function by a constant is therefore an additive character, which changes only the affine coefficient and is exactly the stationary-point translation retained below.

Write $b = m_F(\chi_F) - 1$ and keep t for the ramification break. The next lemma fixes the residue coordinates on the last ramification line. This normalization is used for every odd-prime critical sum below.

Lemma 9.5 (Break-line norm, trace, and character normalization). *Assume that K/F is wildly ramified cyclic of odd prime degree p , with lower break t . Choose a uniformizer ϖ_K , put $\varpi_F = N_{K/F}(\varpi_K)$, choose a generator σ of $\text{Gal}(K/F)$, and set*

$$(175) \quad \beta_0 = \frac{\overline{\sigma(\varpi_K) - \varpi_K}}{\varpi_K^{t+1}} \in k^\times.$$

For $z \in k$ and any lift $\tilde{z} \in \mathcal{O}_F$, one has

$$(176) \quad N_{K/F}(1 + \varpi_K^t \tilde{z}) \equiv 1 + \varpi_F^t (\widetilde{z^p - \beta_0^{p-1} z}) \pmod{\mathfrak{p}_F^{t+1}},$$

and consequently

$$(177) \quad \text{Tr}_{K/F}(\varpi_K^t \tilde{z}) \equiv -\beta_0^{p-1} \varpi_F^t \tilde{z} \pmod{\mathfrak{p}_F^{t+1}}.$$

More generally, if $w \in \mathcal{O}_K^\times$, then

$$(178) \quad \text{Tr}_{K/F}(w \varpi_K^t \tilde{z}) \equiv -\beta_0^{p-1} \varpi_F^t \widetilde{wz} \pmod{\mathfrak{p}_F^{t+1}}.$$

After replacing a generator τ of $S(K/F)$ by a nonzero power, one may and shall arrange that

$$(179) \quad \tau(1 + \varpi_F^t \tilde{z}) = \psi_0(\beta_0^{-p} z) \quad (z \in k).$$

Put

$$(180) \quad \eta_0 = \beta_0^{-p}.$$

If $T = t + 1$ is odd, then, with the critical element $c_\tau = \varpi_F^{t/2}$, the quadratic coefficient of the critical function of τ is exactly η_0 .

Proof. On the critical quotient U_K^t/U_K^{t+1} , the norm is an additive map. Its kernel has order p by Theorem 3.1(ii). The ramification map sends σ^j to $j\beta_0$, so the kernel is the line $\beta_0 \mathbf{F}_p$.

We determine the polynomial, including its scalar. In the exact norm expansion of $1 + \varpi_K^t \tilde{z}$, the degree- p endpoint is

$$N(\varpi_K^t \tilde{z}) = \varpi_F^t \tilde{z}^p.$$

For $2 \leq j \leq p - 1$, Lemma 3.4 gives

$$v_F(s_j(\varpi_K^t \tilde{z})) \geq \left\lfloor \frac{jt + (p-1)(t+1)}{p} \right\rfloor \geq t + 1,$$

because $(j-1)t \geq 1$. Hence the induced norm polynomial is monic, additive, and of the form $z^p + cz$. Its roots are precisely $\beta_0 \mathbf{F}_p$, so

$$z^p + cz = \prod_{j \in \mathbf{F}_p} (z - j\beta_0) = z^p - \beta_0^{p-1}z.$$

This proves (176); subtracting the norm endpoint proves (177). If $w - \tilde{w} \in \mathfrak{p}_K$, then $\varpi_K^t(w - \tilde{w}) \in \mathfrak{p}_K^{t+1}$, and Theorem 3.7(b) gives $\text{Tr}(\mathfrak{p}_K^{t+1}) = \mathfrak{p}_F^{t+1}$. This proves (178).

The image of the polynomial in (176) is

$$\beta_0^p \ker(\text{Tr}_{k/\mathbf{F}_p} : k \rightarrow \mathbf{F}_p).$$

Indeed, after writing $z = \beta_0 y$ it becomes $\beta_0^p(y^p - y)$, and the image of $y \mapsto y^p - y$ is the trace-zero hyperplane. Write the restriction of a nontrivial norm character to the last unit quotient as

$$\tau(1 + \varpi_F^t \tilde{z}) = \psi_0(az).$$

It is trivial on the norm image, so the trace pairing puts a in the annihilator $\beta_0^{-p} \mathbf{F}_p$. It is nonzero because τ has conductor $t+1$. Thus $a = j\beta_0^{-p}$ for some $j \in \mathbf{F}_p^\times$, and replacing τ by $\tau^{j^{-1}}$ proves (179).

If t is even, the polar quotient of the τ -critical function in the coordinate $c_\tau = \varpi_F^{t/2}$ is

$$(x, y) \mapsto \tau(1 + c_\tau^2 \tilde{x}\tilde{y}) = \psi_0(\beta_0^{-p}xy).$$

Thus its quadratic coefficient is $\eta_0 = \beta_0^{-p}$, as claimed. $\square$

Fix from now on the generator τ , the uniformizers, and the coefficient η_0 of Lemma 9.5. After this replacement of the generator, reapply Lemma 6.6 and the appropriate high- or low-conductor stationary-class table; hence the symbols α, α_1 and the twist representatives below refer to choices made for this normalized generator. Whenever χ_F has odd conductor, write $(\eta, \eta\gamma)$ for its coefficient pair in the critical coordinate specified in Lemma 9.9; write $(\eta_0, \eta_0\gamma_0)$ for the corresponding pair of τ when T is odd. When T is even, η_0 is the normalized last-layer coefficient (180); no τ -critical sum occurs.

Remark 9.6 (Intrinsic break-line coefficient). The coefficient η_0 in the break-line formulas is retained as the coefficient determined by the fixed Galois generator and the chosen uniformizer. We do not replace the generator merely to force a cosmetic normal form such as $\eta_0 = \beta_0^{-p}$. All later annihilator, Frobenius-preimage, and coefficient identities are proved for this intrinsic value. This keeps the source of every critical row visible.

Lemma 9.7 (Critical polynomial, translations, and changes of coordinate). *Let E be a local field of odd residue characteristic, let $m_E(\theta) = 2d + 1$, and choose $c \in \mathfrak{p}_E^d \setminus \mathfrak{p}_E^{d+1}$. For a chosen representative β_θ of the stationary numerator class, put*

$$\Phi_\theta(\bar{x}) = \psi_E(\beta_\theta c \tilde{x} / \gamma_\theta) \theta(1 + c \tilde{x})^{-1} \quad (\bar{x} \in k_E).$$

Let

$$A_\theta = \kappa_\theta(c) \in k_E^\times$$

be the polar coefficient defined by Lemma 9.2. Then there is a unique $B_\theta \in k_E$ such that

$$\Phi_\theta(x) = \psi_0(A_\theta x^2 / 2 + B_\theta x).$$

Replacing c by $\tilde{\lambda}c$, where $\lambda \in k_E^\times$ and $\tilde{\lambda}$ is a unit lift, replaces (A_θ, B_θ) by $(\lambda^2 A_\theta, \lambda B_\theta)$. Precomposing the finite function with Frobenius $x \mapsto x^p$ replaces the pair by $(A_\theta^{1/p}, B_\theta^{1/p})$, whereas precomposing with inverse Frobenius $x \mapsto x^{1/p}$ replaces it by (A_θ^p, B_θ^p) . These operations do not change the normalized finite sum, because they merely reindex it.

If a permitted representative is replaced by $\beta_\theta + c_0$ and the elementary stationary factor is translated at the same time, then A_θ is unchanged and B_θ is replaced by

$$B_\theta + \Delta B_\theta(c_0, c),$$

where $\Delta B_\theta(c_0, c)$ is defined by (167). Thus a linear coefficient may never be discarded without simultaneously retaining the corresponding elementary translation factor.

Proof. By Lemma 9.2, the polar quotient of Φ_θ is $\psi_0(A_\theta xy)$. Since the residue characteristic is odd, the function $x \mapsto \psi_0(A_\theta x^2/2)$ has the same polar quotient. Their quotient is therefore an additive character of k_E , and the perfect trace pairing gives a unique $B_\theta \in k_E$ for which it is $\psi_0(B_\theta x)$. This proves existence and uniqueness.

The coordinate-scaling rule follows from $\Phi_{\theta, \beta_\theta, \tilde{\lambda}_c}(x) = \Phi_{\theta, \beta_\theta, c}(\lambda x)$ and (166). For Frobenius, write $A_\theta = A_0^p$ and $B_\theta = B_0^p$. Since $\psi_0(z^p) = \psi_0(z)$,

$$\psi_0(A_\theta x^{2p}/2 + B_\theta x^p) = \psi_0(A_0 x^2/2 + B_0 x),$$

which proves the stated p th-root direction; the inverse direction is equivalent. Finally, (168) shows that changing the representative multiplies the critical function by $\psi_0(\Delta B_\theta(c_0, c)x)$, so it changes only the displayed affine coefficient. Lemma 4.6 provides the matching elementary translation. $\square$

Lemma 9.8 (Higher symmetric terms on the upper critical line). *Assume that K/F is wildly ramified cyclic of odd prime degree p , with break t , and put $T = t+1$. Suppose $m_F(\chi_F) = 2d+1 > 1$, set $b = 2d$ and $a = t - b$, and suppose that the upper conductor is odd, say $m_K(\chi_K) = 2d_K + 1$, with*

$$2d_K = \begin{cases} b, & b < t, \\ t, & b = t, \\ t + p(b - t), & b > t. \end{cases}$$

Let $u \in K^\times$ satisfy $v_K(u) = a$, put $n = N_{K/F}(u)$, and let $x \in \mathfrak{p}_K^{d_K}$. Then, for every $3 \leq j \leq p-1$,

$$(181) \quad s_j(ux) \in \mathfrak{p}_F^T, \quad n s_j(x) \in \mathfrak{p}_F^T.$$

Moreover

$$(182) \quad \sum_{j=2}^p (s_j(ux) - n s_j(x)) \equiv s_2(ux) - n s_2(x) \pmod{\mathfrak{p}_F^T}.$$

Thus the surviving correction on the last ramification quotient is homogeneous quadratic and has no affine part.

Proof. Since $v_F(n) = v_K(u) = a$, Lemma 3.4 gives

$$\begin{aligned} v_F(s_j(ux)) &\geq \left\lfloor \frac{j(d_K + a) + (p-1)T}{p} \right\rfloor, \\ v_F(n s_j(x)) &\geq a + \left\lfloor \frac{j d_K + (p-1)T}{p} \right\rfloor. \end{aligned}$$

It is therefore enough to prove

$$(183) \quad j(d_K + a) \geq T, \quad j d_K + pa \geq T.$$

If $p = 3$, the range $3 \leq j \leq p-1$ is empty and $s_3(ux) = N(u)N(x) = n s_3(x)$, so both assertions of the lemma follow immediately. We may therefore suppose $p \geq 5$. The formulas below show in particular that $d_K + a > 0$; hence both left sides increase with j , and it suffices to take $j = 3$. For $b < t$, put $a = t - b > 0$, so $d_K = (t - a)/2$; for $b > t$, put $q = b - t = -a > 0$, so $d_K = (t + pq)/2$. Direct substitution gives

	$\frac{3(d_K + a) - T}{t + 3a - 2}$	$\frac{3d_K + pa - T}{t + (2p-3)a - 2}$
$b < t$	$\frac{2}{t-2}$	$\frac{2}{t-2}$
$b = t$	$\frac{2}{2}$	$\frac{2}{2}$
$b > t$	$\frac{t + 3(p-2)q - 2}{2}$	$\frac{t + pq - 2}{2}$

Every entry is nonnegative. In the middle row, $b = t = 2d$ and $d \geq 1$, so $t \geq 2$; in the other rows use $t \geq 1$, $a \geq 1$ or $q \geq 1$, and $p \geq 3$. This proves (183) and hence (181).

Finally,

$$s_p(ux) = N(ux) = N(u)N(x) = n s_p(x),$$

so the degree- p terms cancel exactly. Removing them and the terms of degrees $3, \dots, p-1$ proves (182). The remaining term is homogeneous of degree two in x . $\square$

Lemma 9.9 (Odd-prime critical-coordinate transport). *Put*

$$(u, n) = \begin{cases} (\varepsilon_1 \beta_1 / \alpha_1, \varepsilon \beta / \alpha), & b \leq t, \\ (\beta_1 / \alpha_1, \beta / \alpha), & b > t. \end{cases}$$

Then $n = N(u)$. Put $a = t - b$, write

$$u = \varpi_K^a u_0, \quad n = \varpi_F^a n_0, \quad u_0, n_0 \in \mathcal{O}^\times,$$

and set

$$(184) \quad \zeta = \bar{u}_0, \quad d_0 = \zeta^p.$$

Then $\bar{n}_0 = d_0$. When $m_F(\chi_F) = 2d + 1$ is odd, the lower critical coordinate $c_\chi = \varpi_F^d$ gives

$$(185) \quad \eta = \eta_0 d_0.$$

The upper conductor has the same parity as $m_F(\chi_F)$. When this common parity is odd, the phase-preserving upper critical coordinate constructed in the proof gives the complete coefficient pair

$$(186) \quad (A_K, B_K) = \begin{cases} (\eta, \eta\gamma), & b < t, \\ (-\eta_0(d_0^p - d_0), \eta_0 d_0(\gamma - \gamma_0)), & b = t, \\ (-d_0^{p-1} \eta, 0), & b > t. \end{cases}$$

When the common parity is even, the corresponding residual factors are 1 by (172). In the odd rows, (186) is an equality of complete critical functions, including the affine term and the translation of the stationary origin.

Proof. Use the uniformizers fixed in Lemma 9.5. In every range of the statement,

$$v_K(u) = v_F(n) = a = t - b.$$

Thus the decompositions in the statement satisfy $n_0 = N(u_0)$. Since the extension is totally ramified, the residue norm is Frobenius, and therefore

$$(187) \quad \bar{n}_0 = \bar{u}_0^p = \zeta^p = d_0.$$

There is no additional residue scalar when n is viewed as an element of K . Indeed, if σ generates $\text{Gal}(K/F)$, then

$$\frac{\varpi_F}{\varpi_K^p} = \prod_{i=0}^{p-1} \frac{\sigma^i(\varpi_K)}{\varpi_K} \equiv 1 \pmod{\mathfrak{p}_K},$$

because the extension is wildly ramified. Hence, after division by ϖ_K^{pa} , the initial residue of $n = \varpi_F^a n_0$ is d_0 for every integer a , including $a < 0$.

Put

$$(188) \quad \mathcal{A} = \begin{cases} \alpha/\delta, & b \leq t, \\ \alpha/\gamma_F, & b > t. \end{cases}$$

Here δ in the first row is the low-conductor admissible element of (134), not the residue ζ of (184). Corollary 6.3, together with Propositions 6.8 and 6.10, identifies the three stationary classes. Choose the representatives displayed in those tables and, whenever a conductor is odd, translate the corresponding critical function as in Lemma 4.6. For these compatible choices the following are exact equalities in the fields, not merely congruences of classes:

$$(189) \quad \frac{\beta(\tau)}{\gamma_\tau} = \mathcal{A}, \quad \frac{\beta(\chi_F)}{\gamma_\chi} = \mathcal{A}n, \quad \frac{\beta_K}{\gamma_K} = \mathcal{A}(n - u).$$

In the low range the selected ratios are respectively α/δ , $\varepsilon\beta/\delta$, and $\alpha(n-u)/\delta$, whereas in the high range they are α/γ_F , β/γ_F , and $\alpha(n-u)/\gamma_F$. The normalization of the generator τ in Lemma 9.5 is consequently the identity

$$(190) \quad \psi_F(\mathcal{A}\varpi_F^t \tilde{z}) = \psi_0(\eta_0 z) \quad (z \in k).$$

Suppose first that $m_F(\chi_F) = 2d + 1$. Take $c_\chi = \varpi_F^d$. Since $a + 2d = t$, the polar quotient of the χ_F critical function is

$$\begin{aligned} (X, Y) &\longmapsto \psi_F(\mathcal{A}nc_\chi^2 \tilde{X}\tilde{Y}) \\ &= \psi_F(\mathcal{A}\varpi_F^t n_0 \tilde{X}\tilde{Y}) = \psi_0(\eta_0 \bar{n}_0 XY). \end{aligned}$$

Together with (187), this proves (185) in the stated coordinate.

We now calculate the upper quadratic coefficient without suppressing the trace scalar. The parity of $m_K(\chi_K)$ is the parity of $m_F(\chi_F)$: in the low range the conductors are equal, while in the high range

$$m_K(\chi_K) - 1 = t + p(b - t) \equiv b \pmod{2}.$$

Assume this parity is odd and write $m_K(\chi_K) = 2d_K + 1$. Put $r = v_K(n - u)$. The exact conductor of χ_K implies $r = 0$ at the boundary; away from the boundary the two terms have distinct valuations. Hence

$$(191) \quad \begin{array}{c|c|c} & r & 2d_K \\ \hline b < t & t - b & b \\ b = t & 0 & t \\ b > t & p(t - b) & t + p(b - t). \end{array}$$

In every row,

$$(192) \quad r + 2d_K = t.$$

Take $c_K = \varpi_K^{d_K}$ and put $w_0 = \varpi_K^{-r}(n - u) \in \mathcal{O}_K^\times$. In the natural upper residue coordinate Z , define

$$\Phi_K^{\text{nat}}(Z) = \psi_K\left(\frac{\beta_K c_K \tilde{Z}}{\gamma_K}\right) \chi_K(1 + c_K \tilde{Z})^{-1}.$$

For $Z, W \in k$, Lemma 9.7 and (189) give

$$\frac{\Phi_K^{\text{nat}}(Z + W)}{\Phi_K^{\text{nat}}(Z)\Phi_K^{\text{nat}}(W)} = \psi_F\left(\mathcal{A}\text{Tr}_{K/F}((n - u)c_K^2 \tilde{Z}\tilde{W})\right).$$

By (192), the argument of the trace is $w_0 \varpi_K^t \tilde{Z}\tilde{W}$. Formula (178) and the fact that $x \mapsto \psi_F(\mathcal{A}x)$ is trivial on $\mathfrak{p}_F^{t+1}$ therefore give

$$(193) \quad \begin{aligned} \frac{\Phi_K^{\text{nat}}(Z + W)}{\Phi_K^{\text{nat}}(Z)\Phi_K^{\text{nat}}(W)} &= \psi_0\left(-\eta_0 \beta_0^{p-1} \bar{w}_0 ZW\right) \\ &= \psi_0\left(\beta_0^{-1} \text{in}(u - n)ZW\right), \end{aligned}$$

where $\text{in}(y) = \overline{\varpi_K^{-v_K(y)}} y$. The last equality uses $\eta_0 = \beta_0^{-p}$ and $\text{in}(u - n) = -\bar{w}_0$.

The displayed upper coordinate is the inverse-Frobenius reindexing of the natural one:

$$(194) \quad \Phi_{\chi_K}(X) = \Phi_K^{\text{nat}}(X^{1/p}).$$

It is phase preserving because Frobenius permutes k . By Lemma 9.7, the quadratic coefficient in this coordinate is the p th power of the coefficient in (193). Thus

$$(195) \quad \frac{A_K}{\eta_0} = \text{in}(u - n)^p.$$

Equivalently, for $X, Y \in k$,

$$(196) \quad \frac{\Phi_{\chi_K}(X + Y)}{\Phi_{\chi_K}(X)\Phi_{\chi_K}(Y)} = \psi_0(\eta_0 \text{in}(u - n)^p XY).$$

This proves the exact scalar in the upper polar form; no square-class choice or unrecorded uniformizer factor remains.

The initial term of $u - n$ is determined by the two valuations. If $b < t$, then $a > 0$ and $v_K(u) = a < pa = v_K(n)$, so it is ζ . If $b = t$, both terms are units and it is $\zeta - d_0$. If $b > t$, then $a < 0$ and $v_K(n) = pa < a = v_K(u)$, so it is $-d_0$. Consequently

$$A_K = \eta_0 \begin{cases} \zeta^p, & b < t, \\ (\zeta - d_0)^p, & b = t, \\ (-d_0)^p, & b > t, \end{cases} = \begin{cases} \eta, & b < t, \\ -\eta_0(d_0^p - d_0), & b = t, \\ -d_0^{p-1}\eta, & b > t. \end{cases}$$

Here $\zeta^p = d_0$, $\eta = \eta_0 d_0$, and p is odd.

It remains to determine the affine coefficient. We do this from an exact identity of complete critical functions, retaining the stationary ratios (189).

Let

$$P(y) = N(1 + y) - 1, \quad H(y) = P(y) - \text{Tr}(y) = \sum_{j=2}^p s_j(y).$$

For $X \in k$, let $Z = X^{1/p}$, choose a lift $\tilde{Z}$, and put $x = c_K \tilde{Z}$. Thus X is precisely the displayed upper coordinate of (194). Let $r_\chi(X)$ be the normalized class of $P(x)$ on the χ_F critical line and let $r_\tau(X)$ be the normalized class of $P(ux)$ on the τ critical line. A class is defined to be zero if the corresponding conductor is even or if the element lies one step beyond the critical line. Then

$$(197) \quad \Phi_{\chi_K}(X) = \Phi_{\chi_F}(r_\chi(X)) \Phi_\tau(r_\tau(X))^{-1} \psi_F(\mathcal{A}\{H(ux) - nH(x)\}).$$

To verify this identity, first note that $1 + P(ux) = N(1 + ux)$ and hence $\tau(1 + P(ux)) = 1$. Therefore the right-hand side of (197) is

$$\begin{aligned} & \psi_F(\mathcal{A}n P(x)) \chi_F(1 + P(x))^{-1} \psi_F(-\mathcal{A}P(ux)) \psi_F(\mathcal{A}\{H(ux) - nH(x)\}) \\ &= \psi_F(\mathcal{A}\{n \text{Tr}(x) - \text{Tr}(ux)\}) \chi_F(N(1 + x))^{-1} \\ &= \psi_K\left(\frac{\beta_K x}{\gamma_K}\right) \chi_K(1 + x)^{-1} = \Phi_{\chi_K}(X), \end{aligned}$$

where the second equality uses $P = \text{Tr} + H$ and the third uses (189).

We now compute the two classes in (197). Only odd conductors need be considered. If $b < t$, write $m = 2d + 1$, so $b = 2d < t$ and the upper conductor is again m . Here $c_K = \varpi_K^d$ and $c_\chi = \varpi_F^d$. For $x = c_K \tilde{X}^{1/p}$, the norm term is

$$N(x) = c_\chi \tilde{X} \pmod{\mathfrak{p}_F^{d+1}},$$

whereas Lemma 3.4 gives $v_F(s_j(x)) \geq d + 1$ for $1 \leq j < p$. Thus $r_\chi(X) = X$ exactly in the coordinate (194). If the conductor $T = t + 1$ of τ is odd, put $d_\tau = t/2$. Since $v_K(u) = t - b$,

$$v_K(ux) = d + t - b = t - d \geq d_\tau + 1,$$

and the same symmetric-function estimate puts every term of $P(ux)$ in $\mathfrak{p}_F^{d_\tau+1}$. If T is even there is no τ critical line. In either case $r_\tau(X) = 0$.

If $b = t$, then necessarily $m = T = 2d + 1$ in the odd row and $v_K(u) = 0$. In this case

$$c_K = \varpi_K^d, \quad c_\chi = c_\tau = \varpi_F^d.$$

Modulo the next critical ideal,

$$P(x) \equiv N(x) = c_\chi \tilde{X}, \quad P(ux) \equiv N(u)N(x) = c_\tau \widetilde{d_0 X}.$$

Hence $r_\chi(X) = X$ and $r_\tau(X) = d_0 X$ with no further coordinate choice.

Finally suppose $b > t$ and write again $m = 2d + 1$. The upper critical depth is

$$d_K = \frac{t + p(b - t)}{2}.$$

The norm term of $P(x)$ lies beyond the χ_F critical line because

$$d_K - d = \frac{(p - 1)(b - t)}{2} \geq 1.$$

For the remaining symmetric terms, the smallest lower bound occurs at $j = 1$, and

$$2\{d_K + (p-1)T - p(d+1)\} = (p-1)t - 2 \geq 0.$$

Hence Lemma 3.4 gives $P(x) \in \mathfrak{p}_F^{d+1}$, so $r_\chi(X) = 0$. If T is odd, write $t = 2d_\tau$. Then

$$v_K(ux) = d_K + t - b = d_\tau + (p-2)(d - d_\tau) \geq d_\tau + 1,$$

and again all terms of $P(ux)$ lie beyond the τ critical line. Thus $r_\tau(X) = 0$; the assertion is automatic when T is even. We have proved

$$(198) \quad \begin{array}{c|c|c} & r_\chi(X) & r_\tau(X) \\ \hline b < t & X & 0 \\ b = t & X & d_0 X \\ b > t & 0 & 0. \end{array}$$

The polynomial $H(ux) - nH(x)$ has no terms of degree zero or one. Lemma 9.8 shows that, modulo $\mathfrak{p}_F^T$, its degree- p terms cancel exactly and every term of degree $3, \dots, p-1$ vanishes. Since $z \mapsto \psi_F(\mathcal{A}z)$ is trivial on $\mathfrak{p}_F^T$, the correction factor in (197) is therefore homogeneous quadratic and contributes no affine character. The total upper quadratic coefficient has already been computed independently in (195); no identification of the correction factor alone with that coefficient is required. Consequently the affine coefficient of (197) is obtained only from the two lower critical functions in (198). Their affine parts give

$$(199) \quad L_{\chi K}(X) = \begin{cases} \psi_0(\eta\gamma X), & b < t, \\ \psi_0((\eta\gamma - d_0\eta_0\gamma_0)X) = \psi_0(\eta_0 d_0(\gamma - \gamma_0)X), & b = t, \\ 1, & b > t. \end{cases}$$

Together with the quadratic coefficients already computed, this proves (186). Since (197) is an identity of the complete finite functions, the translation of the elementary stationary value has been retained throughout. $\square$

Lemma 9.10 (Successor-half-depth Teichmüller congruence). *Let K/F be wildly ramified cyclic of odd prime degree p , with positive lower break t , and put $T = t + 1$. Suppose that $T > 1$ is odd. For $j \in \mathbf{F}_p$, let $[j] \in \mathcal{O}_F$ be its Teichmüller lift and let the same symbol j on the right denote its natural prime-field lift. Then*

$$(200) \quad [j] - j \in \mathfrak{p}_F^{(T+1)/2}.$$

Proof. In equal characteristic the two lifts agree. In mixed characteristic, $[j] - j$ is divisible by p in the maximal unramified subfield of F . Since $p = \text{Tr}_{K/F}(1)$ and the different exponent is $(p-1)T$, the trace-ideal formula gives

$$v_F(p) \geq \left\lfloor \frac{(p-1)T}{p} \right\rfloor.$$

Write $T = 2r + 1$. Since $T > 1$, one has $r \geq 1$, and for $p \geq 3$,

$$p(r+1) \leq (p-1)(2r+1).$$

Hence

$$\left\lfloor \frac{(p-1)T}{p} \right\rfloor \geq r+1 = \frac{T+1}{2},$$

which proves (200). $\square$

Proposition 9.11 (Vanishing of the actual wild-odd affine translations). *Use the actual high- and low-conductor representatives from Propositions 6.8 and 6.10, and compare them with the literal natural-power representatives used in the finite coefficient table. Whenever the corresponding critical conductor is odd, let d_* be its critical half-depth and let*

$$c_{0,j}^H, \quad c_{0,j}^R, \quad c_{0,j}^T$$

denote the high norm-row, low norm-row, and low boundary-twist representative differences. Then

$$(201) \quad c_{0,j}^H, c_{0,j}^R, c_{0,j}^T \in \mathfrak{p}_F^{d_*+1},$$

and hence all corresponding affine translation coefficients vanish:

$$(202) \quad \Delta_j^H = \Delta_j^R = \Delta_j^T = 0.$$

Thus the actual source rows and the normalized rows used in the coefficient table have identical complete critical functions, not merely the same polar coefficients.

Proof. In every row where this comparison occurs, the odd norm-character conductor is $T = 2d_* + 1$. The actual scalar is the Teichmüller representative $[j]$, whereas the literal power row uses the natural scalar j . Lemma 9.10 gives

$$[j] - j \in \mathfrak{p}_F^{d_*+1}.$$

Multiplication by the selected stationary source coefficient preserves this extra layer and proves the high inclusion in (201). The low norm and boundary-twist rows use the same Teichmüller-to-natural comparison after the common low stationary scalar is inserted, so the identical argument proves the other two inclusions. Corollary 9.3 now gives (202) and the equality of complete critical functions. $\square$

Proposition 9.12 (Complete odd-prime coefficient table). *The following coefficient pairs apply whenever the corresponding conductor is odd; for even conductor the phase is 1 by (172). If $b < t$, then*

$$(203) \quad (A_j, B_j) = (j\eta_0, j\eta_0\gamma_0) \quad (j \in \mathbf{F}_p^\times), \quad (A_0, B_0) = (\eta, \eta\gamma).$$

If $b = t$, then for every $j \in \mathbf{F}_p$,

$$(204) \quad (A_j, B_j) = (\eta_0(j + d_0), \eta_0(j\gamma_0 + d_0\gamma)).$$

If $b > t$, then for every $j \in \mathbf{F}_p$,

$$(205) \quad (A_j, B_j) = (\eta, \eta\gamma).$$

The upper character χ_K has the pair

$$(206) \quad (A_K, B_K) = \begin{cases} (\eta, \eta\gamma), & b < t, \\ (-\eta_0(d_0^p - d_0), \eta_0 d_0(\gamma - \gamma_0)), & b = t, \\ (-d_0^{p-1}\eta, 0), & b > t. \end{cases}$$

Proof. Proposition 9.11 first identifies the actual source rows with the normalized rows: every affine representative-change coefficient is zero. We may therefore compute with the displayed normalized representatives without dropping a translation term.

We next prove equality of the complete lower critical functions, not only equality of their polar quotients. On the dominant critical layer, write all linearized restrictions with one additive denominator Γ . Let β_τ and β_χ denote coefficients for the restrictions of τ and χ_F relative to Γ ; if a character is already trivial on that layer, take the corresponding coefficient to be 0. By Lemma 4.3(c), the coefficient $j\beta_\tau + \beta_\chi$ represents the restriction of $\tau^j\chi_F$ there.

Suppose $b < t$ and $j \neq 0$. A coefficient pair is needed only when T is odd. Put $d_\tau = (T-1)/2$ and choose $c_\tau \in \mathfrak{p}_F^{d_\tau} \setminus \mathfrak{p}_F^{d_\tau+1}$. Since $m_F(\chi_F) \leq T-1$,

$$v_F(c_\tau) = \frac{T-1}{2} \geq \left\lceil \frac{m_F(\chi_F)}{2} \right\rceil.$$

Thus the stationary linearization of χ_F is valid at $c_\tau\tilde{x}$. With the sum representative just chosen,

$$\begin{aligned} \frac{\Phi_{\tau^j\chi_F}(x)}{\Phi_{\tau^j}(x)} &= \psi_F\left(\frac{\beta_\chi c_\tau \tilde{x}}{\Gamma}\right) \chi_F(1 + c_\tau \tilde{x})^{-1} \\ &= 1. \end{aligned}$$

Hence the two complete critical functions agree. Since $\Phi_{\tau^j} = \Phi_{\tau^j}^j$, its pair is $(j\eta_0, j\eta_0\gamma_0)$. The case $j = 0$ is χ_F itself and has pair $(\eta, \eta\gamma)$.

Suppose $b > t$. A coefficient pair is needed only when $m_F(\chi_F)$ is odd. Put $d_\chi = (m_F(\chi_F) - 1)/2$ and choose $c_\chi \in \mathfrak{p}_F^{d_\chi} \setminus \mathfrak{p}_F^{d_\chi+1}$. Since $m_F(\chi_F) > T$,

$$v_F(c_\chi) = \frac{m_F(\chi_F) - 1}{2} \geq \left\lceil \frac{T}{2} \right\rceil.$$

The stationary linearization of every τ^j is therefore valid at $c_\chi \tilde{x}$, and for every $j \in \mathbf{F}_p$,

$$\begin{aligned} \frac{\Phi_{\tau^j \chi_F}(x)}{\Phi_{\chi_F}(x)} &= \psi_F\left(\frac{j\beta_\tau c_\chi \tilde{x}}{\Gamma}\right) \tau^j (1 + c_\chi \tilde{x})^{-1} \\ &= 1. \end{aligned}$$

Thus every complete critical function has the pair $(\eta, \eta\gamma)$.

At $b = t$, the two characters have the common conductor T . Lemma 4.3(c) identifies the stationary class of $\tau^j \chi_F$ with the sum of the two classes, and Lemma 4.8 shows that its leading class does not cancel. We may therefore choose the sum representative at the same conductor and translate the critical functions consistently. The complete critical function of the product is the pointwise product of the two complete critical functions, so the coefficient pairs add:

$$j(\eta_0, \eta_0\gamma_0) + (\eta_0 d_0, \eta_0 d_0 \gamma).$$

This proves the entire lower table, including its affine coefficients, without reusing a cancelled representative at a smaller conductor. The upper table is exactly (186) of Lemma 9.9. $\square$

Proposition 9.13 (Odd-prime finite phase cancellation). *Let R_j be the critical phase of τ^j , T_j that of $\tau^j \chi_F$, and U the upper critical phase. Then*

$$(207) \quad \mathcal{R}_{\text{odd}} := U \frac{\prod_{j \neq 0} R_j}{\prod_j T_j} = 1.$$

Proof. All phases in this proof use the parity convention (172).

If $b < t$, (203) gives $T_j = R_j$ for $j \neq 0$, while the $j = 0$ pair and the upper pair agree, so $T_0 = U$.

Suppose $b > t$. If $m_F(\chi_F)$ is even, then every T_j and U is 1. If it is odd, all T_j are the same phase $T = Q_{\psi_0}(\eta, \eta\gamma)$. In the explicit formula for Q_{ψ_0} , the additive factor has p th power 1, because its value lies in the group of p th roots of unity. Hence Lemma 9.4 gives

$$T^p = q_0^p \nu(\eta)^p = q_0 \nu(-1) \nu(\eta) = q_0 \nu(-\eta) = U,$$

because d_0^{p-1} is a square and the upper affine coefficient is zero. Independently, if the conductor $T = t + 1$ of τ is even then all R_j are 1. If it is odd, then

$$\begin{aligned} \prod_{j \neq 0} R_j &= q_0^{p-1} \nu(\eta_0)^{p-1} \nu\left(\prod_{j \neq 0} j\right) \psi_0\left(-\frac{\eta_0 \gamma_0^2}{2} \sum_{j \neq 0} j\right) \\ &= q_0^{p-1} \nu(-1) = 1. \end{aligned}$$

Here $\sum_{j \neq 0} j = 0$, $\prod_{j \neq 0} j = -1$, and $q_0^{p-1} = \nu(-1)$ by Lemma 9.4. Hence $U \prod R_j = \prod T_j$ in every parity pattern above the break.

Finally suppose $b = t$. All four kinds of character have conductor T . If T is even, every residual phase is 1. If T is odd, the actual boundary coefficient pairs carry one common nonzero scalar. Proposition 5.8 shows that removing this scalar does not change the complete affine-pencil quotient, while Proposition 9.11 shows that no affine representative-change term remains. We may therefore use the normalized pairs of Proposition 9.12. Put $d_0 = \zeta^p$ as in (184). Here $d_0 = \bar{n}$, and Lemma 4.8 together with (51) gives $j + d_0 \neq 0$ for every $j \in \mathbf{F}_p$. The calculation of $\prod_{j \neq 0} R_j$ above depends only on τ and therefore also gives

$$(208) \quad \prod_{j \neq 0} R_j = 1.$$

For the Gaussian prefactors in $\prod_j T_j$, put

$$A_{\text{prod}} = \prod_{j \in \mathbf{F}_p} \eta_0(j + d_0) = \eta_0^p(d_0^p - d_0), \quad A_K = -\eta_0(d_0^p - d_0).$$

Then

$$\frac{A_{\text{prod}}}{A_K} = -\eta_0^{p-1}.$$

Since η_0^{p-1} is a square,

$$\nu(A_{\text{prod}}) = \nu(-1)\nu(A_K).$$

Together with $q_0^p = q_0\nu(-1)$ from Lemma 9.4, this gives

$$(209) \quad q_0^p \nu(A_{\text{prod}}) = q_0\nu(-1)^2 \nu(A_K) = q_0\nu(A_K).$$

Thus the complete Gaussian prefactors agree, although their quadratic character parts alone differ by $\nu(-1)$. Put $e = \gamma - \gamma_0$. The additive part is

$$\begin{aligned} \sum_{j \in \mathbf{F}_p} \frac{B_j^2}{A_j} &= \eta_0 \sum_j \frac{(\gamma_0(j + d_0) + d_0 e)^2}{j + d_0} \\ &= \eta_0 d_0^2 e^2 \sum_j \frac{1}{j + d_0} = \frac{\eta_0 d_0^2 e^2}{d_0 - d_0^p}. \end{aligned}$$

The first two expanded sums vanish because $\sum_j (j + d_0) = 0$ and $p = 0$ in k ; the last equality is the logarithmic derivative of $\prod_j (X + j) = X^p - X$. For the upper pair (A, B) ,

$$\frac{B^2}{A} = \frac{\eta_0 d_0^2 e^2}{d_0 - d_0^p}.$$

Thus $U = \prod_j T_j$. Together with (208), this proves (207) in every parity case. $\square$

9.2.1. *Exact stationary-factor assembly.* We now assemble the complete stationary factors. Let $b = m_F(\chi_F) - 1$ and put

$$(210) \quad a = t - b.$$

In the low range $b \leq t$, use the notation of theorem 6.10 and set

$$(211) \quad A = \alpha/\delta, \quad u = \varepsilon_1 \beta_1 / \alpha_1, \quad n = N(u) = \varepsilon \beta / \alpha.$$

In the high range $b > t$, use theorem 6.8 and set

$$(212) \quad A = \alpha/\gamma_F, \quad u = \beta_1 / \alpha_1, \quad n = N(u) = \beta / \alpha.$$

In both cases

$$(213) \quad v_K(u) = v_F(n) = a.$$

The character linearizations needed for the correction units will be proved only after their exact depths have been established.

Represent $\mathbf{F}_p^\times$ by its Teichmüller lifts. Define

$$(214) \quad u^\vee = -n/u, \quad z_0 = \frac{N(1 + u^\vee)}{1 + N(u^\vee)}, \quad z_j = \frac{N(u + j)}{n + j} \quad (j \neq 0).$$

If $a \neq 0$, the denominators are nonzero by valuation. If $a = 0$, then $m_F(\chi_F) = T$. On the last layer the stationary class of $\tau^j \chi_F$ is represented, up to the common nonzero scalar, by $n + j$. Lemma 4.8, equivalently the noncancellation criterion (51), therefore gives $\bar{n} + j \neq 0$ for every $j \in \mathbf{F}_p$ (for $j = 0$ this also follows from $n \in \mathcal{O}_F^\times$). Thus $\bar{n} \notin -\mathbf{F}_p$; in particular every $n + j$ and $1 - n^{p-1}$ is a unit. Put $x_0 = z_0 - 1$ and $x_j = z_j - 1$.

Lemma 9.14 (Parameter-ratio identities). *One has*

$$(215) \quad N(u + j) = (n + j)z_j, \quad N(n - u) = \left(\prod_{j \in \mathbf{F}_p} (n + j) \right) z_0.$$

Proof. The first identity is the definition. Since p is odd, $N(u^\vee) = -n^{p-1}$ and

$$N(1 + u^\vee) = -\frac{N(n - u)}{n}.$$

Also $\prod_j(n + j) = n^p - n$. Substitution in the definition of z_0 gives the second identity. $\square$

Lemma 9.15 (Depth of the correction units). *Let*

$$R(c) = \left\lfloor \frac{c + (p-1)T}{p} \right\rfloor \quad (c \geq 0).$$

Then

$$(216) \quad z_j - 1 \in \mathfrak{p}_F^{R(|a|)} \subseteq \mathfrak{p}_F^{s_0} \quad (j \neq 0),$$

and

$$(217) \quad z_0 - 1 \in \mathfrak{p}_F^{R((p-1)|a|)} \subseteq \mathfrak{p}_F^{\lceil m/2 \rceil}.$$

Thus $z_j - 1$ lies in the linearization domain of τ and $z_0 - 1$ lies in the linearization domain of χ_F .

Proof. For $w \in K$ with $1 + N(w) \neq 0$, put $Z(w) = N(1 + w)/(1 + N(w))$. If $v_K(w) = c > 0$, then

$$Z(w) - 1 = \frac{s_1(w) + \cdots + s_{p-1}(w)}{1 + N(w)},$$

so [theorem 3.4](#) gives $v_F(Z(w) - 1) \geq R(c)$. If $c < 0$, the exact identity $Z(w) = Z(w^{-1})$ reduces to the positive case. If $c = 0$, the minimal-conductor condition used above says that $1 + N(w)$ is a unit in each occurrence of $Z(w)$: for $w = u/j$ this is equivalent to $\bar{n} + j \neq 0$, and for $w = u^\vee$ to $1 - \bar{n}^{p-1} \neq 0$. The strong symmetric estimate then gives

$$v_F(s_i(w)) \geq \left\lfloor \frac{(p-1)T}{p} \right\rfloor = R(0) \quad (1 \leq i \leq p-1),$$

so the same conclusion holds when $c = 0$.

Now $z_j = Z(u/j)$, and $z_0 = Z(u^\vee)$ with $v_K(u^\vee) = (p-1)a$. This proves the first inclusions.

We use the elementary integer inequality

$$(218) \quad \left\lfloor \frac{(p-1)M}{p} \right\rfloor \geq \left\lceil \frac{M}{2} \right\rceil \quad (p \geq 3, M \geq 2).$$

If M is even, the real inequality $(p-1)M/p \geq M/2$ already has an integral right side. If M is odd, then $M \geq 3$, and

$$\frac{(p-1)M}{p} \geq \frac{M+1}{2} \iff (p-2)M \geq p,$$

which follows from $M \geq 3$ and $p \geq 3$.

Since $R(|a|) \geq R(0)$, applying (218) with $M = T$ gives $R(|a|) \geq \lceil T/2 \rceil = s_0$. For z_0 , if $a \geq 0$, then $m = T - a$ and

$$R((p-1)a) = \left\lfloor \frac{(p-1)(T+a)}{p} \right\rfloor \geq \left\lceil \frac{T+a}{2} \right\rceil \geq \left\lceil \frac{T-a}{2} \right\rceil = \left\lceil \frac{m}{2} \right\rceil.$$

If $a < 0$, put $c = -a$, so $m = T + c$; applying (218) with $M = T + c$ gives

$$R((p-1)c) = \left\lfloor \frac{(p-1)(T+c)}{p} \right\rfloor \geq \left\lceil \frac{T+c}{2} \right\rceil = \left\lceil \frac{m}{2} \right\rceil.$$

This proves (217). $\square$

Proposition 9.16 (Correction-depth character linearization). *With the notation of (211)–(214), one has*

$$(219) \quad \chi_F(z_0) = \psi_F(An(z_0 - 1)), \quad \tau^j(z_j) = \psi_F(Aj(z_j - 1)) \quad (j \neq 0).$$

Proof. The stationary linearization of τ is valid on $\mathfrak{p}_F^{s_0}$, and that of χ_F is valid on $\mathfrak{p}_F^{[m/2]}$. Lemma 9.15 puts $z_j - 1$ and $z_0 - 1$ in precisely these two domains. The coefficients of the two linearizations are A and An , respectively. Applying the linearizations and taking the j th power of the first identity gives (219). $\square$

Proposition 9.17 (Exact assembly of the complete stationary quotient). *With the orientation (17), substitution of the complete Lamprecht factors gives*

$$(220) \quad \mathcal{E}_{K/F}(\chi_F) = \mathcal{R}_{\text{odd}} \left[\psi_F(A \operatorname{Tr}_{K/F}(u)) \chi_F(z_0) \prod_{j \neq 0} \tau^j(z_j) \right]^{-1}.$$

Consequently

$$(221) \quad \mathcal{E}_{K/F}(\chi_F) = \mathcal{R}_{\text{odd}} \psi_F(-AX),$$

where

$$(222) \quad X = \operatorname{Tr}_{K/F}(u) + n(z_0 - 1) + \sum_{j \neq 0} j(z_j - 1).$$

Proof. Write every local constant by (63). Corollary 6.3 identifies the pullback class, Lemma 4.3(c) identifies every twist class, and Propositions 6.8 and 6.10 give the displayed representatives of those classes. At the boundary $m = T$, Lemma 4.8 proves that no leading twist class cancels, so every representative below is used at its actual conductor. Choose the normalized representatives displayed below. Whenever the conductor is odd, translate the associated critical function as prescribed by Lemma 4.6. Thus the calculation uses no canonical field element and discards neither an elementary factor nor a Hasse correction.

In the high range $b > t$, the admissible-element and stationary-parameter pairs are

$$\begin{array}{l|l} \chi_K & (\gamma_F, \alpha(n-u)), \\ \tau^j & (\gamma_F/\alpha, j) \quad (j \neq 0), \\ \tau^j \chi_F & (\gamma_F, \alpha(n+j)) \quad (j \in \mathbf{F}_p). \end{array}$$

In the low range $b \leq t$, they are

$$\begin{array}{l|l} \chi_K & (\delta/\varepsilon_1, (\alpha/\varepsilon_1)(n-u)), \\ \tau^j & (\delta, \alpha j) \quad (j \neq 0), \\ \chi_F & (\delta/\varepsilon, (\alpha/\varepsilon)n), \\ \tau^j \chi_F & (\delta, \alpha(n+j)) \quad (j \neq 0). \end{array}$$

Each ordered pair consists of an admissible element and the selected representative of the corresponding stationary numerator class. The tables are precisely the high- and low-conductor stationary-class tables with their common scalar made visible; their use is legitimate because the complete stationary factor is representative-independent.

The factors contributed by the admissible elements cancel exactly. In the high range their quotient is

$$\frac{\chi_K(\gamma_F) \prod_{j \neq 0} \tau^j(\gamma_F/\alpha)}{\prod_j (\tau^j \chi_F)(\gamma_F)} = 1,$$

and in the low range it is

$$\frac{\chi_K(\delta/\varepsilon_1) \prod_{j \neq 0} \tau^j(\delta)}{\chi_F(\delta/\varepsilon) \prod_{j \neq 0} (\tau^j \chi_F)(\delta)} = 1.$$

Here $N(\varepsilon_1) = \varepsilon$ and α is a norm.

After the common scalar is cancelled, the multiplicative stationary values contribute

$$(223) \quad H = \chi_F \left(\frac{\prod_j (n+j)}{N(n-u)} \right) \prod_{j \neq 0} \tau^j \left(\frac{n+j}{j} \right).$$

By Lemma 9.14, the first ratio is z_0^{-1} . Moreover

$$\frac{n+j}{j} = N \left(\frac{u+j}{j} \right) z_j^{-1},$$

so the norm character kills the norm and

$$(224) \quad H = \chi_F(z_0)^{-1} \prod_{j \neq 0} \tau^j(z_j)^{-1}.$$

The additive stationary values have signed sum

$$\mathrm{Tr}_{K/F}(n - u) + \sum_{j \neq 0} j - \sum_{j \in \mathbb{F}_p} (n + j) = -\mathrm{Tr}_{K/F}(u),$$

using the Teichmuller sum in Lemma 6.4. Their additive factor is therefore $\psi_F(-A \mathrm{Tr}_{K/F}(u))$. Together with (224) and the complete residual quotient $\mathcal{R}_{\mathrm{odd}}$, this is precisely (220).

Finally Proposition 9.16 evaluates all correction characters at their now-verified depths. Substitution gives (221) and (222). $\square$

9.2.2. *The norm–trace formula and its valuation.* Let $s_r = s_r(u)$ be the elementary symmetric functions of the conjugates of u , with $s_1 = \mathrm{Tr}_{K/F}(u)$ and $s_p = n$. Put

$$(225) \quad B = s_1 + \sum_{r=1}^{p-2} (-n)^r s_{p-r}.$$

Proposition 9.18 (Closed norm–trace formula). *The element X in (222) satisfies the exact identity*

$$(226) \quad X = \frac{pB}{1 - n^{p-1}}.$$

Proof. For $j \neq 0$,

$$x_j = \frac{N(u + j) - (n + j)}{n + j} = \frac{\sum_{r=1}^{p-1} j^{p-r} s_r}{n + j}.$$

Because $j^{p-1} = 1$,

$$(227) \quad \frac{1}{n + j} = \frac{\sum_{q=0}^{p-2} (-n)^q j^{p-2-q}}{1 - n^{p-1}}.$$

Indeed, multiplying the numerator by $n + j$ telescopes to $1 - n^{p-1}$. Summing over the $(p - 1)$ Teichmuller roots and using

$$\sum_{j \neq 0} j^r = \begin{cases} p - 1, & p - 1 \mid r, \\ 0, & p - 1 \nmid r, \end{cases}$$

shows that only the indices $r + q = 1$ and $r + q = p$ survive. Hence

$$(228) \quad \sum_{j \neq 0} j x_j = \frac{(p - 1)B}{1 - n^{p-1}}.$$

On the other hand,

$$N(n - u) - (n^p - n) = \sum_{r=1}^{p-1} (-1)^r s_r n^{p-r}.$$

Since $z_0 = N(n - u)/(n^p - n)$,

$$\begin{aligned} n x_0 &= -\frac{\sum_{r=1}^{p-1} (-1)^r s_r n^{p-r}}{1 - n^{p-1}} \\ &= \frac{s_1 n^{p-1} + \sum_{q=1}^{p-2} (-n)^q s_{p-q}}{1 - n^{p-1}} \\ &= \frac{B}{1 - n^{p-1}} - s_1. \end{aligned}$$

Adding s_1 , this identity, and (228) proves (226). This is an identity in the field F itself. In equal characteristic the same finite sums have $p = 0$ and $p - 1 = -1$ in F , so no lifting to characteristic zero is being used. $\square$

Lemma 9.19 (The mixed-characteristic p -valuation forced by ramification). *Assume $\text{char } F = 0$. Put $e = v_F(p)$ and*

$$e_0 = \left\lfloor \frac{(p-1)T}{p} \right\rfloor = T - \lceil T/p \rceil.$$

Then $e \geq e_0$ and, for odd p , $2e_0 \geq T$.

Proof. Since $p = \text{Tr}_{K/F}(1)$, the trace-ideal formula in Theorem 3.7(b) gives $e \geq e_0$. For $p \geq 3$ and $T \geq 2$, $\lceil T/p \rceil \leq T/2$, hence $2e_0 = 2T - 2\lceil T/p \rceil \geq T$. $\square$

Proposition 9.20 (Vanishing or depth of the norm–trace error). *One has*

$$(229) \quad X \in \mathfrak{p}_F^T, \quad \psi_F(A\mathfrak{p}_F^T) = 1.$$

More precisely, if $\text{char } F = p$, then $X = 0$; if $\text{char } F = 0$, the first inclusion follows from the ramification valuation of p . In either case $\psi_F(-AX) = 1$.

Proof. The denominator $1 - n^{p-1}$ in (226) is nonzero. If $v_F(n) \neq 0$, this follows from valuations; if $v_F(n) = 0$, Lemma 4.8 and (51) show that its residue is nonzero, so it is a unit.

Suppose first that $\text{char } F = p$. In the exact identity (226), the integer p is zero in F . Therefore

$$X = \frac{pB}{1 - n^{p-1}} = 0,$$

which proves the first assertion in equal characteristic.

Now assume $\text{char } F = 0$. Let $a = v_K(u) = v_F(n)$ and use the notation e, e_0 of Lemma 9.19.

If $a > 0$, the denominator in (226) is a unit. By theorem 3.4, every $s_j(u)$ has valuation at least e_0 , and the powers of n only increase valuation. Hence $v_F(pB) \geq e + e_0 \geq 2e_0 \geq T$.

If $a = 0$, the denominator is again a unit by Lemma 4.8 and (51), and the same estimate applies.

If $a < 0$, put $c = -a > 0$, $y = u^{-1}$, and for $1 \leq r \leq p-1$ set

$$(230) \quad L_r(c) = \left\lfloor \frac{rc + (p-1)T}{p} \right\rfloor.$$

Then $v_F(1 - n^{p-1}) = (p-1)a$, so division by the denominator is multiplication, up to a unit, by n^{1-p} . The reciprocal identity is

$$(231) \quad s_j(u) = n s_{p-j}(y) \quad (1 \leq j \leq p-1).$$

For the s_1 term,

$$v_F(pn^{1-p}s_1(u)) = v_F(pn^{2-p}s_{p-1}(y)) \geq e + (p-2)c + L_{p-1}(c) \geq e + e_0 \geq T.$$

For a term $n^{p-j}s_j(u)$ with $2 \leq j \leq p-1$,

$$v_F(pn^{1-j}s_j(u)) = v_F(pn^{2-j}s_{p-j}(y)) \geq e + (j-2)c + L_{p-j}(c) \geq e + e_0 \geq T.$$

Thus $X \in \mathfrak{p}_F^T$ also in mixed characteristic.

Finally τ has conductor T and $\tau(1+x) = \psi_F(Ax)$ on a layer containing $\mathfrak{p}_F^T$. Since τ is trivial on $1 + \mathfrak{p}_F^T$, one has $\psi_F(A\mathfrak{p}_F^T) = 1$. The conclusion follows. $\square$

Corollary 9.21 (Wild odd-prime non-stable case). *For odd p , every non-stable wildly ramified case satisfies $\mathcal{E}_{K/F}(\chi_F) = 1$.*

Proof. Combine theorems 9.13, 9.17 and 9.20. $\square$

9.3. Wild quadratic extensions. Assume $p = 2$ and K/F is wildly ramified quadratic. Let $S(K/F) = \{1, \tau\}$. The stationary calculation in this subsection is made under the standing hypotheses that χ_F has minimal conductor in its $S(K/F)$ -orbit and that $m_F(\chi_F) > 1$. Proposition 9.32 removes these restrictions by twist invariance and the endpoint calculation. Under these standing hypotheses, keep

$$b = m_F(\chi_F) - 1, \quad a = t - b = T - m_F(\chi_F), \quad T = t + 1.$$

If $b \leq t$, use the low-parameter notation and put

$$A = \frac{\alpha}{\delta}, \quad u = \frac{\varepsilon_1 \beta_1}{\alpha_1}, \quad n = N(u) = \frac{\varepsilon \beta}{\alpha}.$$

If $b > t$, use Proposition 6.9 and put

$$A = \frac{\alpha \delta_h}{\gamma_F}, \quad u = \frac{\beta_1}{\alpha_1}, \quad n = N(u) = \frac{\beta}{\alpha \delta_h}.$$

In both ranges $v_K(u) = v_F(n) = a$, and on every unit layer used below

$$\tau(1+z) = \psi_F(Az), \quad \chi_F(1+z) = \psi_F(Anz).$$

Let u^σ be the nontrivial conjugate of u and put

$$(232) \quad s = \text{Tr}_{K/F}(u), \quad u^\vee = -n/u = -u^\sigma,$$

$$z_0 = \frac{N(1+u^\vee)}{1+n}, \quad z_1 = \frac{N(1+u)}{1+n}, \quad x = z_0 - 1, \quad y = z_1 - 1.$$

If $a \neq 0$, then $1+n \neq 0$ by valuation. If $a = 0$, then $m_F(\chi_F) = T$, and on the last layer the stationary class of $\tau\chi_F$ is represented, up to the common nonzero scalar, by $1+n$. Lemma 4.8 and (51) exclude its cancellation. Thus $1+n$ is a unit.

Lemma 9.22 (Exact quadratic norm correction). *One has*

$$(233) \quad x = -\frac{s}{1+n}, \quad y = \frac{s}{1+n}, \quad X := s + nx + y = \frac{2s}{1+n}.$$

Proof. The exact quadratic norm identity gives

$$N(1+u) = 1+s+n, \quad N(1+u^\vee) = 1-s+n.$$

Division by $1+n$ gives the first two formulae. Substituting them in $s+nx+y$ gives $2s/(1+n)$. $\square$

Lemma 9.23 (Quadratic graded trace). *Let K/F be wildly ramified quadratic and let $T = v_K(\mathfrak{D}_{K/F}) = t+1$. If $r \geq 0$ and $r+T$ is odd, put $q = (r+T-1)/2$. Then trace induces a k -linear isomorphism*

$$(234) \quad \text{Tr}_{K/F} : \mathfrak{p}_K^r / \mathfrak{p}_K^{r+1} \xrightarrow{\sim} \mathfrak{p}_F^q / \mathfrak{p}_F^{q+1}.$$

In particular:

(a) *If T is odd and $d_\tau = (T-1)/2$, then $v_F(2) = d_\tau$. For every $w \in \mathcal{O}_K^\times$ and every lift $\tilde{w} \in \mathcal{O}_F$ of its residue,*

$$(235) \quad \text{Tr}(w) \equiv 2\tilde{w} \pmod{\mathfrak{p}_F^{d_\tau+1}}.$$

(b) *If $m = 2d+1 > T$ and $r = m-T$, then trace identifies $\mathfrak{p}_K^r / \mathfrak{p}_K^{r+1}$ with $\mathfrak{p}_F^d / \mathfrak{p}_F^{d+1}$.*

Proof. For a quadratic extension, (35) reads

$$\text{Tr}(\mathfrak{p}_K^j) = \mathfrak{p}_F^{\lfloor (j+T)/2 \rfloor}.$$

If $r+T = 2q+1$, then

$$\text{Tr}(\mathfrak{p}_K^r) = \mathfrak{p}_F^q, \quad \text{Tr}(\mathfrak{p}_K^{r+1}) = \mathfrak{p}_F^{q+1}.$$

Thus trace induces a surjection in (234). Both source and target are one-dimensional k -vector spaces, and the map is k -linear because the residue extension is trivial. It is therefore an isomorphism.

If T is odd, apply the isomorphism with $r = 0$. The class of 1 is nonzero and maps to the class of $\text{Tr}(1) = 2$, proving $v_F(2) = d_\tau$. Since $w - \tilde{w} \in \mathfrak{p}_K$, the trace-ideal formula at $j = 1$ gives

$$\text{Tr}(w) - 2\tilde{w} \in \text{Tr}(\mathfrak{p}_K) = \mathfrak{p}_F^{(T+1)/2} = \mathfrak{p}_F^{d_\tau+1},$$

which proves (a). In particular, a wildly ramified quadratic extension with T odd is necessarily of mixed characteristic; in equal characteristic two the equality $v_F(2) = d_\tau$ is impossible. If $m = 2d+1$ and $r = m-T$, then $r+T = m$ and $q = d$, proving (b). $\square$

9.4. A normalized quadratic refinement in characteristic two. Put

$$\psi_0(z) = (-1)^{\mathrm{Tr}_{k/\mathbf{F}_2}(z)}.$$

An odd critical layer supplies a nowhere-zero unimodular function $h : k \rightarrow \mathbf{C}^\times$ satisfying

$$(236) \quad h(r+s) = h(r)h(s)\psi_0(rs).$$

If no odd critical layer occurs, there is no Hasse factor and this subsection is unnecessary.

Lemma 9.24 (Normalized refinement). *After multiplying h by an additive character, one obtains a unimodular function $\mathbf{q} : k \rightarrow \mathbf{C}^\times$ such that*

$$(237) \quad \mathbf{q}(r+s) = \mathbf{q}(r)\mathbf{q}(s)\psi_0(rs), \quad \mathbf{q}(r^2) = \mathbf{q}(r).$$

Every other function with the same polar character is uniquely

$$(238) \quad h_\lambda(z) = \mathbf{q}(z)\psi_0(\lambda z) \quad (\lambda \in k).$$

If

$$H(\lambda) = \mathrm{ph} \left(\sum_{z \in k} h_\lambda(z) \right), \quad H_0 = H(0),$$

then

$$(239) \quad H(\lambda) = H_0 \overline{\mathbf{q}(\lambda)}, \quad H_0^2 = \mathbf{q}(1), \quad \mathbf{q}(r)^2 = \psi_0(r).$$

Proof. The quotient

$$\ell(z) = \frac{h(z^2)}{h(z)}$$

is additive. Indeed, using (236) and Frobenius invariance of ψ_0 ,

$$\ell(r+s) = \frac{h((r+s)^2)}{h(r+s)} = \frac{h(r^2)}{h(r)} \frac{h(s^2)}{h(s)} = \ell(r)\ell(s).$$

The trace pairing on k is perfect, so there is a unique $c \in k$ such that $\ell(z) = \psi_0(cz)$. Let $f = [k : \mathbf{F}_2]$. For every $z \in k$, telescoping around the Frobenius orbit gives

$$\begin{aligned} 1 &= \frac{h(z^{2^f})}{h(z)} = \prod_{i=0}^{f-1} \frac{h(z^{2^{i+1}})}{h(z^{2^i})} \\ &= \psi_0 \left(c \sum_{i=0}^{f-1} z^{2^i} \right) = \psi_0 \left(c \mathrm{Tr}_{k/\mathbf{F}_2}(z) \right). \end{aligned}$$

Choose z of absolute trace 1. Then $\psi_0(c) = 1$, or equivalently $\mathrm{Tr}_{k/\mathbf{F}_2}(c) = 0$.

Consider the $\mathbf{F}_2$ -linear map

$$L : k \longrightarrow k, \quad L(d) = d + d^{2^{f-1}}.$$

Its image is contained in the trace-zero hyperplane. Its kernel is $\mathbf{F}_2$: indeed, $d^{2^{f-1}} = d$ means that d is fixed by Frob^{f-1} , whose fixed field in $\mathbf{F}_{2^f}$ is $\mathbf{F}_{2^{\gcd(f, f-1)}} = \mathbf{F}_2$. Thus $|\mathrm{im} L| = |k|/2$, which is the size of the trace-zero hyperplane. Hence there is $d \in k$ such that

$$(240) \quad c = d + d^{2^{f-1}}.$$

Replace $h(z)$ by

$$\mathbf{q}(z) = h(z)\psi_0(dz).$$

The polar character is unchanged. Moreover

$$\begin{aligned} \frac{\mathbf{q}(z^2)}{\mathbf{q}(z)} &= \psi_0(cz)\psi_0(dz^2 + dz) \\ &= \psi_0((c + d^{2^{f-1}} + d)z) = 1, \end{aligned}$$

where $\psi_0(dz^2) = \psi_0(d^{2^{f-1}}z)$ and (240) were used. This proves $\mathfrak{q}(z^2) = \mathfrak{q}(z)$. If h' has the same polar character as $\mathfrak{q}$, then $h'/\mathfrak{q}$ is an additive character of k , hence is uniquely $z \mapsto \psi_0(\lambda z)$. This proves (238).

From (237),

$$\mathfrak{q}(z)\psi_0(\lambda z) = \mathfrak{q}(z + \lambda)\mathfrak{q}(\lambda)^{-1},$$

so translation of the finite sum gives the first formula in (239). Setting $r = s$ in (237) gives

$$\mathfrak{q}(r)^2 = \psi_0(r^2) = \psi_0(r),$$

and therefore $\overline{\mathfrak{q}(r)} = \mathfrak{q}(r)\psi_0(r)$.

Put $S_0 = \sum_{r \in k} \mathfrak{q}(r)$. Its polar bicharacter is $(r, s) \mapsto \psi_0(rs)$, which is nondegenerate by the perfect trace pairing. Lemma 4.4 therefore gives

$$|S_0|^2 = |k|,$$

so $S_0 \neq 0$. Finally, translation by 1 gives

$$\overline{S_0} = \sum_r \mathfrak{q}(r)\psi_0(r) = \frac{1}{\mathfrak{q}(1)} \sum_r \mathfrak{q}(r+1) = \frac{S_0}{\mathfrak{q}(1)}.$$

Taking phases yields $H_0^{-1} = H_0/\mathfrak{q}(1)$, and hence $H_0^2 = \mathfrak{q}(1)$. $\square$

Lemma 9.25 (Correction on an odd critical line). *Let $m_E(\theta) = 2d + 1$, choose $c_\theta \in \mathfrak{p}_E^d \setminus \mathfrak{p}_E^{d+1}$, and write the corresponding critical function in normalized characteristic-two form as*

$$h_\lambda(z) = \mathfrak{q}(z)\psi_0(\lambda z).$$

If $z_\theta = 1 + w$ with $w \in \mathfrak{p}_E^d$ and $c = \overline{w/c_\theta} \in k_E$, then

$$(241) \quad \theta(z_\theta)^{-1} = \psi_E\left(-\frac{\beta_\theta w}{\gamma_\theta}\right) h_\lambda(c).$$

Thus the linear approximation and the complete value of the critical function must be translated together.

Proof. Choose a lift $\tilde{c} \in \mathcal{O}_E$ of c and write

$$w = c_\theta \tilde{c} + r, \quad r \in \mathfrak{p}_E^{d+1}.$$

Then

$$1 + w = (1 + c_\theta \tilde{c})(1 + y), \quad y = \frac{r}{1 + c_\theta \tilde{c}} \in \mathfrak{p}_E^{d+1}.$$

The odd-conductor stationary linearization gives $\theta(1 + y) = \psi_E(\beta_\theta y / \gamma_\theta)$. By the definition of the critical function,

$$\theta(1 + c_\theta \tilde{c})^{-1} = \psi_E\left(-\frac{\beta_\theta c_\theta \tilde{c}}{\gamma_\theta}\right) h_\lambda(c).$$

Consequently

$$\theta(1 + w)^{-1} = \psi_E\left(-\frac{\beta_\theta (c_\theta \tilde{c} + y)}{\gamma_\theta}\right) h_\lambda(c).$$

Finally

$$c_\theta \tilde{c} + y - w = -c_\theta \tilde{c} y \in \mathfrak{p}_E^{2d+1}.$$

Since $\gamma_\theta \mathcal{O}_E = \mathfrak{p}_E^{2d+1+n_E(\psi_E)}$ and β_θ is a unit, the additive character is trivial on the resulting difference. Replacing $c_\theta \tilde{c} + y$ by w proves (241). $\square$

Proposition 9.26 (Branch matrix for characteristic-two critical refinements). *Put $m = m_F(\chi_F) = b + 1$ and $T = t + 1$. A normalized quadratic refinement is needed exactly when an actual positive-polar critical function occurs. In particular, no refinement is chosen in precisely the following three all-even rows:*

$range$	$m \pmod{2}$	$T \pmod{2}$
$b < t$	0	0
$b = t$	0	0
$b > t$	0	0.

(242)

In every other row, a refinement is attached only to the odd critical source or sources that actually occur. At the boundary the noncancellation assertion $1 + \rho \neq 0$ is required only in the odd row $b = t$, T odd.

Proof. For even conductor, the finite stationary-layer function is the constant 1 by the even Lamprecht linearization and has no positive polar part. For odd conductor, the critical function has a nondegenerate polar character and is described by a normalized characteristic-two refinement. Checking the three possible ranges $b < t$, $b = t$, and $b > t$ gives exactly the rows in (242); all other parity rows retain at least one odd lower or upper critical source. At the boundary $m = T$, so the only odd boundary is the common odd row. There minimality excludes loss of the final conductor layer and gives $1 + \rho \neq 0$. $\square$

9.5. The four critical functions. By Proposition 9.26, no quadratic refinement is present in an all-even row. Whenever the corresponding conductor is odd, let

$$\gamma_0, \quad \gamma, \quad \gamma_1, \quad \gamma'$$

be the residual linear coefficients in (238) for $\tau, \chi_F, \tau\chi_F, \chi_K$, respectively. If a conductor is even, its finite stationary-layer function is the constant 1 by the even Lamprecht linearization, and the corresponding linear coefficient is omitted. Thus every formula below involving one of these coefficients is asserted only in the parity row in which that coefficient exists. At the boundary $b = t$, the element u in (232) is a unit. Put

$$(243) \quad \rho = \bar{u} \in k^\times.$$

Then $n = N(u)$ has residue $\bar{n} = \rho^2$. On the common critical line, the exact stationary ratios are A for τ and An for χ_F . Hence Lemma 9.2, applied in one common critical coordinate, shows that their polar coefficients have ratio $\bar{n} = \rho^2$. We fix a lift $\tilde{\rho} \in \mathcal{O}_F^\times$ and take the χ_F critical element to be $c_\chi = c_\tau/\tilde{\rho}$ whenever the two lower functions are compared. Thus the normalized χ_F variable is $z \mapsto \rho z$. Put $r = \sqrt{\rho}$; the square root is unique because Frobenius is an automorphism of k .

At the boundary, the critical coefficient of $\tau\chi_F$ is a nonzero scalar times $1 + \rho^2$. It is nonzero because minimality prevents $\tau\chi_F$ from losing its last conductor layer. Since $1 + \rho^2 = (1 + \rho)^2$ in characteristic two, one has $1 + \rho \neq 0$.

Proposition 9.27 (Wild-quadratic critical functions). *There are phase-preserving coordinates for every critical function that occurs. The lower assertions below are made whenever the relevant lower conductors are odd, and each assertion about the upper function is made whenever the upper conductor is odd. In an even-conductor row the corresponding critical function is the constant 1, and any displayed relation involving its linear coefficient is omitted.*

- (i) If $b < t$, the lower product has $\gamma_1 = \gamma_0$, and before the final Frobenius reindexing the upper critical function is

$$\mathbf{q}(z)\psi_0(\gamma z).$$

- (ii) If $b > t$, the lower product has $\gamma_1 = \gamma$, and the raw upper function is

$$\mathbf{q}(z)\psi_0((1 + \gamma_0)z).$$

- (iii) If $b = t$, the two lower functions in the common coordinate are

$$\mathbf{q}(z)\psi_0(\gamma_0 z), \quad \mathbf{q}(\rho z)\psi_0(\rho\gamma z),$$

and the raw upper function is

$$(244) \quad \mathbf{q}(z)\psi_0(\gamma z) \mathbf{q}(\rho z)\psi_0(\rho\gamma_0 z) \psi_0(\rho z).$$

The upper normalized coordinate is obtained from the displayed raw coordinate by precomposition with Frobenius $z \mapsto z^2$.

Proof. Put $m = b + 1$. Corollary 6.3, Proposition 6.10, and Proposition 6.9 identify the stationary classes in the two ranges. Choose the representatives displayed in those propositions and translate

every odd critical function as in Lemma 4.6. With the element A fixed at the beginning of this subsection, the resulting ratios are exact equalities in the fields:

$$(245) \quad \frac{\beta(\chi_F)}{\gamma_F} = An, \quad \frac{\beta_K}{\gamma_K} = A(n-u), \quad n = N(u).$$

For the selected representatives, in the low range these equalities are

$$\frac{\beta}{\delta/\varepsilon} = \frac{\alpha}{\delta} \frac{\varepsilon\beta}{\alpha}, \quad \frac{(\alpha/\varepsilon_1)(n-u)}{\delta/\varepsilon_1} = \frac{\alpha}{\delta}(n-u),$$

and in the high range they are

$$\frac{\beta}{\gamma_F} = \frac{\alpha\delta_h}{\gamma_F} \frac{\beta}{\alpha\delta_h}, \quad \frac{\alpha\delta_h(n-u)}{\gamma_F} = A(n-u).$$

We first establish the lower-function assertions, independently of whether an upper critical function occurs. Suppose $b < t$ and T is odd. Put $d_\tau = (T-1)/2$ and choose $c_\tau \in \mathfrak{p}_F^{d_\tau} \setminus \mathfrak{p}_F^{d_\tau+1}$. Since $m = b+1 < T$,

$$d_\tau \geq \left\lceil \frac{m}{2} \right\rceil,$$

so the stationary linearization of χ_F is valid on the τ -critical line. Choose the sum representative for the stationary class of $\tau\chi_F$. Then, for $z \in k$,

$$\frac{\Phi_{\tau\chi_F}(z)}{\Phi_\tau(z)} = \psi_F(An c_\tau \tilde{z}) \chi_F(1 + c_\tau \tilde{z})^{-1} = 1.$$

Thus the complete lower critical functions agree and $\gamma_1 = \gamma_0$.

Suppose $b > t$ and m is odd. Put $d_\chi = (m-1)/2$ and choose $c_\chi \in \mathfrak{p}_F^{d_\chi} \setminus \mathfrak{p}_F^{d_\chi+1}$. Since $m > T$,

$$d_\chi \geq \left\lceil \frac{T}{2} \right\rceil,$$

so the stationary linearization of τ is valid on the χ_F -critical line. With the sum representative for $\tau\chi_F$ one obtains

$$\frac{\Phi_{\tau\chi_F}(z)}{\Phi_{\chi_F}(z)} = \psi_F(Ac_\chi \tilde{z}) \tau(1 + c_\chi \tilde{z})^{-1} = 1.$$

Hence the complete lower critical functions agree and $\gamma_1 = \gamma$.

At the boundary $b = t$, all relevant lower conductors have the same parity. In the odd row, the common-coordinate convention $c_\chi = c_\tau/\tilde{\rho}$ gives the two lower functions

$$\mathfrak{q}(z)\psi_0(\gamma_0 z), \quad \mathfrak{q}(\rho z)\psi_0(\rho\gamma z),$$

as asserted.

We now assume that the upper conductor is odd and compute its critical function. Let c_K be an upper critical element. For $Z \in k$, choose a lift $\tilde{Z} \in \mathcal{O}_F$, put

$$x_Z = c_K \tilde{Z}, \quad P(y) = N(1+y) - 1 = \text{Tr}(y) + N(y),$$

and let $p_{\text{nat}}(Z)$ and $q_{\text{nat}}(Z)$ be the classes of $P(x_Z)$ and $P(ux_Z)$ on the χ_F - and τ -critical lines, respectively. Thus, whenever the corresponding lower conductor is odd,

$$P(x_Z) \equiv c_\chi \widetilde{p_{\text{nat}}(Z)}, \quad P(ux_Z) \equiv c_\tau \widetilde{q_{\text{nat}}(Z)}$$

modulo the next critical ideal. If an element lies in that next ideal, its class is 0; if a lower conductor is even, the corresponding critical function is 1 and its class is immaterial.

Let

$$e_\chi \equiv m_F(\chi_F) \pmod{2}, \quad e_\tau \equiv T \pmod{2}, \quad e_\chi, e_\tau \in \{0, 1\},$$

and define the lower finite functions by

$$\Phi_{\chi_F}(z) = \begin{cases} h_\gamma(z), & e_\chi = 1, \\ 1, & e_\chi = 0, \end{cases} \quad \Phi_\tau(z) = \begin{cases} h_{\gamma_0}(z), & e_\tau = 1, \\ 1, & e_\tau = 0. \end{cases}$$

In an even row these are literal constant functions: the even stationary linearization cancels the additive and multiplicative factors on the stationary layer. Thus no undefined characteristic-two critical function is being used.

Write $\Phi_K^{\text{nat}}(Z)$ for the upper critical function in this natural coordinate. The defining upper function and (245) give

$$\begin{aligned}\Phi_K^{\text{nat}}(Z) &= \psi_F(A \operatorname{Tr}((n-u)x_Z)) \chi_F(1+P(x_Z))^{-1} \\ &= \Phi_{\chi_F}(p_{\text{nat}}(Z)) \psi_F(-A\{\operatorname{Tr}(ux_Z) + nN(x_Z)\}) \\ &= \Phi_{\chi_F}(p_{\text{nat}}(Z)) \psi_F(-AP(ux_Z)).\end{aligned}$$

For the second equality we used the exact stationary-layer identity

$$\Phi_{\chi_F}(p_{\text{nat}}(Z)) = \psi_F(AnP(x_Z)) \chi_F(1+P(x_Z))^{-1}$$

and cancelled $n \operatorname{Tr}(x_Z)$; the last equality uses $nN(x_Z) = N(ux_Z)$. Since $1+P(ux_Z) = N(1+ux_Z)$ and τ is trivial on norms, the corresponding exact stationary-layer identity is

$$\Phi_{\tau}(q_{\text{nat}}(Z)) = \psi_F(AP(ux_Z)).$$

If $e_{\tau} = 1$, Lemma 9.24 gives $\Phi_{\tau}(q)^2 = \psi_0(q)$; if $e_{\tau} = 0$, then $\Phi_{\tau} \equiv 1$. Hence in both cases

$$\Phi_{\tau}(q)^{-1} = \Phi_{\tau}(q) \psi_0(q)^{e_{\tau}}.$$

Therefore

$$(246) \quad \Phi_K^{\text{nat}}(Z) = \Phi_{\chi_F}(p_{\text{nat}}(Z)) \Phi_{\tau}(q_{\text{nat}}(Z)) \psi_0(q_{\text{nat}}(Z))^{e_{\tau}}.$$

We now calculate the two classes, including their scalars. Write $m_K = 2d_K + 1$ whenever the upper conductor is odd and take $v_K(c_K) = d_K$. The trace-ideal formula gives

$$(247) \quad v_F(\operatorname{Tr}(c_K \tilde{Z})) \geq \left\lfloor \frac{d_K + T}{2} \right\rfloor, \quad v_F(N(c_K \tilde{Z})) = d_K.$$

Thus, in each case in which $P(c_K \tilde{Z})$ reaches a lower critical line, the trace term lies in the next ideal and the initial term is $N(c_K) \tilde{Z}^2$. Consequently its natural class is

$$(248) \quad p_{\text{nat}}(Z) = \lambda_{\chi} Z^2, \quad \lambda_{\chi} = \frac{\overline{N(c_K)}}{c_{\chi}},$$

whenever the χ_F line is reached. Similarly, whenever the τ line is reached and the trace term is one step deeper,

$$(249) \quad q_{\text{nat}}(Z) = \lambda_{\tau} Z^2, \quad \lambda_{\tau} = \frac{\overline{nN(c_K)}}{c_{\tau}}.$$

Both displayed scalars are nonzero when defined. Replacing c_K by ξc_K , with $\xi \in \mathcal{O}_K^{\times}$, multiplies both scalars by $\bar{\xi}^2$. Since Frobenius is an automorphism of k , we may therefore normalize either nonzero scalar to 1 without changing any valuation or the ratio of the two scalars.

Put $a = T - m = v_K(u)$. Suppose first that $b < t$. Then $m_K = m$ and, in the odd row, $d_K = d_{\chi} = (m-1)/2$. Formula (247) and (248) show that $p_{\text{nat}}(Z) = \lambda_{\chi} Z^2$. Scale c_K so that $\lambda_{\chi} = 1$. If T is odd, put $d_{\tau} = (T-1)/2$. Since the odd conductors $m < T$ differ by at least two,

$$v_K(uc_K) = d_K + a = d_{\tau} + \frac{a}{2} \geq d_{\tau} + 1.$$

The norm and trace terms of $P(uc_K \tilde{Z})$ then lie in $\mathfrak{p}_F^{d_{\tau}+1}$, so $q_{\text{nat}}(Z) = 0$; if T is even there is no τ -critical factor. Define the raw variable by $z = Z^2$. Then

$$(250) \quad (p(z), q(z)) = (z, 0) \quad (b < t).$$

Suppose next that $b > t$ and that the upper critical line occurs. Then T is odd. With $d_{\tau} = (T-1)/2$ and $m_K = 2m - T$, one has

$$d_K = m - d_{\tau} - 1, \quad v_K(uc_K) = a + d_K = d_{\tau}.$$

The trace of $uc_K \tilde{Z}$ lies in $\mathfrak{p}_F^{d_\tau+1}$, while its norm has valuation d_τ ; hence (249) applies. Scale c_K so that $\lambda_\tau = 1$. If m is odd and $d_\chi = (m-1)/2$, then

$$d_K - d_\chi = \frac{m-T}{2} \geq 1,$$

so $P(c_K \tilde{Z})$ is beyond the χ_F critical line; if m is even there is no such critical factor. Again putting $z = Z^2$ gives

$$(251) \quad (p(z), q(z)) = (0, z) \quad (b > t).$$

Finally suppose $b = t$. Then $a = 0$, $m_K = m = T$, and in the odd row $d_K = d_\chi = d_\tau$. Both initial terms are norm terms. Equations (248) and (249) give

$$p_{\text{nat}}(Z) = \lambda_\chi Z^2, \quad q_{\text{nat}}(Z) = \lambda_\tau Z^2.$$

The common lower coordinates were fixed by $c_\chi = c_\tau/\tilde{\rho}$, and $\bar{n} = \rho^2$. Therefore

$$(252) \quad \frac{\lambda_\tau}{\lambda_\chi} = \bar{n} \frac{\overline{c_\chi}}{c_\tau} = \rho^2 \rho^{-1} = \rho.$$

Scale c_K so that $\lambda_\chi = 1$. This single scaling preserves (252), and hence simultaneously normalizes the two maps to $p_{\text{nat}}(Z) = Z^2$ and $q_{\text{nat}}(Z) = \rho Z^2$. With the raw variable $z = Z^2$ one obtains

$$(253) \quad (p(z), q(z)) = (z, \rho z) \quad (b = t).$$

Combining the three rows gives

$$(254) \quad \begin{array}{c|cc} & p(z) & q(z) \\ \hline b < t & z & 0 \\ b > t & 0 & z \\ b = t & z & \rho z. \end{array}$$

Define $\Phi_{\chi_K}^{\text{raw}}(z) = \Phi_K^{\text{nat}}(z^{1/2})$; this is well defined because Frobenius is bijective. Substitution of (254) into (246) gives

$$(255) \quad \Phi_{\chi_K}^{\text{raw}}(z) = \begin{cases} \Phi_{\chi_F}(z), & b < t, \\ \Phi_\tau(z)\psi_0(z), & b > t, \\ \Phi_{\chi_F}(z)\Phi_\tau(\rho z)\psi_0(\rho z), & b = t. \end{cases}$$

Indeed, in the first row $q(z) = 0$, while in the second and third rows an upper critical function can occur only when T is odd, so $e_\tau = 1$. The lower assertions were proved before the upper transport calculation, and the upper formulae follow by inserting $h_\lambda(z) = \mathfrak{q}(z)\psi_0(\lambda z)$ in (255). Since the chosen raw variable is $z = Z^2$, the normalized upper function is

$$\Phi_{\chi_K}(Z) = \Phi_{\chi_K}^{\text{raw}}(Z^2),$$

which is exactly precomposition of the raw function with Frobenius. $\square$

Proposition 9.28 (Quadratic coefficient and linear-residue table). *In each parity row in which the indicated coefficients exist, the complete critical functions satisfy*

$$(256) \quad \begin{array}{c|cc} b < t & \gamma_1 = \gamma_0, & (\gamma')^2 = \gamma, \\ b > t & \gamma_1 = \gamma, & (\gamma')^2 = 1 + \gamma_0, \\ b = t & \gamma_1 = \frac{\gamma_0 + \rho\gamma + r}{1 + \rho}, & (\gamma')^2 = \frac{\gamma + \rho\gamma_0 + \rho + r}{1 + \rho}. \end{array}$$

Proof. Only the boundary algebra requires calculation. Multiplying the two lower functions in Proposition 9.27 and using (237) gives

$$\mathfrak{q}((1 + \rho)z)\psi_0((\gamma_0 + \rho\gamma)z + \rho z^2).$$

Since $\psi_0(\rho z^2) = \psi_0(rz)$, normalization by the coordinate $(1 + \rho)z$ gives the displayed formula for γ_1 .

For the upper boundary function, (244) becomes

$$\mathfrak{q}((1 + \rho)z)\psi_0((\gamma + \rho\gamma_0 + \rho)z + \rho z^2).$$

Replacing ρz^2 by rz and normalizing by $(1 + \rho)z$ gives the raw affine coefficient

$$L = \frac{\gamma + \rho\gamma_0 + \rho + r}{1 + \rho}.$$

The upper normalized function is obtained by precomposition with Frobenius. If $L_0^2 = L$, then

$$\mathfrak{q}(z^2)\psi_0(Lz^2) = \mathfrak{q}(z)\psi_0(L_0z),$$

by (237) and Frobenius invariance of the absolute trace. Thus Frobenius replaces L by its unique square root γ' , so $(\gamma')^2 = L$. The two nonboundary rows are the same argument with $L = \gamma$ and $L = 1 + \gamma_0$. $\square$

9.6. Stationary-factor assembly. The elementary stationary value and the odd critical phase are canonical only as a product. We therefore assemble the four complete stationary factors before simplifying their residual phases.

Proposition 9.29 (Wild-quadratic stationary assembly). *Use the compatible stationary-class representatives of Proposition 6.9 in the high range and of Proposition 6.10 in the low range. Use the critical coordinates of Proposition 9.27, and keep each elementary stationary value together with the critical phase translated with it. Then the complete error term has the following form.*

If $b < t$, or if $b + 1$ is even, or if $b > t$ and $b + 1$ is odd, then

$$(257) \quad \mathcal{E}_{K/F}(\chi_F) = \psi_F(-AX).$$

At the only exceptional boundary, namely $b = t$ and $T = t + 1$ odd, put

$$(258) \quad A_\rho = \frac{\rho}{1 + \rho}, \quad B = \frac{\rho^2}{1 + \rho^2}, \quad C = \frac{\rho}{1 + \rho^2}.$$

Then the complete elementary-plus-Hasse quotient is

$$(259) \quad \mathcal{E}_{K/F}(\chi_F) = \mathfrak{q}(A_\rho)\mathfrak{q}(B)\mathfrak{q}(C)\psi_0(a_1),$$

where

$$(260) \quad a_1 = \gamma_1 A_\rho + \gamma B + \gamma_0 C + C + A_\rho.$$

Here C is the elementary correction $\psi_F(-AX) = \psi_0(C)$, and the last term A_ρ is the affine term which occurs when the raw upper and lower Hasse quotients are compared.

Proof. We first assemble the elementary stationary factors without making any linear approximation. Corollary 6.3 identifies the pullback class, Lemma 4.3(c) identifies the twist class, and Propositions 6.9 and 6.10 provide simultaneous representatives. At $m = T$, Lemma 4.8 excludes cancellation of the twist's leading class. Choose these representatives and retain the translated critical functions required by Lemma 4.6. In the high range put $c = \alpha\delta_h$; then $A = c/\gamma_F$, $\beta = cn$, and the four pairs (admissible element, selected stationary representative) are

$$(261) \quad \begin{array}{c|c} \chi_K & (\gamma_F, c(n - u)), \\ \tau & (\gamma_F/c, 1), \\ \chi_F & (\gamma_F, cn), \\ \tau\chi_F & (\gamma_F, c(n + 1)). \end{array}$$

In the low range the selected pairs are

$$(262) \quad \begin{array}{c|c} \chi_K & (\delta/\varepsilon_1, (\alpha/\varepsilon_1)(n - u)), \\ \tau & (\delta/\alpha, 1), \\ \chi_F & (\delta/\varepsilon, (\alpha/\varepsilon)n), \\ \tau\chi_F & (\delta, \alpha(n + 1)). \end{array}$$

These are the representatives of the stationary classes already proved, with their common scalar factors displayed. They are not asserted to be canonical field elements; the accompanying translated critical functions make each complete stationary factor independent of the selection.

The scalar and admissible-element contributions cancel exactly. For example, in the high range the quotient of the character values at the admissible elements is $\tau(c)^{-1}$, whereas the quotient of the scalar parts of the four selected stationary representatives is $\tau(c)$. In the low range the χ_F -parts cancel because $N(\delta/\varepsilon_1) = \delta^2/\varepsilon$ and $N(\alpha/\varepsilon_1) = \alpha^2/\varepsilon$, and the remaining τ -parts are trivial because α is a norm.

After these cancellations the multiplicative stationary values are

$$(263) \quad \chi_F(z_0)^{-1} \tau(z_1)^{-1}.$$

Indeed,

$$n - u = -u(1 + u^\vee), \quad N(n - u) = nN(1 + u^\vee) = n(1 + n)z_0,$$

so the χ_F -quotient is $\chi_F(z_0)^{-1}$. Also $N(1 + u) = (1 + n)z_1$ and τ is trivial on norms, whence $\tau(1 + n) = \tau(z_1)^{-1}$. The signed sum of the additive contributions of the selected stationary representatives in either (261) or (262) is

$$\mathrm{Tr}_{K/F}(n - u) + 1 - n - (n + 1) = -s.$$

Consequently, before the odd critical corrections are inserted, the complete quotient is

$$(264) \quad R_H \psi_F(-As) \chi_F(1 + x)^{-1} \tau(1 + y)^{-1},$$

where

$$(265) \quad R_H = \frac{H(\gamma')^{e_K} H(\gamma_0)^{e_\tau}}{H(\gamma)^{e_\chi} H(\gamma_1)^{e_{\tau\chi}}}$$

and e_θ is 1 or 0 according as the conductor of θ is odd or even.

Suppose an odd critical factor $h_\lambda(z) = \mathfrak{q}(z)\psi_0(\lambda z)$ receives a correction $1 + w$ whose normalized critical class is c . Lemma 9.25 gives

$$(266) \quad \theta(1 + w)^{-1} = \psi_E\left(-\frac{\beta_\theta w}{\gamma_\theta}\right) \mathfrak{q}(c)\psi_0(\lambda c).$$

Thus the correction factor is $\mathfrak{q}(c)$, rather than its complex conjugate; this orientation follows from the occurrence of $\theta(\beta)^{-1}$ in Lamprecht's formula. Inserting the two linear parts in (264) gives

$$(267) \quad \psi_F(-A(s + nx + y)) = \psi_F(-AX)$$

by Lemma 9.22. It remains to evaluate R_H and the two finite correction values.

Let $\epsilon(r) \in \{0, 1\}$ denote the parity of the integer r and put $m = b + 1$. The conductor formulas give

$$(268) \quad \begin{array}{c|c|c|c|c} & e_K & e_\tau & e_\chi & e_{\tau\chi} \\ \hline b < t & \epsilon(m) & \epsilon(T) & \epsilon(m) & \epsilon(T) \\ b = t & \epsilon(T) & \epsilon(T) & \epsilon(T) & \epsilon(T) \\ b > t & \epsilon(T) & \epsilon(T) & \epsilon(m) & \epsilon(m). \end{array}$$

Here the last row uses $m_K = 2m - T$ and $m_F(\tau\chi_F) = m$ from Lemma 4.8. In the middle row $b = t$, (46) gives $m_K = T$, while Lemma 4.8 gives $m_F(\tau\chi_F) = T$.

If $b < t$, Proposition 9.27 identifies the upper critical function with that of χ_F and the lower product with that of τ ; hence $R_H = 1$. Moreover the trace-ideal formula gives

$$v_F(s) \geq \left\lfloor \frac{T - m + T}{2} \right\rfloor.$$

This is at least one step beyond every odd critical line of χ_F and τ , because $m < T$. Thus neither x nor y supplies a critical correction, and the residual quotient is 1.

Assume $b > t$. If m is even, the χ_F and $\tau\chi_F$ factors are even. If T is also even, $R_H = 1$. If T is odd, Frobenius invariance of the finite sum and $(\gamma')^2 = 1 + \gamma_0$ give

$$R_H = H(1 + \gamma_0)H(\gamma_0) = 1.$$

Indeed,

$$H(1 + \gamma_0)H(\gamma_0) = H_0^2 \overline{\mathfrak{q}(1 + \gamma_0)\mathfrak{q}(\gamma_0)} = 1$$

by (237) and $H_0^2 = \mathfrak{q}(1)$. The same trace bound, now using $v_F(1+n) = T-m$, places y at least one step beyond the τ critical line, so there is no correction. Thus the residual quotient is again 1.

Now let $b > t$ and m be odd. In both parities of T one obtains

$$(269) \quad R_H = \frac{1}{\mathfrak{q}(1)\psi_0(\gamma)}.$$

For T even this is simply $H(\gamma)^{-2}$; for T odd, the numerator in (265) is $H(1+\gamma_0)H(\gamma_0) = 1$, and the denominator is again $H(\gamma)^2 = \mathfrak{q}(1)\psi_0(\gamma)$. Put $c = m - T > 0$ and $w = u^{-1}$. Then $v_K(w) = c$ and $c + T = m$ is odd. Lemma 9.23(b) shows that

$$c_\chi := -\mathrm{Tr}(w) \in \mathfrak{p}_F^{(m-1)/2} \setminus \mathfrak{p}_F^{(m+1)/2}$$

is a valid critical element for χ_F . We must also check that this critical element is phase preserving. Since

$$v_F(nc_\chi^2) = (T-m) + (m-1) = T-1,$$

the map

$$\lambda : k \longrightarrow \mathbf{C}^\times, \quad \lambda(z) = \psi_F(An c_\chi^2 \tilde{z})$$

is the nontrivial character induced by τ on U_F^{T-1}/U_F^T . Indeed, $T \geq 2$ and hence $T-1 \geq \lceil T/2 \rceil$, so the stationary linearization of τ is valid on U_F^{T-1} ; nontriviality follows from the exact conductor T of τ .

Put

$$r = \frac{w}{\mathrm{Tr}(w)}.$$

Then $\mathrm{Tr}(r) = 1$ and, since $N(w) = n^{-1}$,

$$N(r) = \frac{N(w)}{\mathrm{Tr}(w)^2} = \frac{1}{nc_\chi^2}, \quad N(r)^{-1} = nc_\chi^2.$$

Moreover,

$$v_K(r) = v_K(w) - v_K(\mathrm{Tr}(w)) = (m-T) - (m-1) = 1-T.$$

For every $z \in k$ and every lift $\tilde{z} \in \mathcal{O}_F$, the element $1 + (\tilde{z}/N(r))r$ is a unit of K : indeed,

$$v_K\left(\frac{\tilde{z}}{N(r)}r\right) \geq 2(T-1) + (1-T) = T-1 > 0.$$

The exact quadratic norm identity therefore gives

$$\begin{aligned} N_{K/F}\left(1 + \frac{\tilde{z}}{N(r)}r\right) &= 1 + \frac{\tilde{z}}{N(r)}\mathrm{Tr}(r) + \frac{\tilde{z}^2}{N(r)^2}N(r) \\ &= 1 + nc_\chi^2(\tilde{z} + \tilde{z}^2). \end{aligned}$$

Since τ is trivial on norms, its stationary linearization gives

$$\lambda(z + z^2) = 1 \quad (z \in k).$$

The image of the Artin-Schreier map $z \mapsto z^2 + z$ is $\ker \mathrm{Tr}_{k/\mathbf{F}_2}$. Thus λ is a nontrivial additive character trivial on the trace-zero hyperplane. There is exactly one such character, namely $z \mapsto (-1)^{\mathrm{Tr}_{k/\mathbf{F}_2}(z)} = \psi_0(z)$. Consequently

$$(270) \quad \psi_F(An c_\chi^2 \tilde{z}) = \psi_0(z),$$

so $c_\chi = -\mathrm{Tr}(w)$ is phase preserving and may be used in the normalized critical-function calculation. Since

$$\frac{s}{n} = \mathrm{Tr}(u^{-1}) = \mathrm{Tr}(w)$$

and $x = -s/(1+n)$, one has the exact identity

$$\frac{x}{c_\chi} = \frac{n}{1+n} = 1 - \frac{1}{1+n}.$$

Here $v_F(n) = T - m = -c$, so $v_F((1+n)^{-1}) = c > 0$. Therefore the normalized correction class is

$$(271) \quad x' = \overline{x/c_\chi} = 1.$$

The correction in (266) is then $\mathfrak{q}(1)\psi_0(\gamma)$, which is the inverse of (269). Hence the residual quotient is 1 in every nonboundary case. Combining this with (267) proves (257).

It remains to assume $b = t$ and T odd. All four critical factors are present. Put

$$L = \frac{\gamma + \rho\gamma_0 + \rho + r}{1 + \rho}, \quad A_\rho = \frac{\rho}{1 + \rho}.$$

Since $(\gamma')^2 = L$ and the finite sum is Frobenius-invariant, $H(\gamma') = H(L)$. Hence

$$\begin{aligned} R_H &= \frac{\overline{\mathfrak{q}(L)\mathfrak{q}(\gamma_0)}}{\overline{\mathfrak{q}(\gamma)\mathfrak{q}(\gamma_1)}} \\ &= \mathfrak{q}(\gamma)\mathfrak{q}(\gamma_1)\overline{\mathfrak{q}(L)\mathfrak{q}(\gamma_0)}. \end{aligned}$$

Set $X_0 = \gamma + \gamma_1$ and $Y_0 = L + \gamma_0$. In characteristic two, substitution of the boundary formulae gives

$$\begin{aligned} X_0 + Y_0 &= \gamma + \gamma_0 + \frac{\gamma_0 + \rho\gamma + r + \gamma + \rho\gamma_0 + \rho + r}{1 + \rho} \\ &= \frac{\rho}{1 + \rho} = A_\rho. \end{aligned}$$

Thus $X_0 = Y_0 + A_\rho$. The refinement law therefore gives

$$(272) \quad R_H = \mathfrak{q}(A_\rho)\psi_0(E), \quad E = Y_0A_\rho + \gamma\gamma_1 + L\gamma_0.$$

Let $a_\rho = r/(1+r)$, so $a_\rho^2 = A_\rho$. A direct expansion in characteristic two gives

$$(273) \quad E + \gamma_1A_\rho + A_\rho = (a_\rho(\gamma + \gamma_0) + A_\rho)^2 + (a_\rho(\gamma + \gamma_0) + A_\rho).$$

The right side has absolute trace zero. Thus

$$(274) \quad R_H = \mathfrak{q}(A_\rho)\psi_0(\gamma_1A_\rho + A_\rho).$$

We now determine the two correction classes directly from the graded trace. Put $d_\tau = (T-1)/2$. By Lemma 9.23(a), $v_F(2) = d_\tau$, so we may choose

$$(275) \quad c_\tau = 2, \quad c_\chi = 2/\tilde{\rho}.$$

Since $u - \tilde{\rho} \in \mathfrak{p}_K$, the same lemma gives

$$s - 2\tilde{\rho} = \text{Tr}(u - \tilde{\rho}) \in \mathfrak{p}_F^{d_\tau+1}.$$

Consequently

$$(276) \quad \overline{s/c_\tau} = \rho.$$

Moreover, our choice $c_\tau = 2$ gives

$$(277) \quad \overline{2/c_\tau} = 1.$$

Together with $\bar{n} = \rho^2$, Lemma 9.22 gives

$$(278) \quad x' = \overline{x/c_\chi} = \frac{\rho^2}{1 + \rho^2} = B, \quad y' = \overline{y/c_\tau} = \frac{\rho}{1 + \rho^2} = C.$$

We next prove the exact normalization of the polar character; the valuation of c_τ alone does not determine its residual scalar. Since this is the odd boundary and the endpoint conductors have already been removed, $T \geq 3$. Hence $T-1 \geq \lceil T/2 \rceil$, so the stationary linearization of τ is valid on $4\mathcal{O}_F$. Since $c_\tau = 2$ and $v_F(4) = T-1$, the map

$$\lambda : k \longrightarrow \mathbf{C}^\times, \quad \lambda(z) = \psi_F(Ac_\tau^2\tilde{z}) = \tau(1 + 4\tilde{z})$$

is the character induced by τ on U_F^{T-1}/U_F^T . It is nontrivial because τ has exact conductor T . For every $z \in k$, the element $1 + 2\tilde{z}$ lies in $F^\times \subset K^\times$, and hence

$$N_{K/F}(1 + 2\tilde{z}) = (1 + 2\tilde{z})^2 = 1 + 4(\tilde{z} + \tilde{z}^2).$$

The character τ is trivial on norms, so its stationary linearization gives

$$\lambda(z + z^2) = 1.$$

The image of the Artin–Schreier map $z \mapsto z^2 + z$ is exactly $\ker \text{Tr}_{k/\mathbf{F}_2}$: it is contained there, its kernel is $\mathbf{F}_2$, and therefore its image has $|k|/2$ elements. Thus λ is a nontrivial additive character trivial on the trace-zero hyperplane. There is only one such character, namely $z \mapsto (-1)^{\text{Tr}_{k/\mathbf{F}_2}(z)} = \psi_0(z)$. Consequently

$$(279) \quad \psi_F(Ac_\tau^2 \tilde{z}) = \psi_0(z) \quad (z \in k).$$

At the boundary $X = 2y = c_\tau y$. Since $y/c_\tau \equiv C \pmod{\mathfrak{p}_F}$, one has $X \equiv c_\tau^2 \tilde{C} \pmod{\mathfrak{p}_F^T}$ for any lift $\tilde{C} \in \mathcal{O}_F$ of C ; the character $x \mapsto \psi_F(Ax)$ is trivial on $\mathfrak{p}_F^T$. Thus

$$(280) \quad \psi_F(-AX) = \psi_0(C),$$

where the sign disappears after reduction to the residue field of characteristic two.

The two factors from (266) are $\mathfrak{q}(B)\psi_0(\gamma B)$ and $\mathfrak{q}(C)\psi_0(\gamma_0 C)$. Multiplying them with (274) and (280) yields

$$\mathcal{E}_{K/F}(\chi_F) = \mathfrak{q}(A_\rho)\mathfrak{q}(B)\mathfrak{q}(C)\psi_0(\gamma_1 A_\rho + \gamma B + \gamma_0 C + C + A_\rho),$$

which is (259) and (260). $\square$

9.7. Final quadratic cancellation.

Lemma 9.30 (Artin–Schreier parity of the quadratic break). *Assume $\text{char } F = 2$ and let K/F be ramified quadratic. Then its unique lower ramification break t is odd. Consequently $T = t + 1$ is even.*

Proof. Because K/F is separable quadratic in characteristic two, it has an Artin–Schreier presentation

$$K = F(y), \quad y^2 - y = a.$$

Replacing y by $y - c$ replaces a by $a - (c^2 - c)$. Choose a uniformizer π_F . If the leading pole order $r = -v_F(a) > 0$ is even, write its leading residue coefficient as $\bar{d}^2$, possible because the finite residue field is perfect, and take a lift $d \in \mathcal{O}_F$. Subtracting

$$(d\pi_F^{-r/2})^2 - d\pi_F^{-r/2}$$

cancels the term of pole order r and introduces only terms of strictly smaller pole order. Repeating this finite process gives an equivalent equation for which either $v_F(a) \geq 0$ or

$$v_F(a) = -t < 0 \quad \text{with } t \text{ odd.}$$

The first alternative cannot give a ramified quadratic extension: its reduction is either irreducible, in which case the extension is unramified, or has a root, in which case Hensel’s lemma (the derivative is -1) removes the extension. Thus the second alternative holds.

Normalize v_K by $v_K(K^\times) = \mathbf{Z}$. Since $v_K|_F = 2v_F$ and $v_K(y) < 0$, the equation gives

$$2v_K(y) = v_K(a) = -2t, \quad v_K(y) = -t.$$

Hence

$$\pi_K = \pi_F^{(t+1)/2} y$$

is a uniformizer of K . For the nontrivial automorphism $\sigma(y) = y + 1$,

$$\sigma(\pi_K) - \pi_K = \pi_F^{(t+1)/2},$$

whose K -valuation is $t + 1$. Therefore the lower break is exactly t , and it is odd. $\square$

Lemma 9.31 (Depth outside the odd boundary). *If $\text{char } F = 2$, then $X = 0$. If $\text{char } F = 0$ and $a \neq 0$, then $X \in \mathfrak{p}_F^T$; the same is true when $a = 0$ and T is even. Hence $\psi_F(AX) = 1$ in all these cases.*

Proof. In equal characteristic two, the exact identity (233) gives

$$X = \frac{2s}{1+n} = 0;$$

the denominator is nonzero by the minimality condition stated before Lemma 9.22.

Assume now $\text{char } F = 0$ and put $e = v_F(2)$. Since $2 = \text{Tr}_{K/F}(1)$, the trace-ideal formula gives $e \geq \lfloor T/2 \rfloor$. Also $v_F(s) \geq \lfloor (a+T)/2 \rfloor$. If $a > 0$, $1+n$ is a unit and

$$v_F(X) \geq e + \left\lfloor \frac{a+T}{2} \right\rfloor \geq T.$$

If $a < 0$, put $c = -a$; then $v_F(1+n) = a = -c$ and

$$v_F(X) \geq e + \left\lfloor \frac{T-c}{2} \right\rfloor + c = e + \left\lfloor \frac{T+c}{2} \right\rfloor \geq T.$$

If $a = 0$, the bound is $e + \lfloor T/2 \rfloor$, which is at least T when T is even. Finally $\psi_F(\text{Ap}_F^T) = 1$ because τ has conductor T and is linearized by A . $\square$

Proposition 9.32 (Wild quadratic cancellation). *For every wildly ramified quadratic extension K/F of nonarchimedean local fields and every quasi-character χ_F of $F^\times$,*

$$\mathcal{E}_{K/F}(\chi_F) = 1.$$

Proof. By Lemma 2.14, replace χ_F by a member of minimal conductor in its $S(K/F)$ -orbit. The cases $m_F(\chi_F) = 0$ and $m_F(\chi_F) = 1$ were proved in Subsection 9.1. Hence assume $m_F(\chi_F) > 1$; all stationary parameters and assembly formulas of this subsection are now defined.

Suppose first that $\text{char } F = 2$. By Lemma 9.30, T is even, so the exceptional boundary $b = t$, T odd cannot occur. Equation (257) therefore applies, and Lemma 9.31 gives $X = 0$. Hence $\mathcal{E}_{K/F}(\chi_F) = 1$.

Assume now $\text{char } F = 0$. Outside the boundary $b = t$, T odd, equation (257) and Lemma 9.31 give $\mathcal{E}_{K/F}(\chi_F) = \psi_F(-AX) = 1$.

It remains to take $b = t$ and T odd. Use the notation (258). Then $B + C = A_\rho$. By (259),

$$(281) \quad \mathcal{E}_{K/F}(\chi_F) = \mathfrak{q}(A_\rho)\mathfrak{q}(B)\mathfrak{q}(C)\psi_0(a_1).$$

Since $B + C = A_\rho$, the refinement identity gives

$$\mathfrak{q}(B)\mathfrak{q}(C) = \mathfrak{q}(A_\rho)\psi_0(BC), \quad BC = \frac{\rho^3}{1+\rho^4}.$$

Since $\mathfrak{q}(A_\rho)^2 = \psi_0(A_\rho)$, (281) is $\psi_0(a_2)$ with

$$a_2 = a_1 + A_\rho + \frac{\rho^3}{1+\rho^4}.$$

Substituting $\gamma_1 = (\gamma_0 + \rho\gamma + r)/(1+\rho)$ cancels the terms in γ and γ_0 and gives

$$(282) \quad a_2 = \frac{r^7 + r^3 + r^2}{1+r^8} = \frac{\rho^3}{(1+\rho^2)^2} + \frac{\rho(1+r)}{(1+\rho)^2}.$$

The first term on the right is the square of $r^3/(1+\rho^2)$. Frobenius invariance of ψ_0 therefore reduces its phase to that of

$$a_4 = \frac{r^3}{1+\rho^2} + \frac{\rho(1+r)}{(1+\rho)^2} = \frac{\rho}{1+\rho^2} = \frac{\rho}{(1+\rho)^2}.$$

With $c = (1+\rho)^{-1}$, $a_4 = c + c^2$, so $\text{Tr}_{k/\mathbf{F}_2}(a_4) = 0$ and $\psi_0(a_4) = 1$. This proves the boundary case and hence the proposition. $\square$

10. COMPLETION OF THE FIRST MAIN LEMMA

Proof of Theorem 1.1. Let F be a nonarchimedean local field and let K/F be cyclic of prime degree ℓ . If K/F is unramified, apply Proposition 7.1. If it is ramified and tame, then it is totally ramified; the odd-degree and quadratic cases are Propositions 7.3 and 7.4.

Suppose that K/F is wild. Then $\ell = p$, the residue characteristic, and K/F is totally ramified with one lower break t . By Lemma 2.14, replace χ_F by a member of minimal conductor in its $S(K/F)$ -orbit. Conductors 0 and 1 are treated in Subsection 9.1. For odd p , the stable range is Proposition 8.1, and the remaining non-stable range is Corollary 9.21. For $p = 2$, the wild quadratic calculation is Proposition 9.32. These cases exhaust all cyclic extensions of prime degree. Hence $\mathcal{E}_{K/F}(\chi_F) = 1$, which is exactly (1). $\square$

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